NIC26

Own the Vibe

GitHub Copilot and M365 Copilot

For ITPros

Ståle Hansen
CEO, Principal Cloud Architect
Copilot MVP & RD

Simon Skotheimsvik
Senior Cloud Architect
Intune MVP

NIC26

CloudWay

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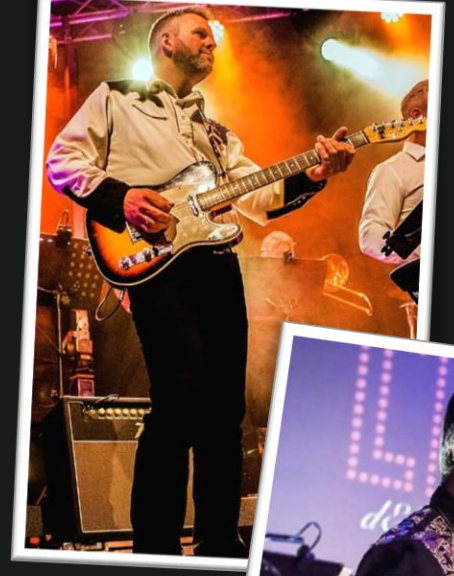
CloudWay



NIC26

UNITED STATES PATENT OFFICE

Application April 23, 1951, Serial No. 14,931

Aug. 14, 1951

Des. 164,227

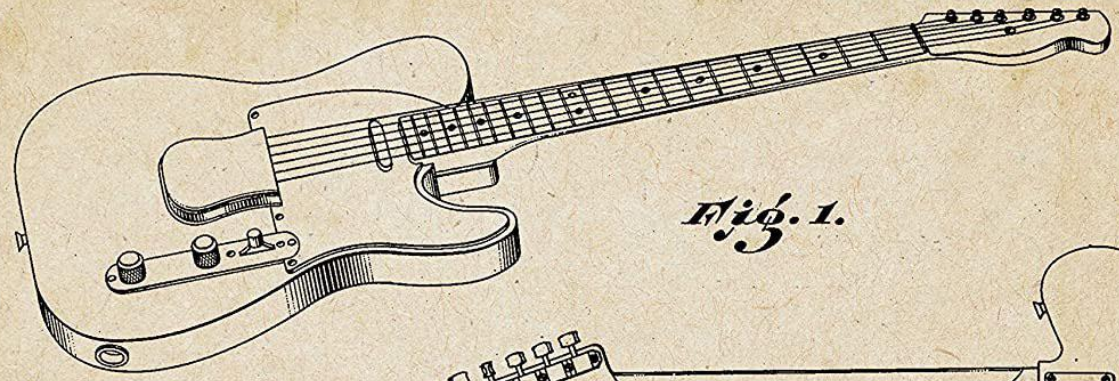


Fig. 1.

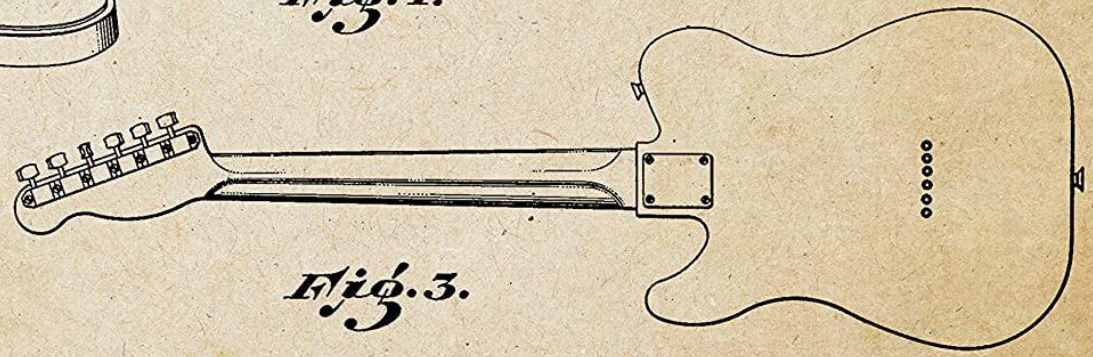


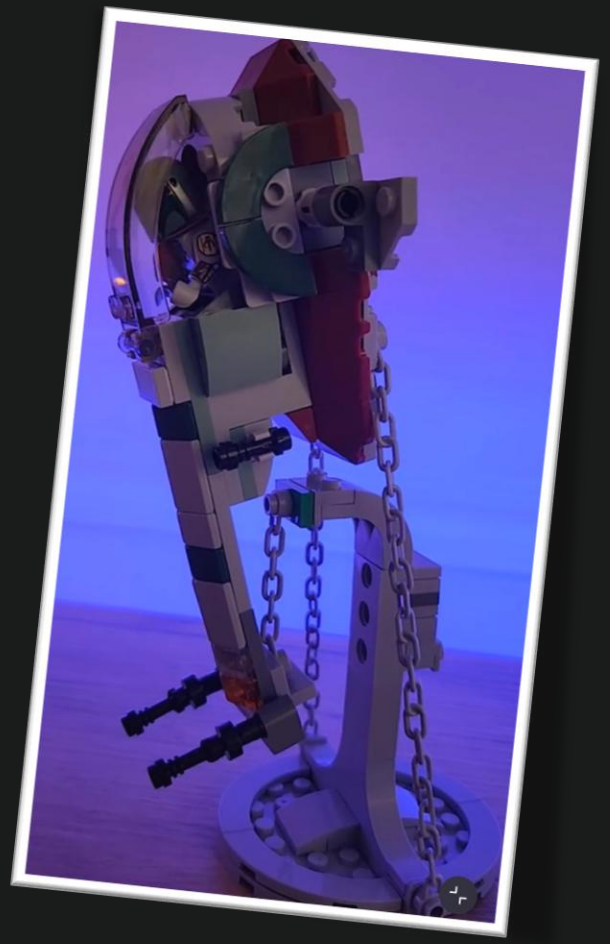
Fig. 3.

Fig. 2.

CLARENCE L. FENDER,
INVENTOR.

Lyon & Lyon
ATTORNEYS.

NIC26



G.K. CHRISTIANSEN

567913

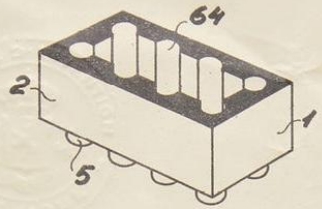


Fig. 9

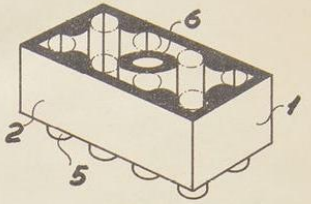


Fig. 10

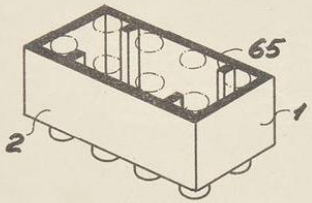


Fig. 11

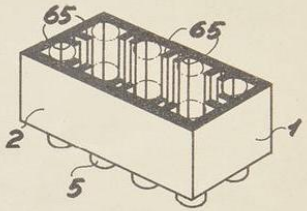


Fig. 12

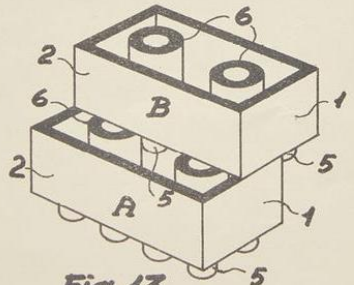


Fig. 13

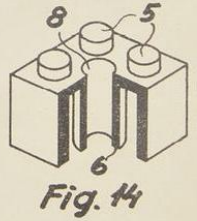


Fig. 14

Bruxelles, le 21 mai 1958
P.Pon. de G.K. CHRISTIANSEN
OFFICE KIRKPATRICK- C.T.PLUCKER 25 IX 1958

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Today we will vibe!

NIC26

The Birth of Vibe Coding

Andrej Karpathy

- Co-founder of OpenAI, former Tesla AI lead

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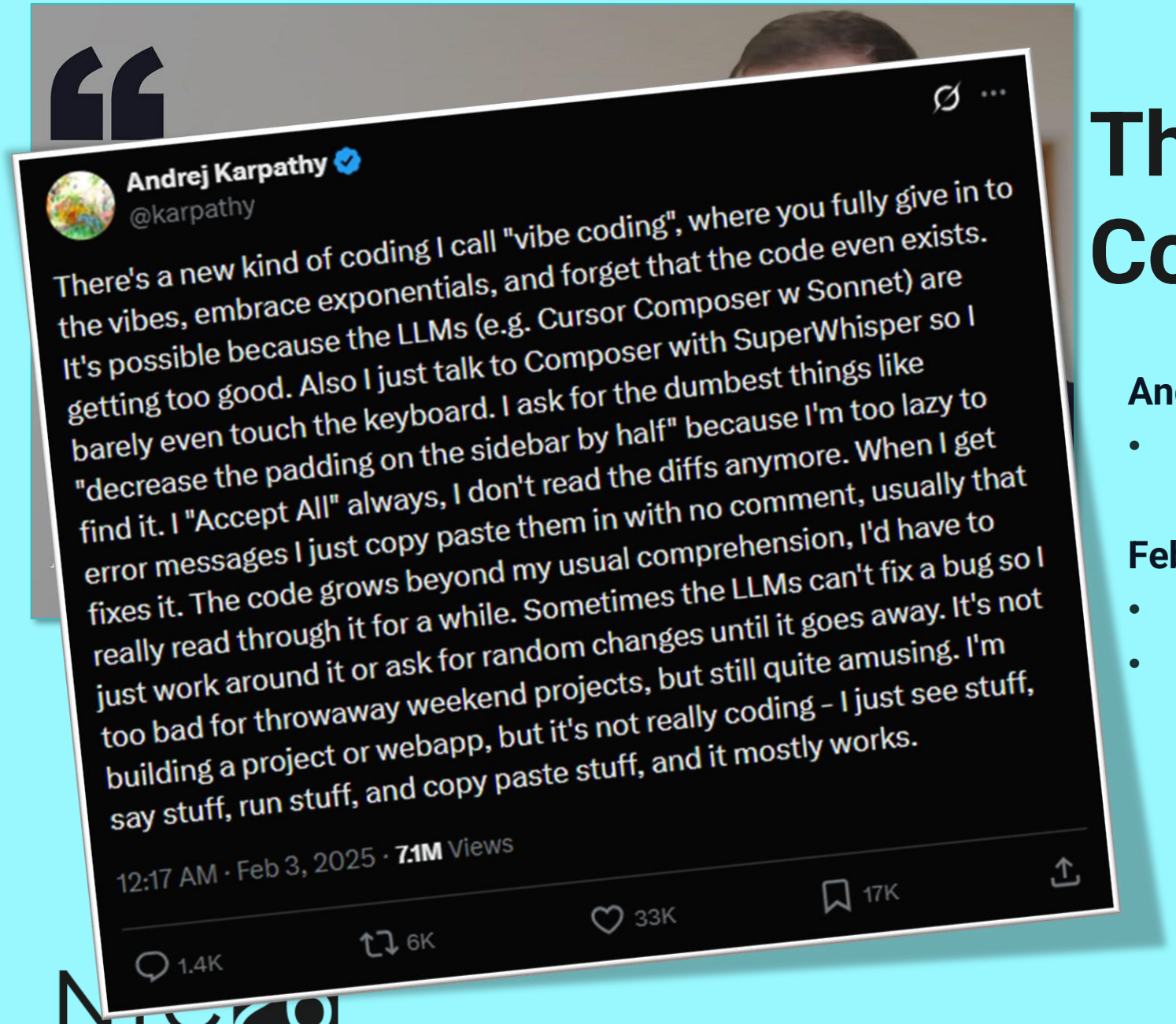


The Birth of Vibe Coding

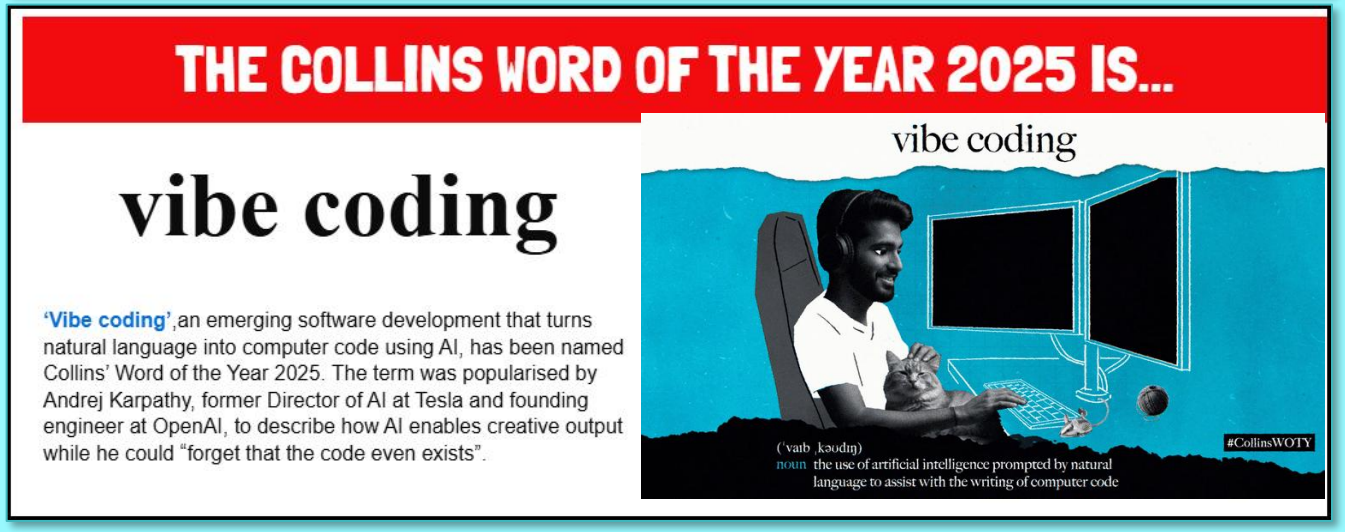
Andrej Karpathy

- Co-founder of OpenAI, former Tesla AI lead

February 3, 2025

- Posted on X about a “new kind of coding”
- 7.1+ million views, global discussion

https://x.com/karpathy/status/1886192184808149383



The Birth of Vibe Coding

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- **Collins Word of the Year 2025: Vibe Coding**
- From a “shower thought” tweet to defining a movement

Vibe Coding Defined

Describe what you want in natural language

- AI generates the code for you
- You run it, test it, iterate through conversation

The key shift:

- You focus on **WHAT** the software should do
- AI handles **HOW** each line is written

“The hottest new programming language is English”

— Andrej Karpathy

NIC26

Vibe coding

Vibe coding
&
Vibe productivity

NIC26

ITPro Scenario:

create a change request change all departments to a unified standard

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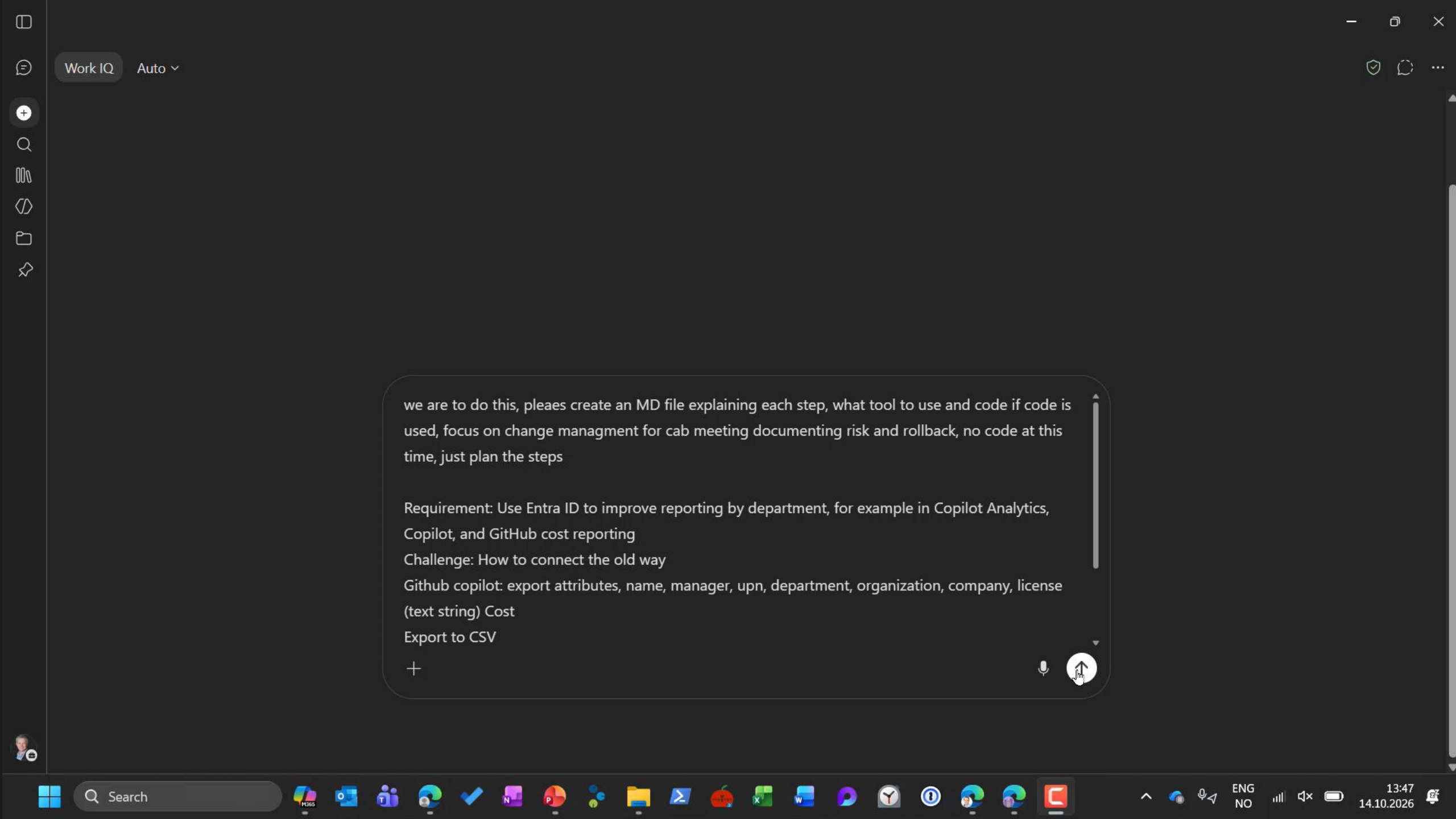
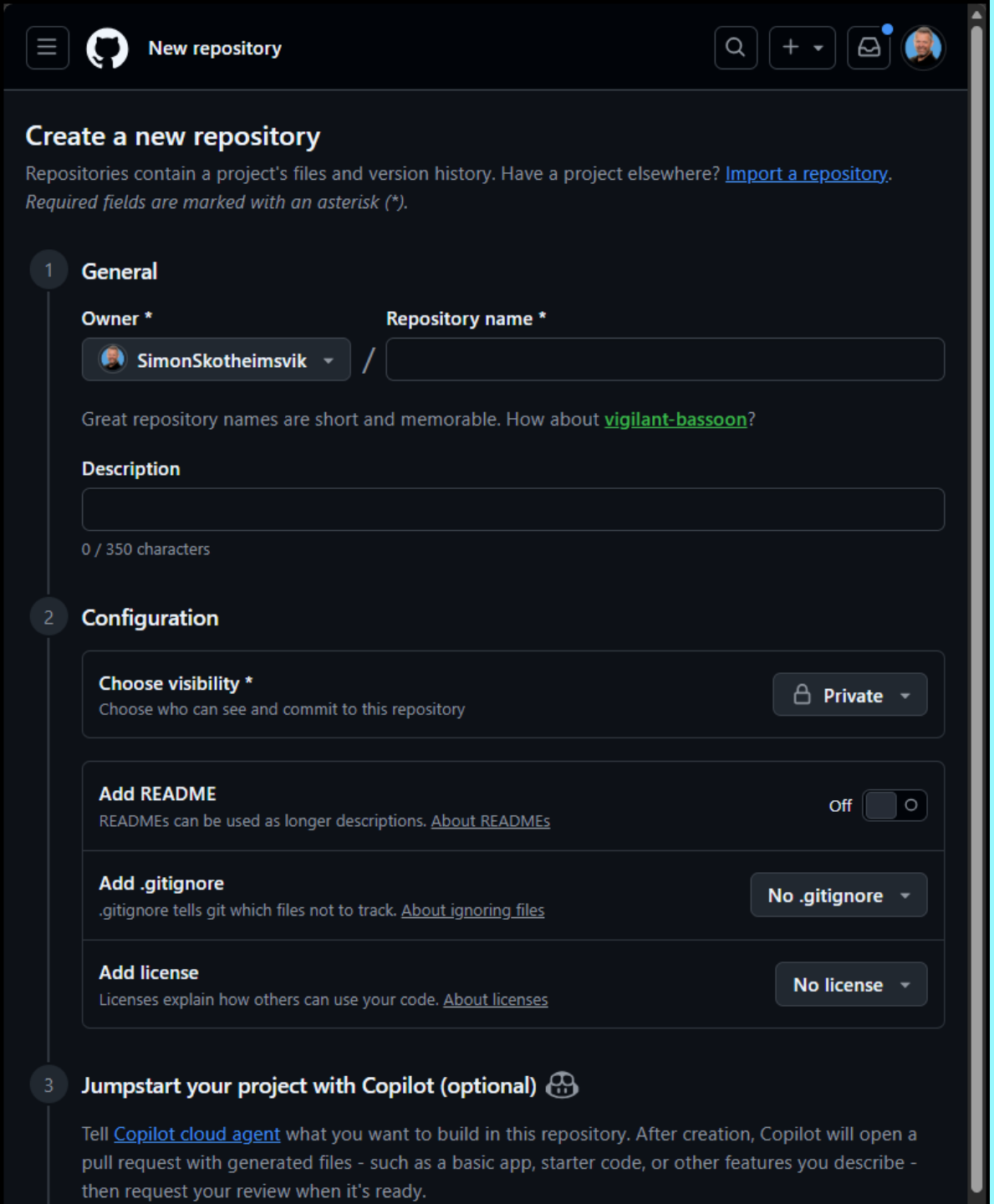
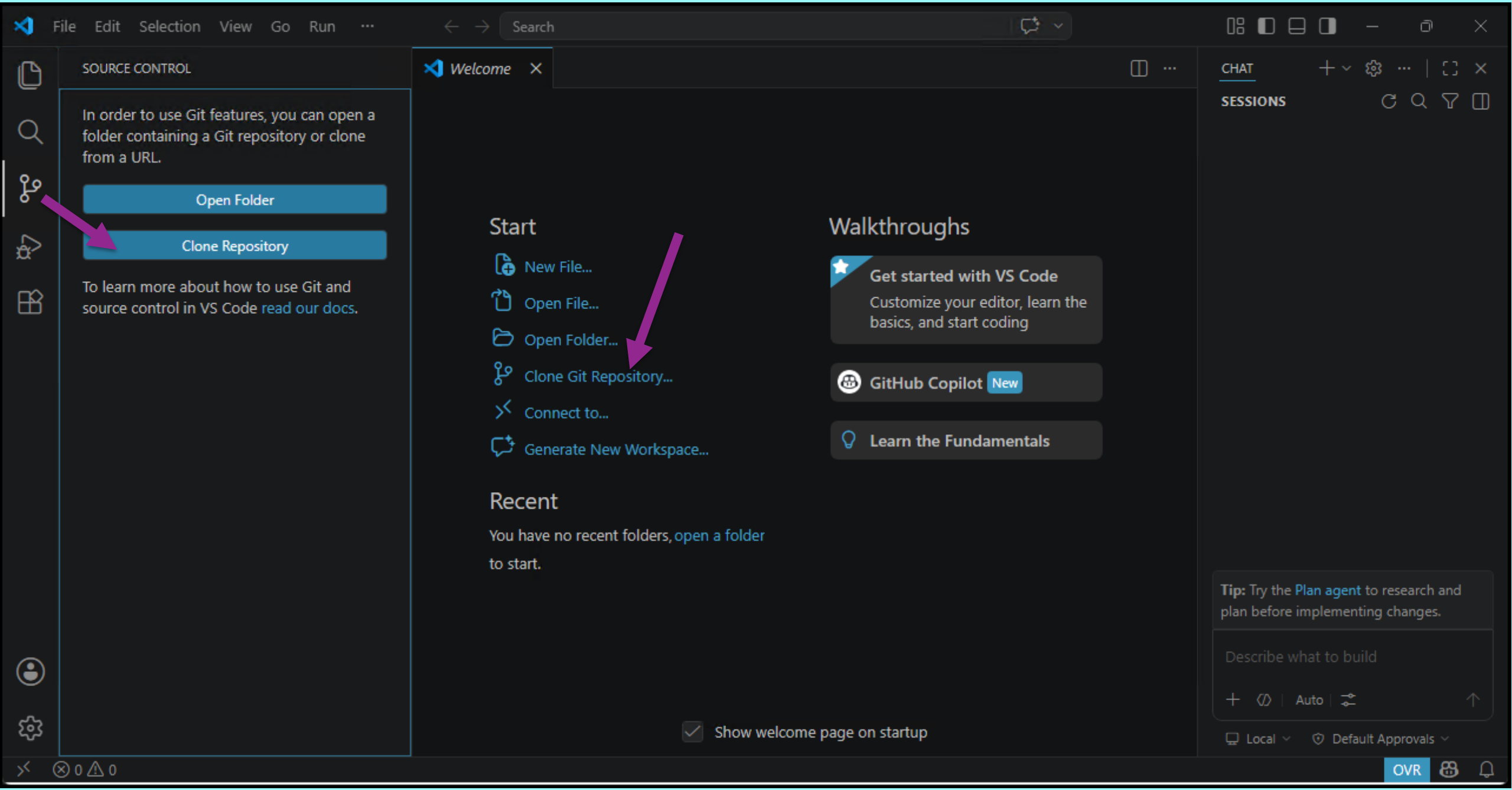

Work IQ
Auto
we are to do this, pleaes create an MD file explaining each step, what tool to use and code if code is used, focus on change managment for cab meeting documenting risk and rollback, no code at this time, just plan the steps
Requirement: Use Entra ID to improve reporting by department, for example in Copilot Analytics, Copilot, and GitHub cost reporting
Challenge: How to connect the old way
Github copilot: export attributes, name, manager, upn, department, organization, company, license (text string) Cost
Export to CSV
Search
ENG
NO
13:47
14.10.2026

If you are doing code in
M365 Copilot
you are doing it wrong

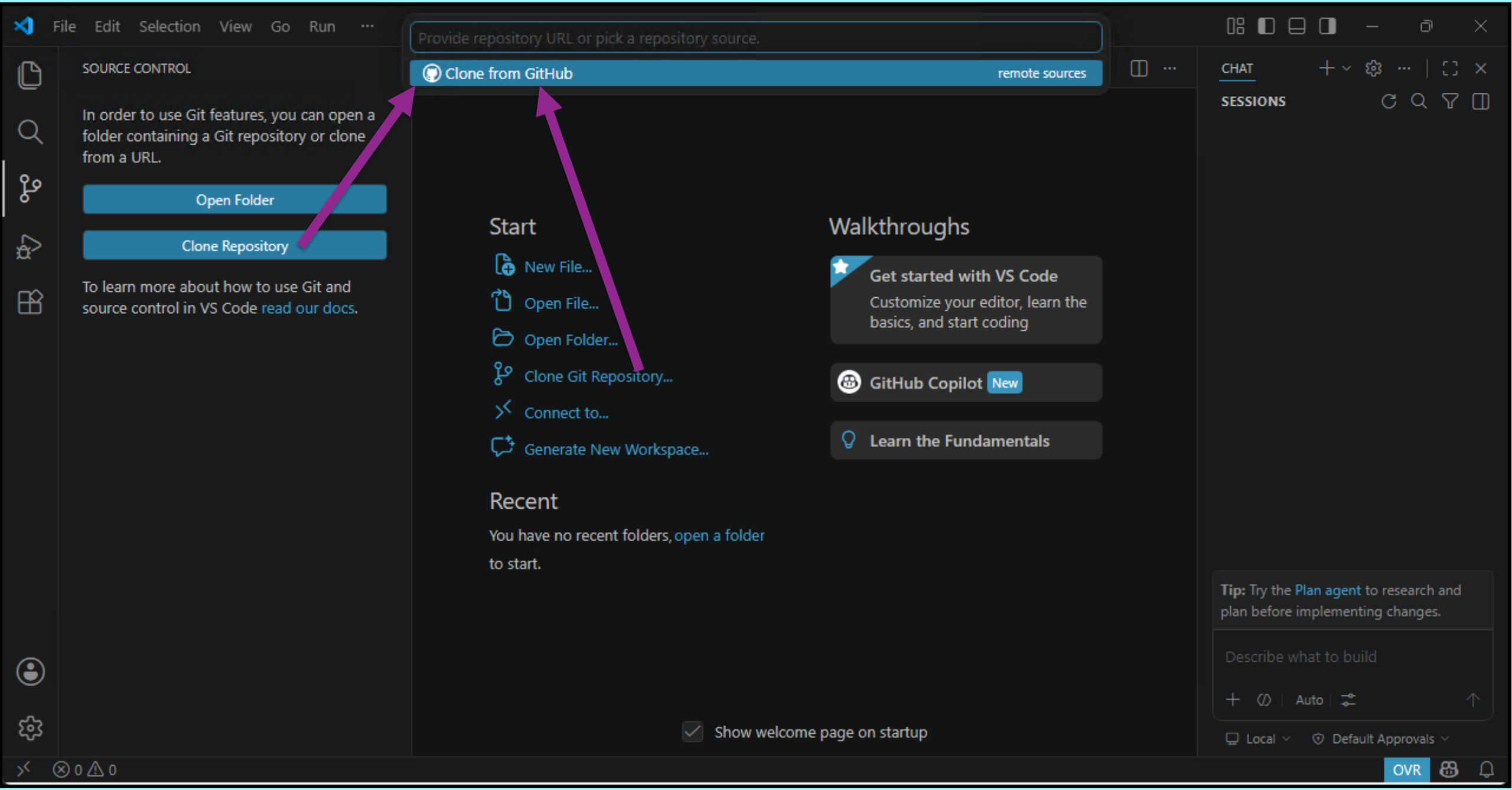


GitHub Code Repositories

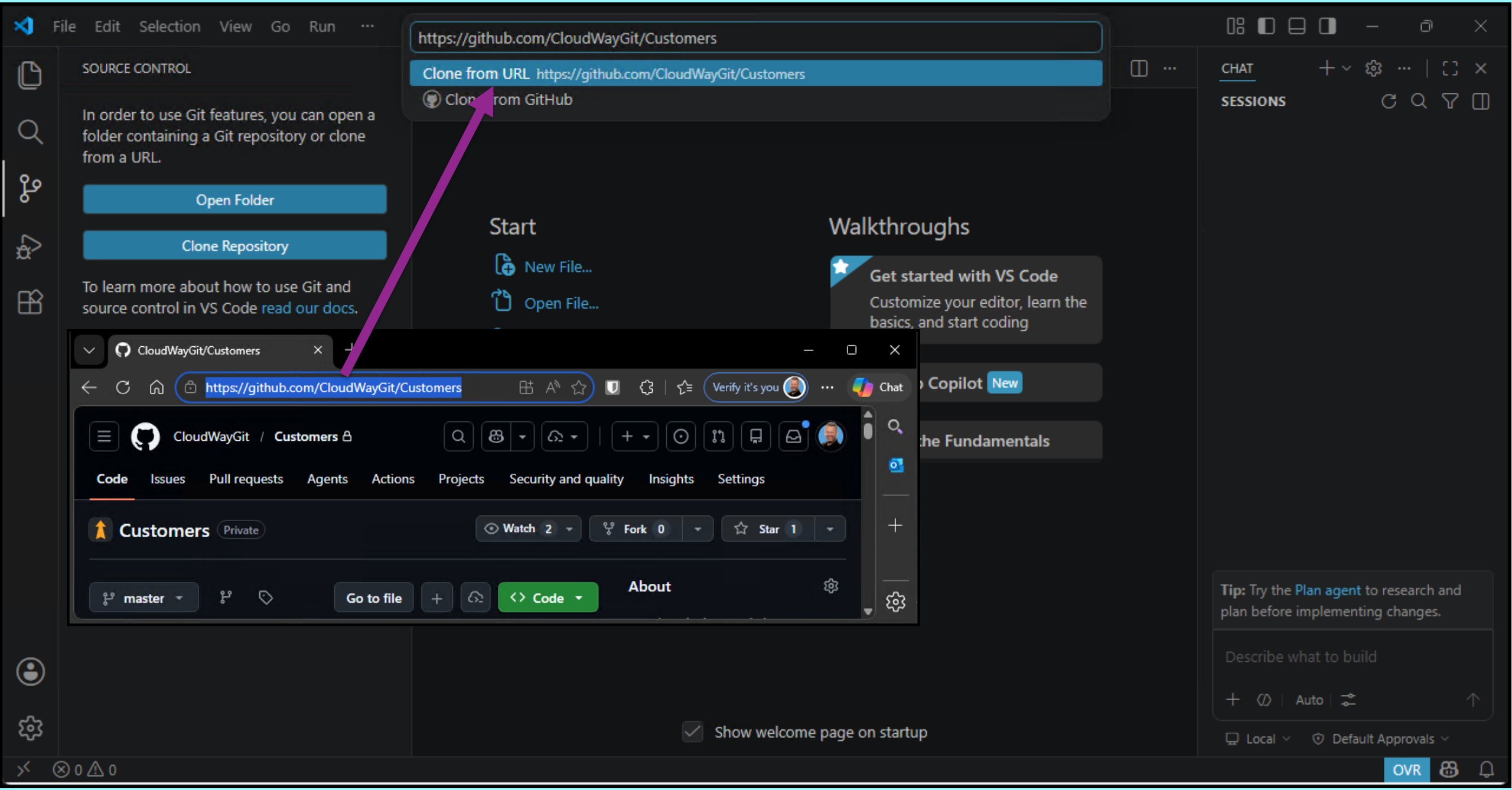
- A GitHub repository (often shortened to "repo") is the fundamental unit of project storage on GitHub
- Serving as a centralized workspace for code, documentation, and version history.
- Think of it as a smart project folder that not only stores files but also tracks every change made over time
- github.com/new



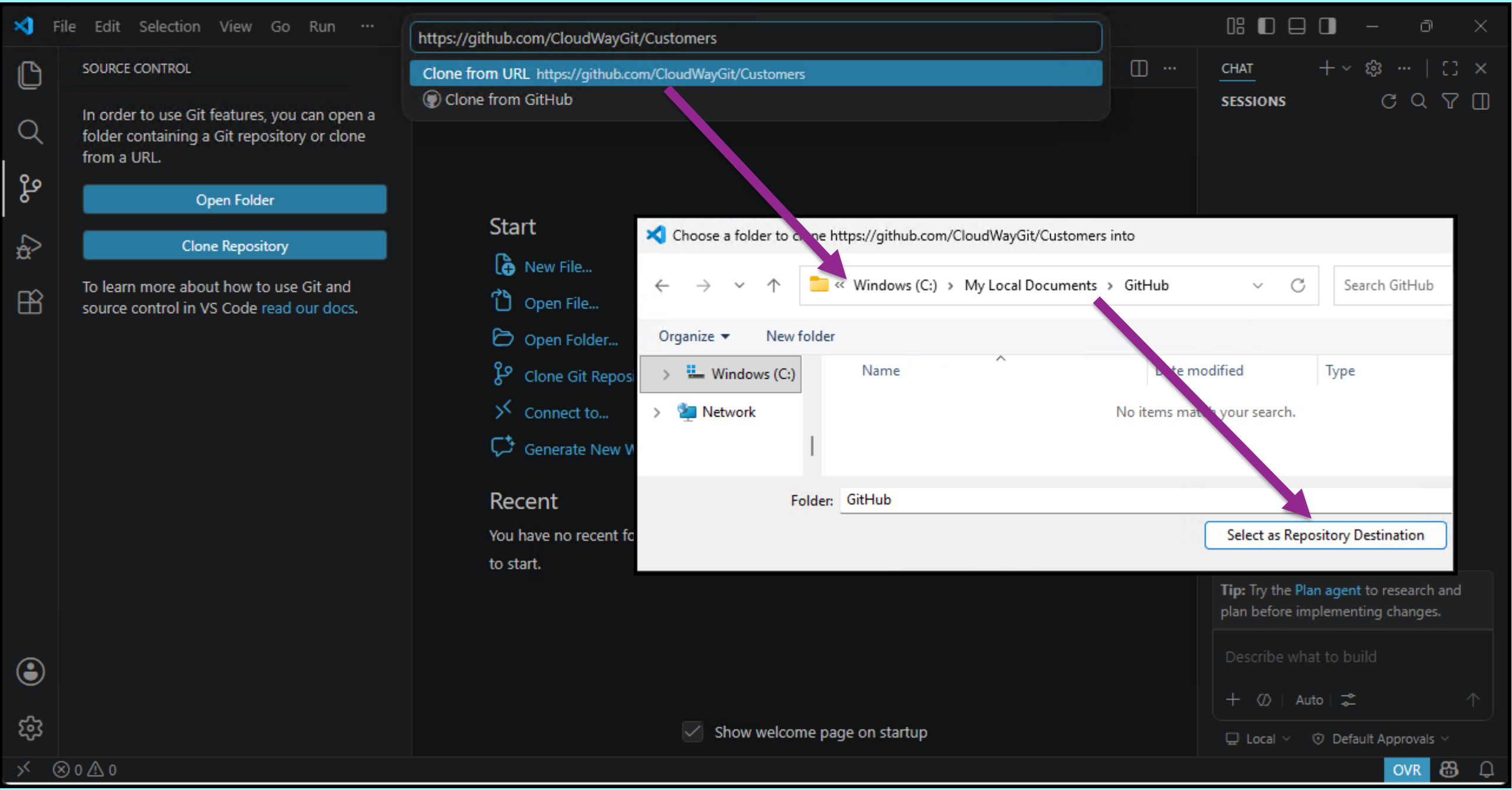
VSCode -https://code.visualstudio.com/download - Ctrl+Shift+G to open the Source Control blade



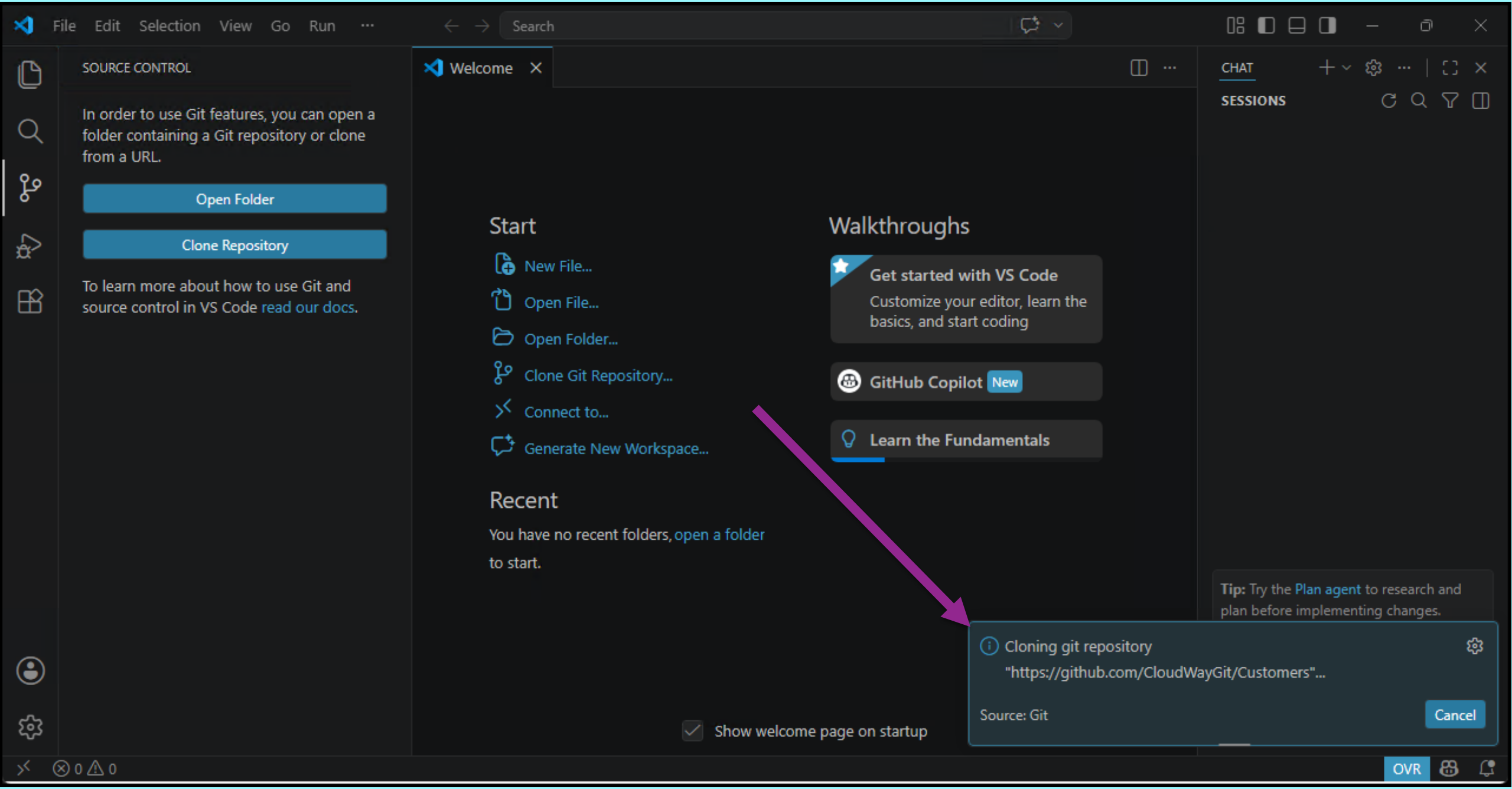
Ctrl+Shift+P to clone a repo. Search “Git Clone” if not showing up



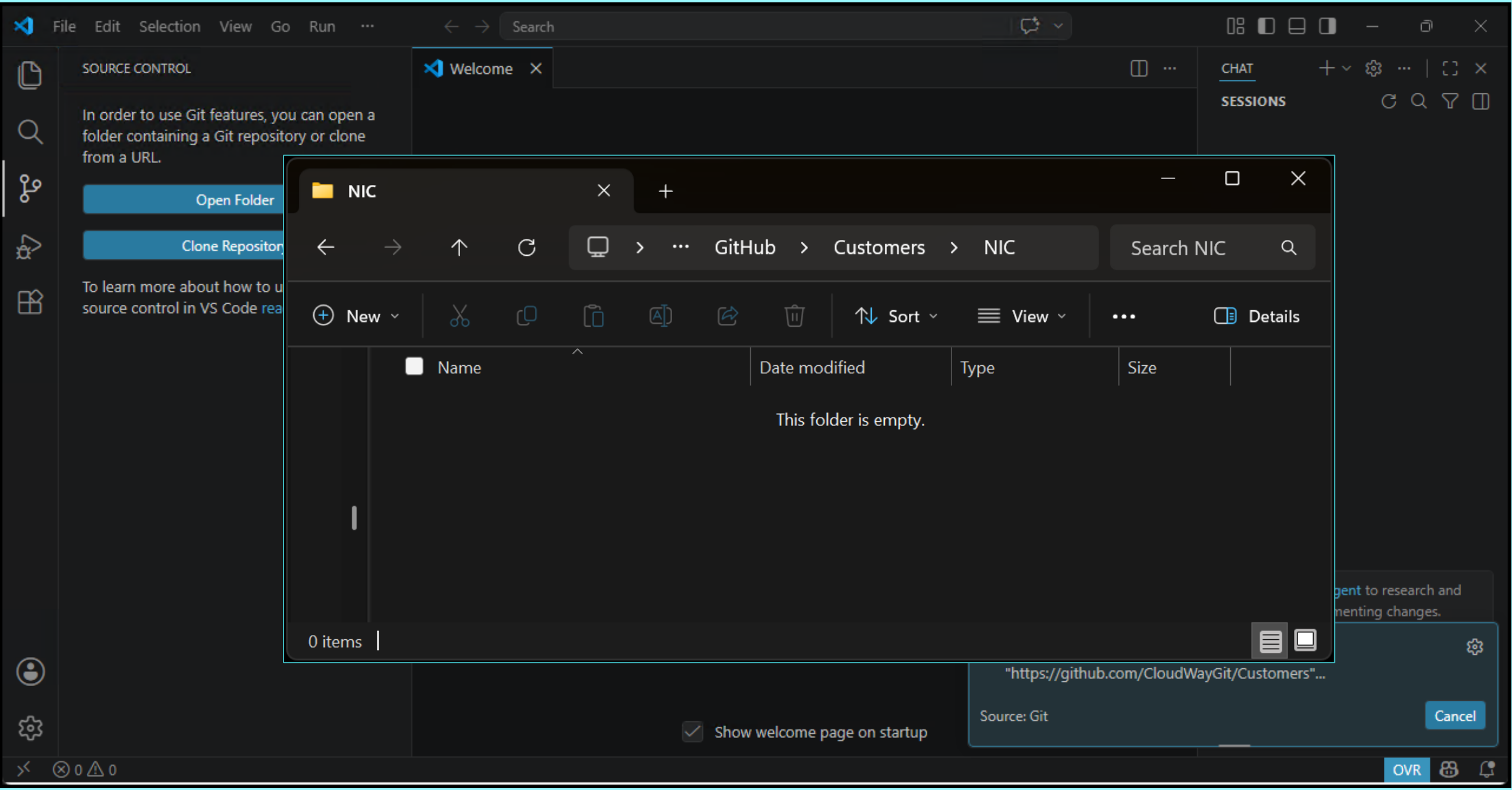
Can use the URL for the repo



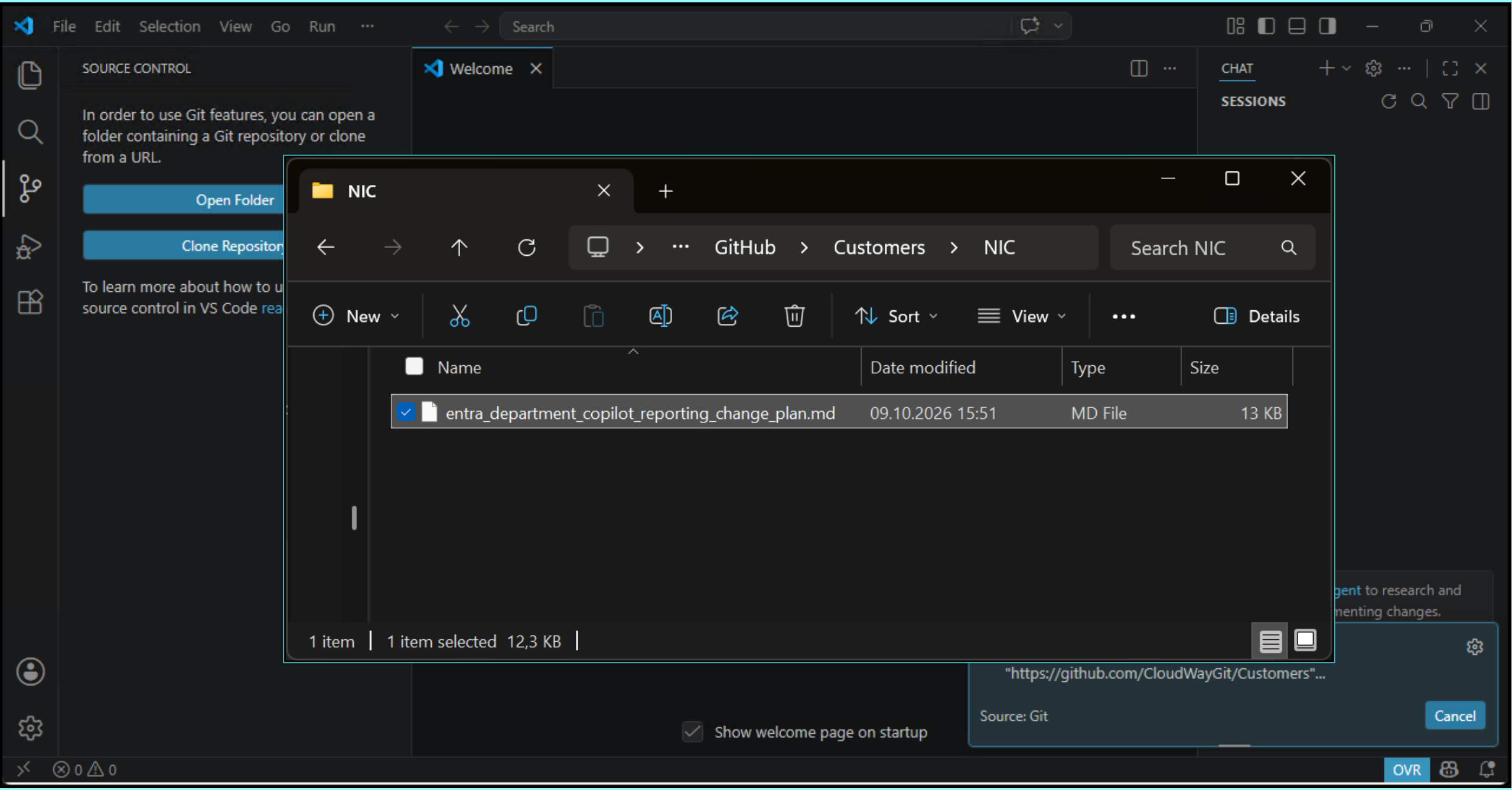
Select a local folder as the repository destination. A new folder with the repo name will be created



The content is now cloned locally



Content is visible in File Explorer – Copy the MD file here



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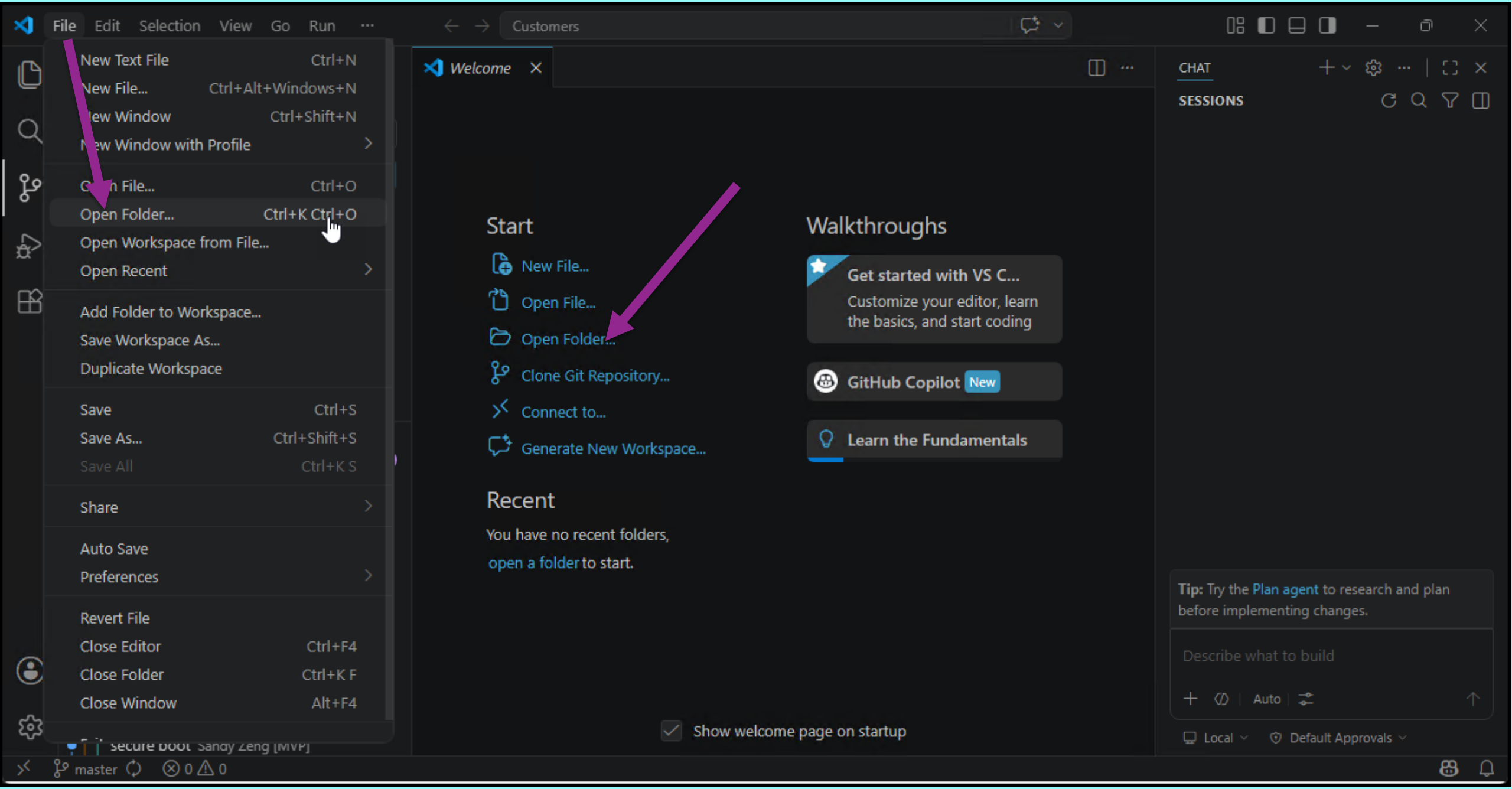
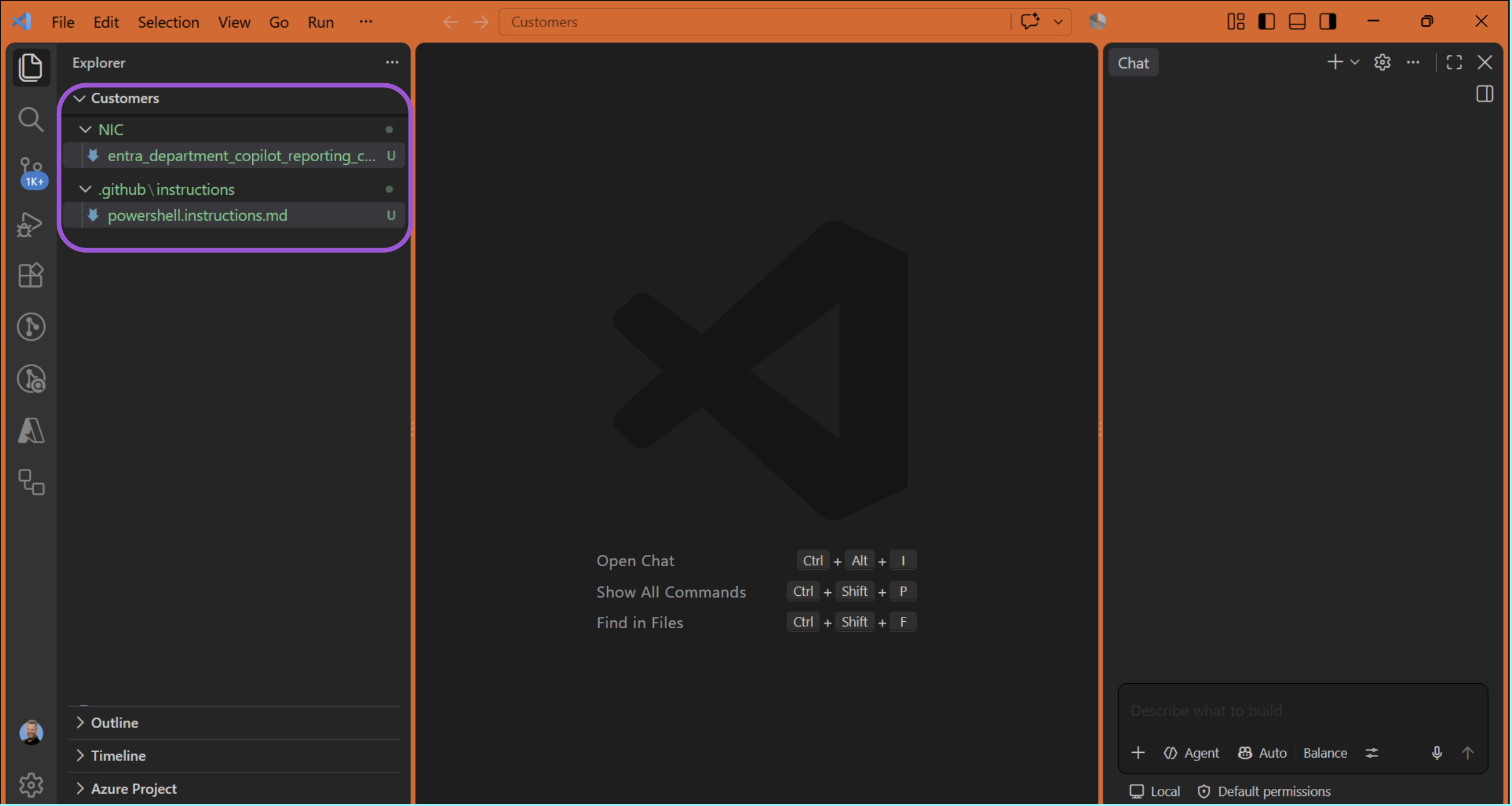
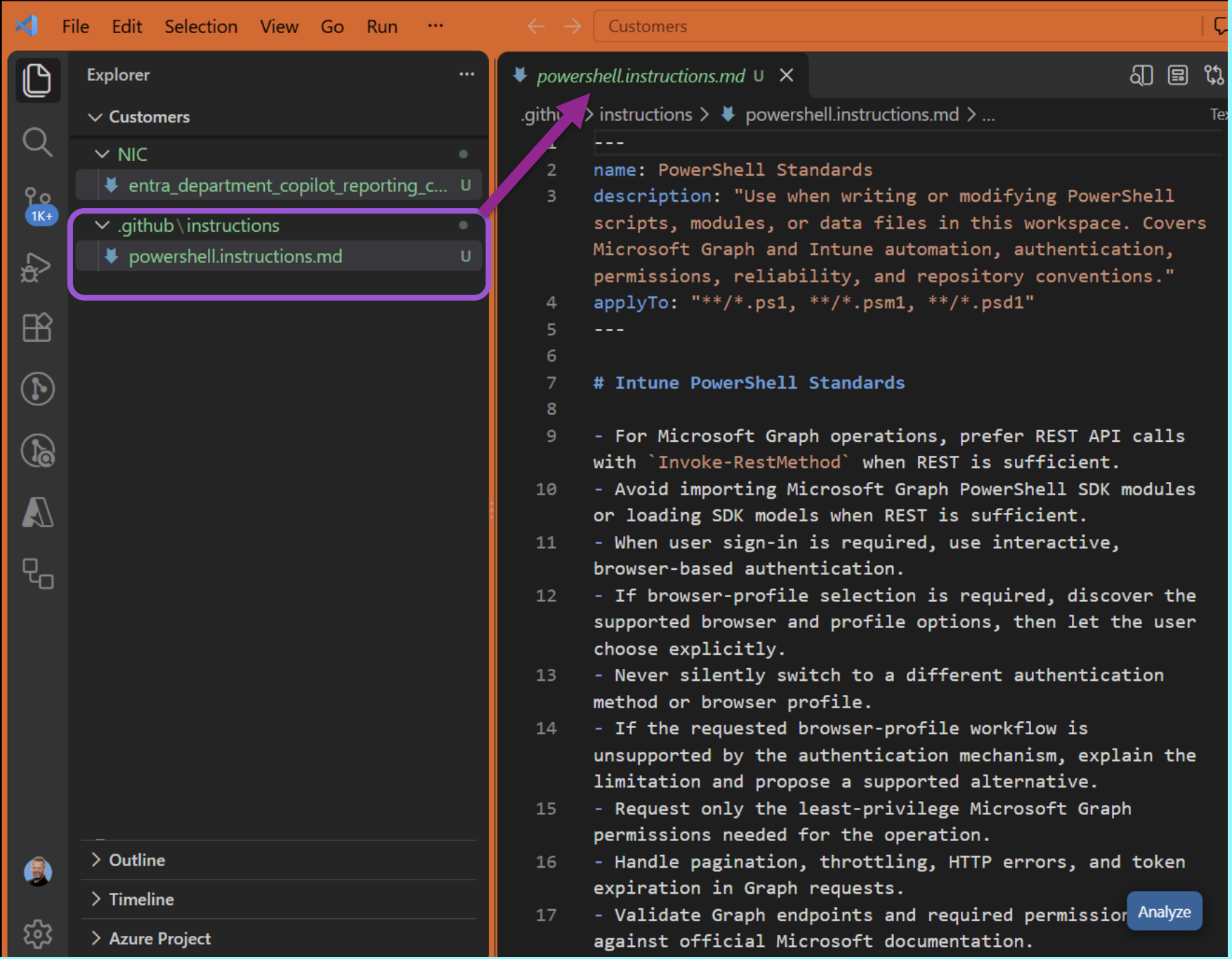


Open the folder in VSCode



The folder structure is now available in the Explorer (Ctrl+Shift+E)



Instructions

- .github/copilot-instructions.md
- Define your coding standards, conventions, and preferences
- Applied automatically to every request
- Also helpful in projects with multiple people involved
- Markdown files (MD)

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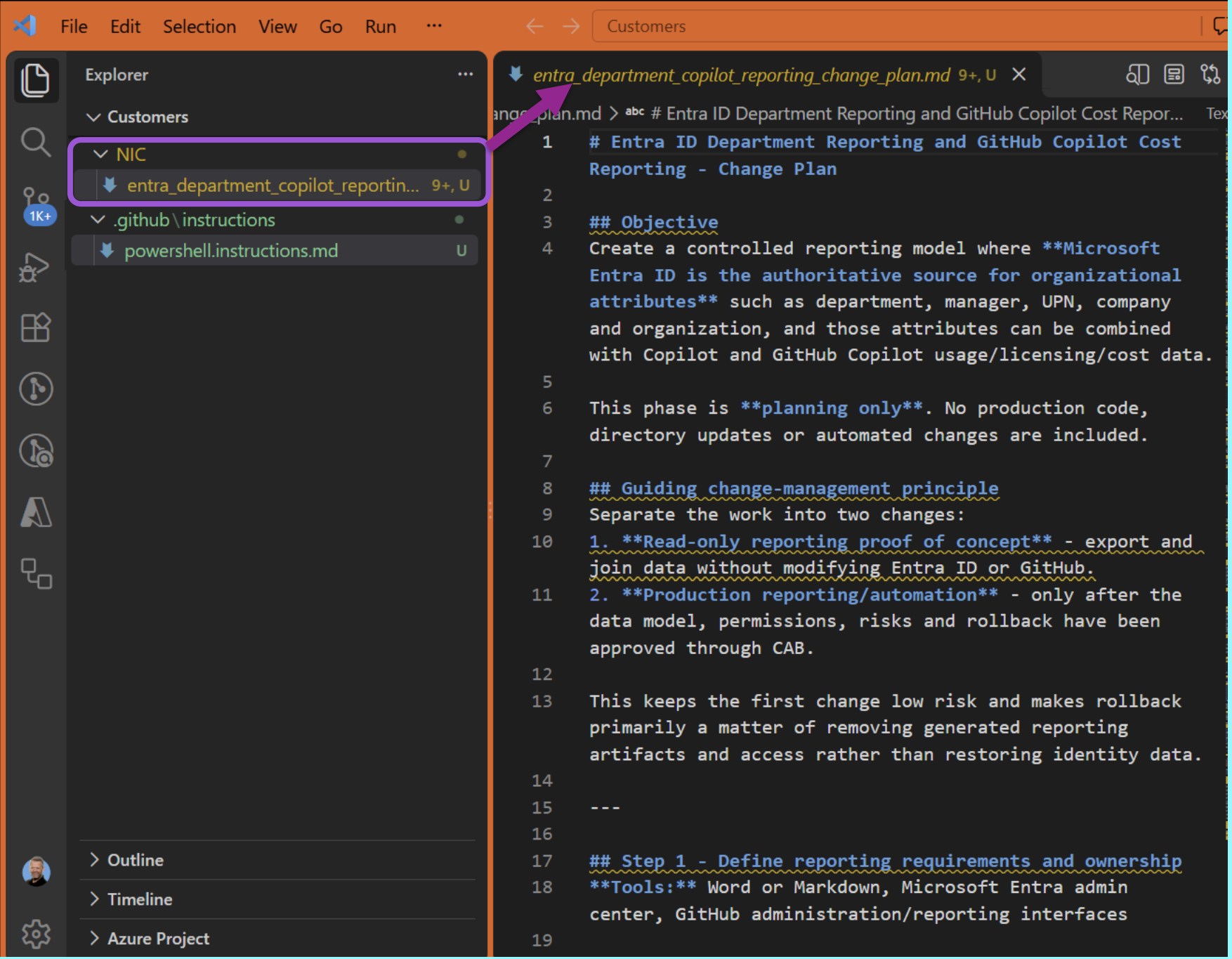
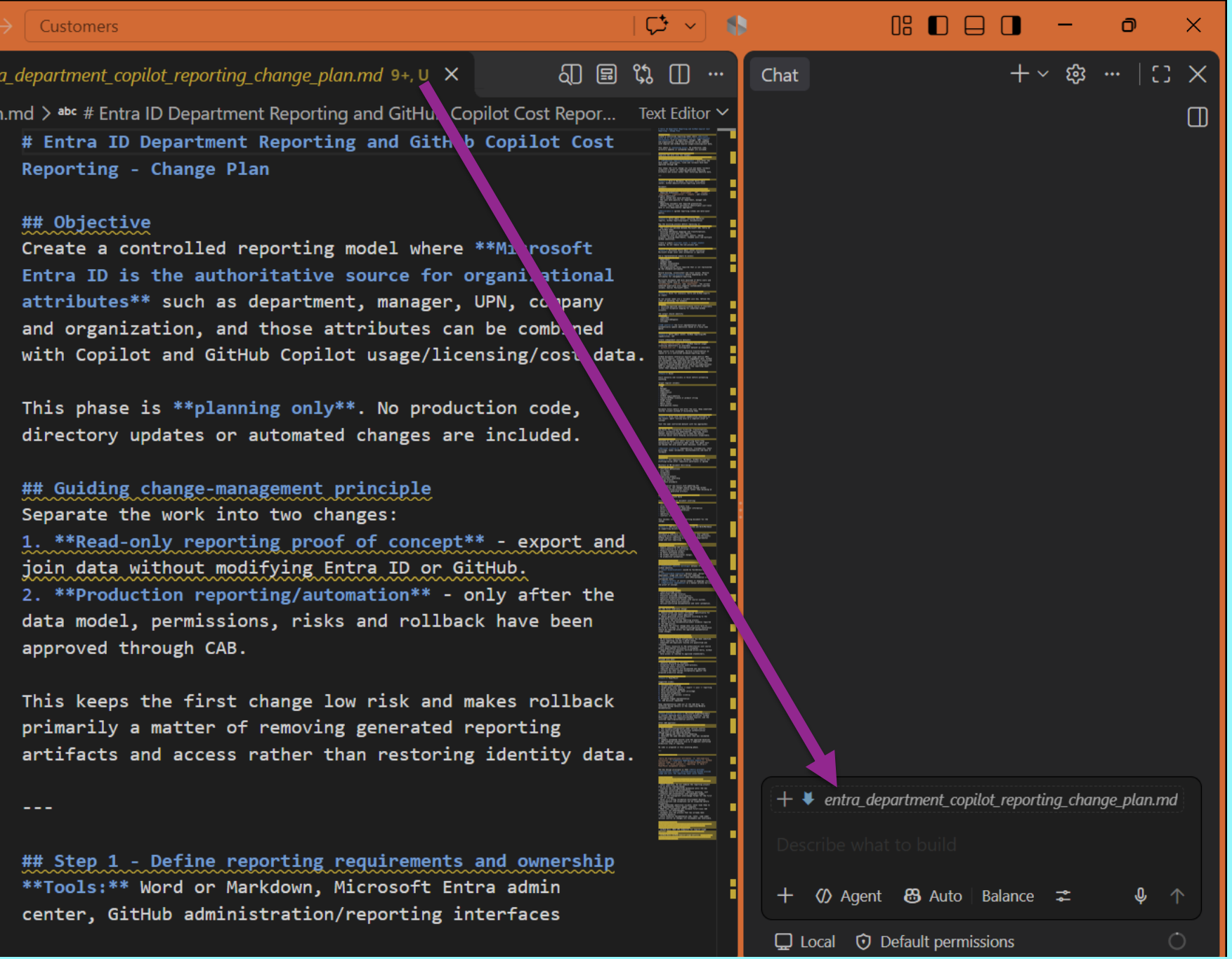


File Context

- Starting a prompt from a file, gives GitHub Copilot context

- GitHub Copilot will also take context from
  - Other open files
  - The open repo



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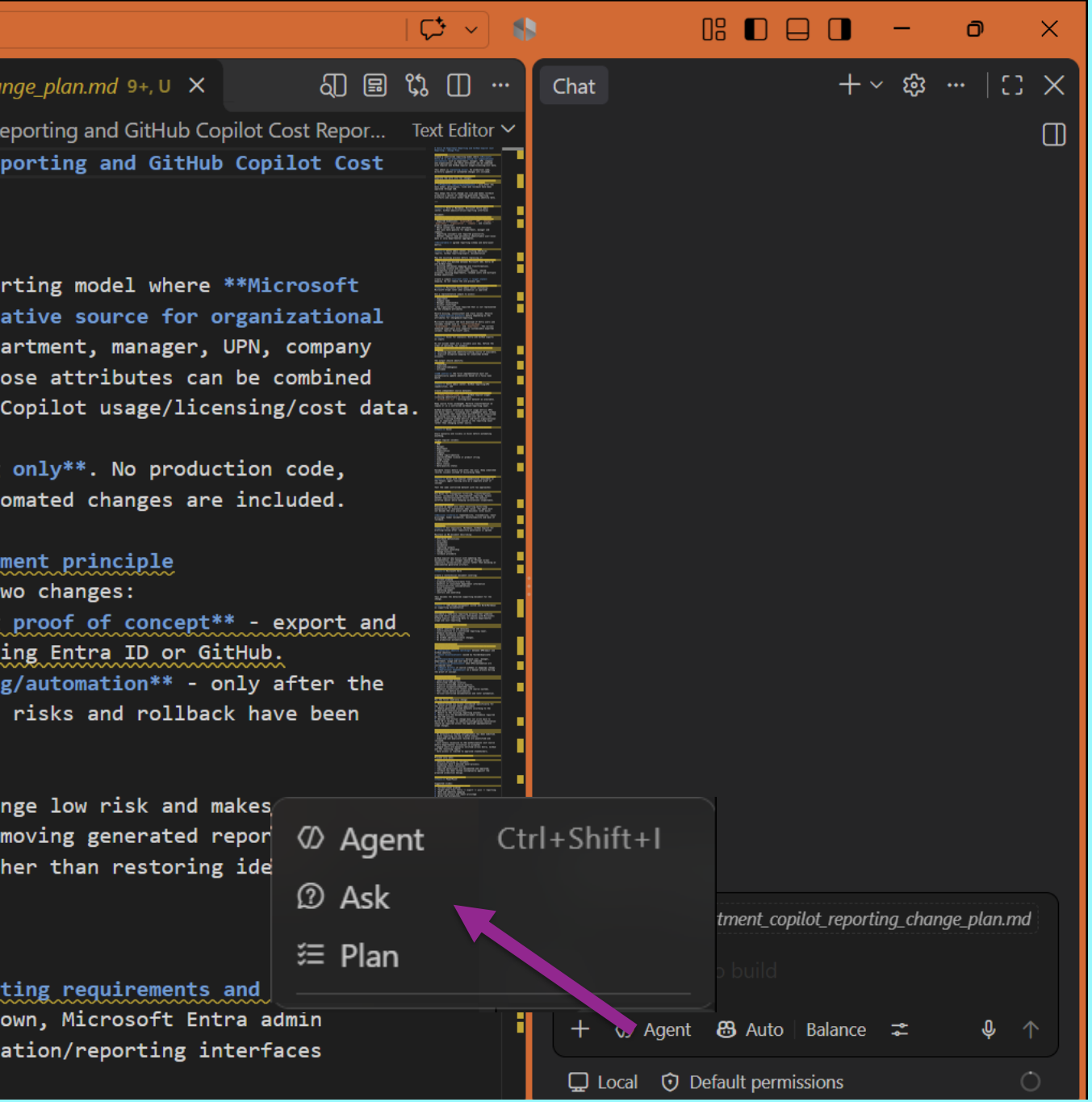
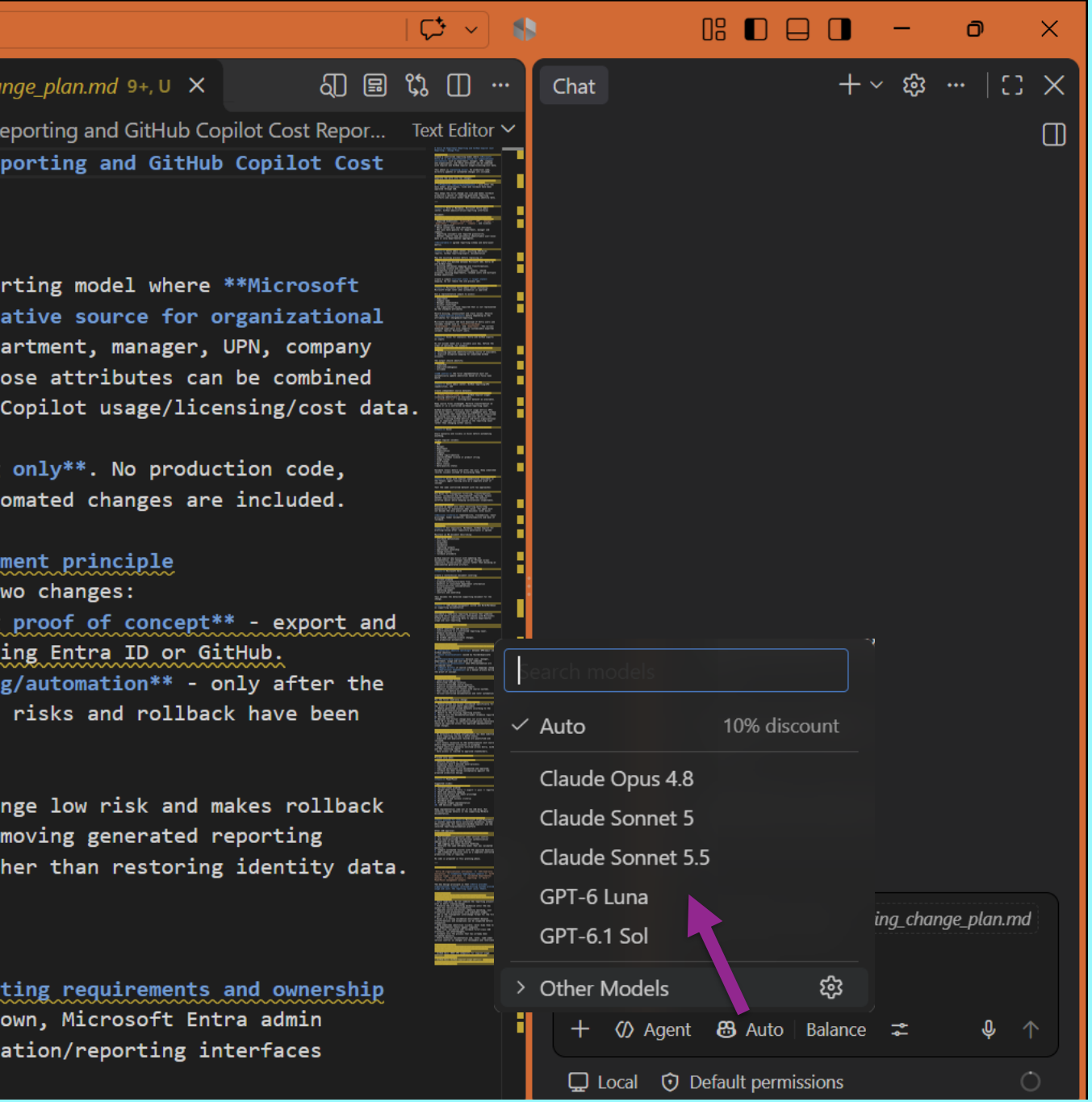


VS Code Chat Modes

- **Ask Mode**
- Q&A chatbot — answers questions, shows code
- Never touches your files
- Use for: “How does this work?” questions

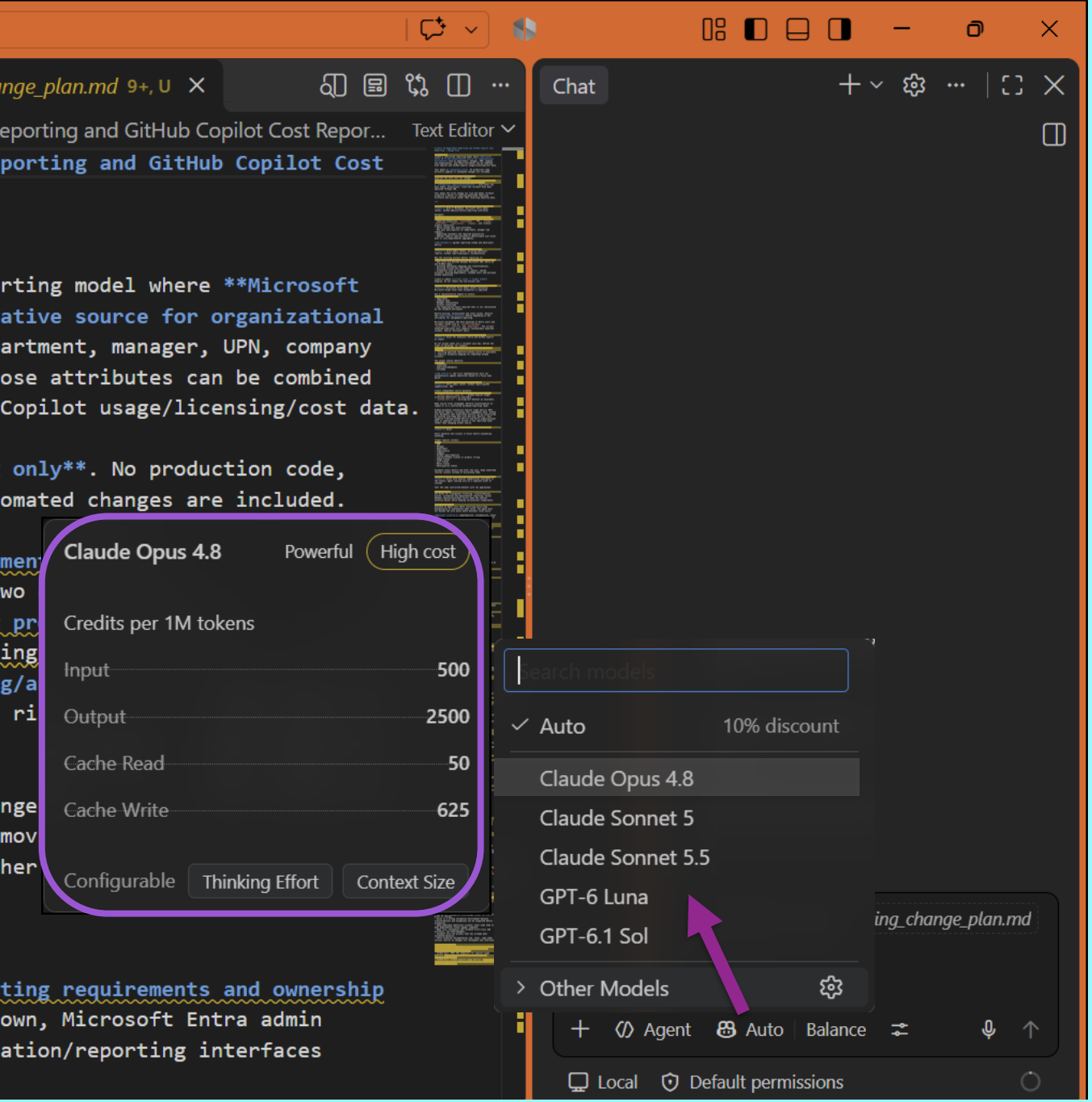
- **Plan Mode**
- Creates a step-by-step plan before making changes
- Outlines approach and gets your approval first
- Great for complex tasks where you want oversight

- **Agent Mode**
- Autonomous multi-file, multi-step execution
- Plans, edits, runs commands, fixes errors
- The game changer for IT Pros



Select Models

- Choose which model powers Copilot Chat and agent mode



Select Models

- Choose which model powers Copilot Chat and agent mode
- Badges give the trade-off at a glance
  - Powerful, high cost
  - Input is cheaper than output
    - Plan well to get a good input
- "Auto" balances cost, speed, and quality
  - 10% discount on the credits it spends

https://platform.openai.com/tokenizer

Select Models

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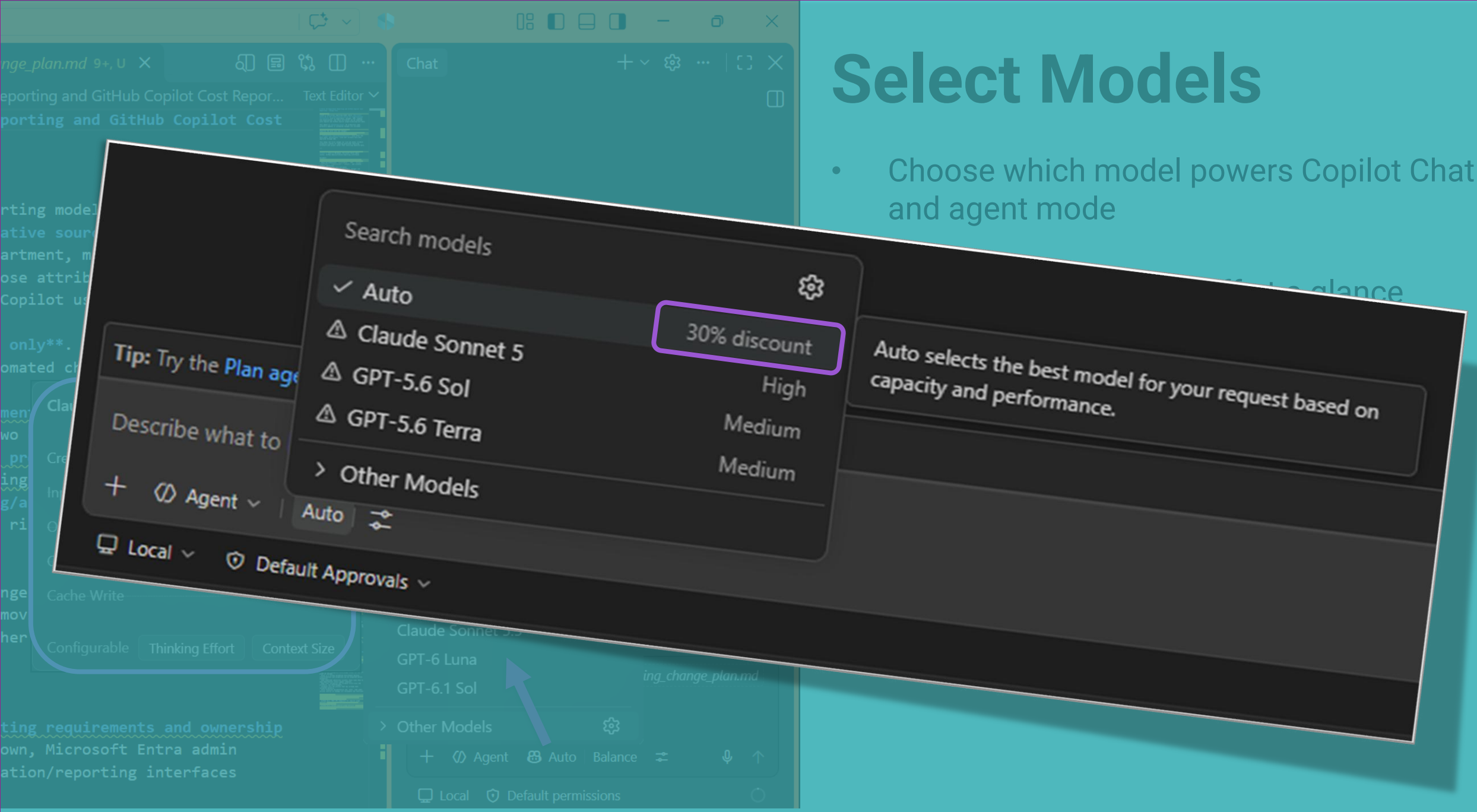
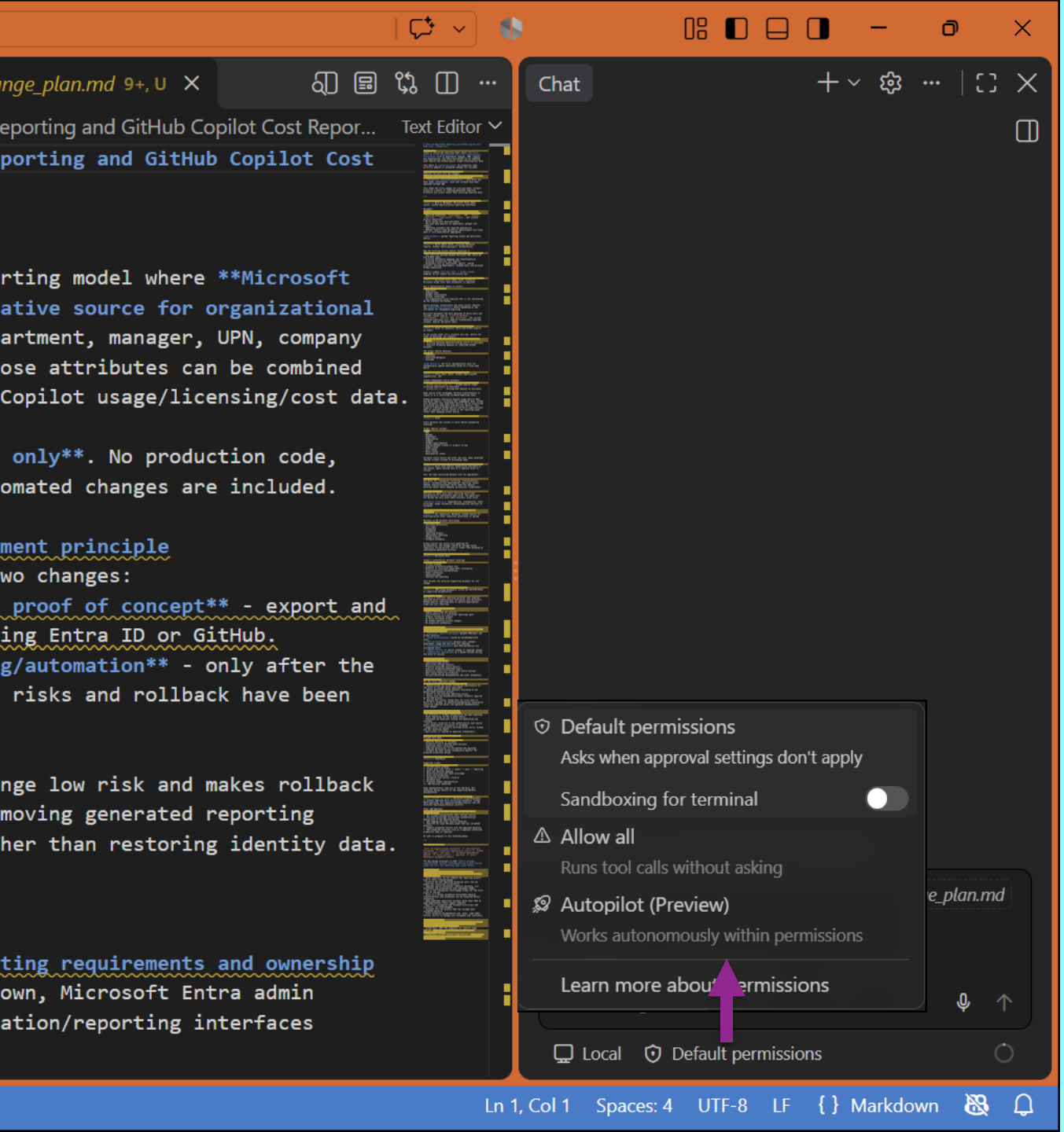
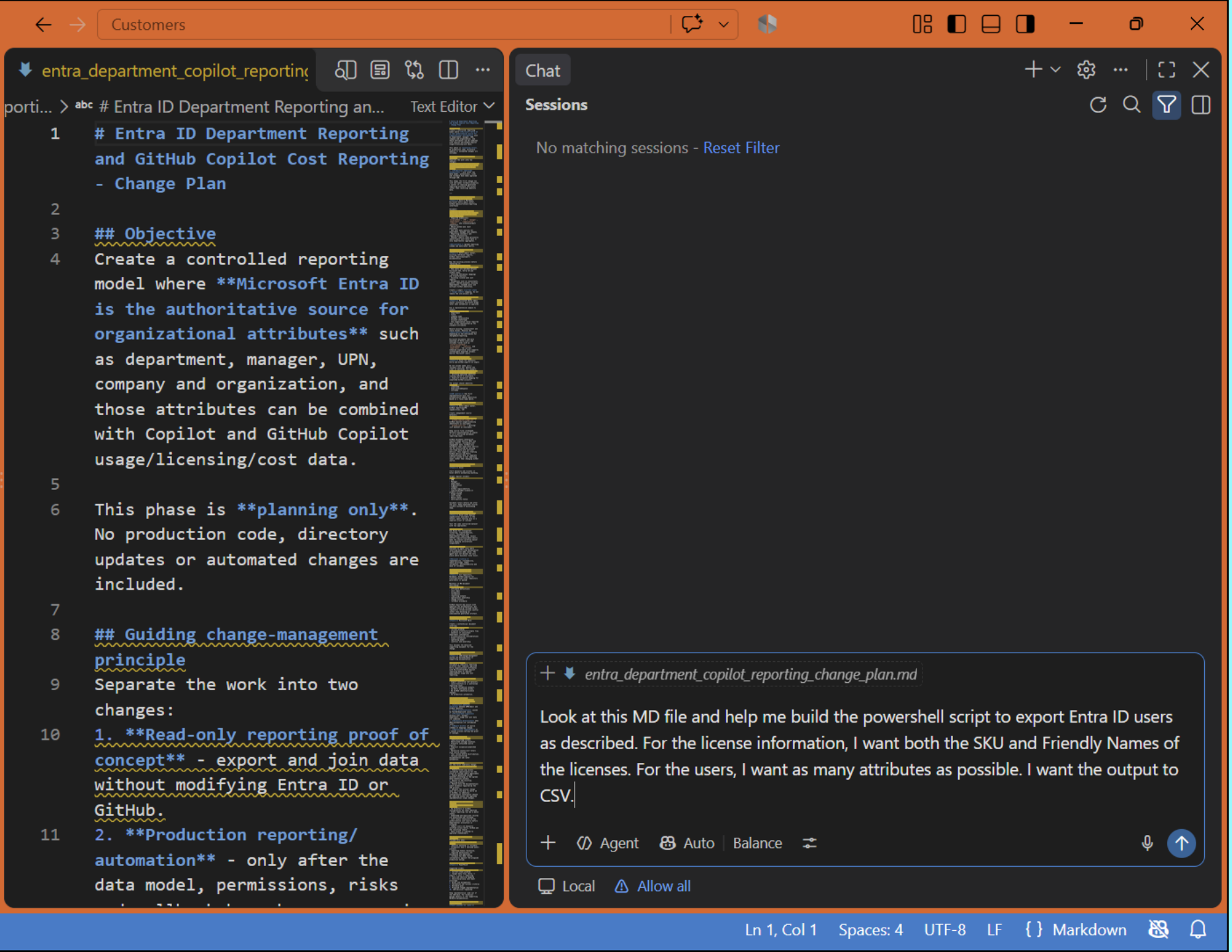


https://github.com/features/copilot/plans



Permissions

- **Default Permissions** – Copilot follows your configured security and permission settings and asks for approval when required.
- **Allow all** – All Copilot tool actions are automatically approved without asking you first.
- **Autopilot (Preview)** – Copilot works autonomously, making decisions and iterating on tasks from start to finish without manual approval.



You describe the goal

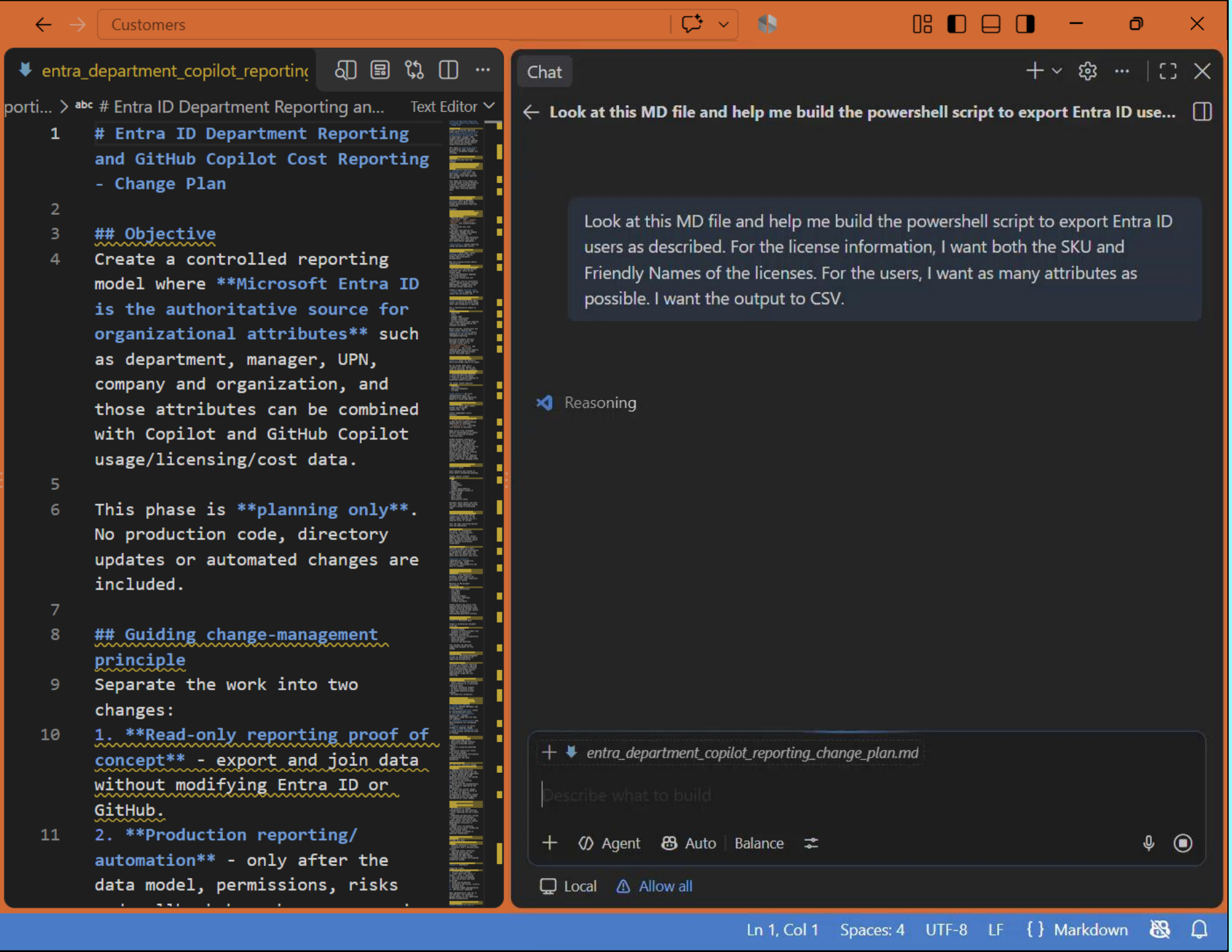
- Copilot determines how to get there

What Agent Mode does

- Analyzes your entire codebase for context
- Plans multi-step solutions
- Edits multiple files simultaneously
- Runs terminal commands and tests
- Self-corrects when errors occur

How to activate

- Ctrl+Shift+I → select “Agent” from the dropdown



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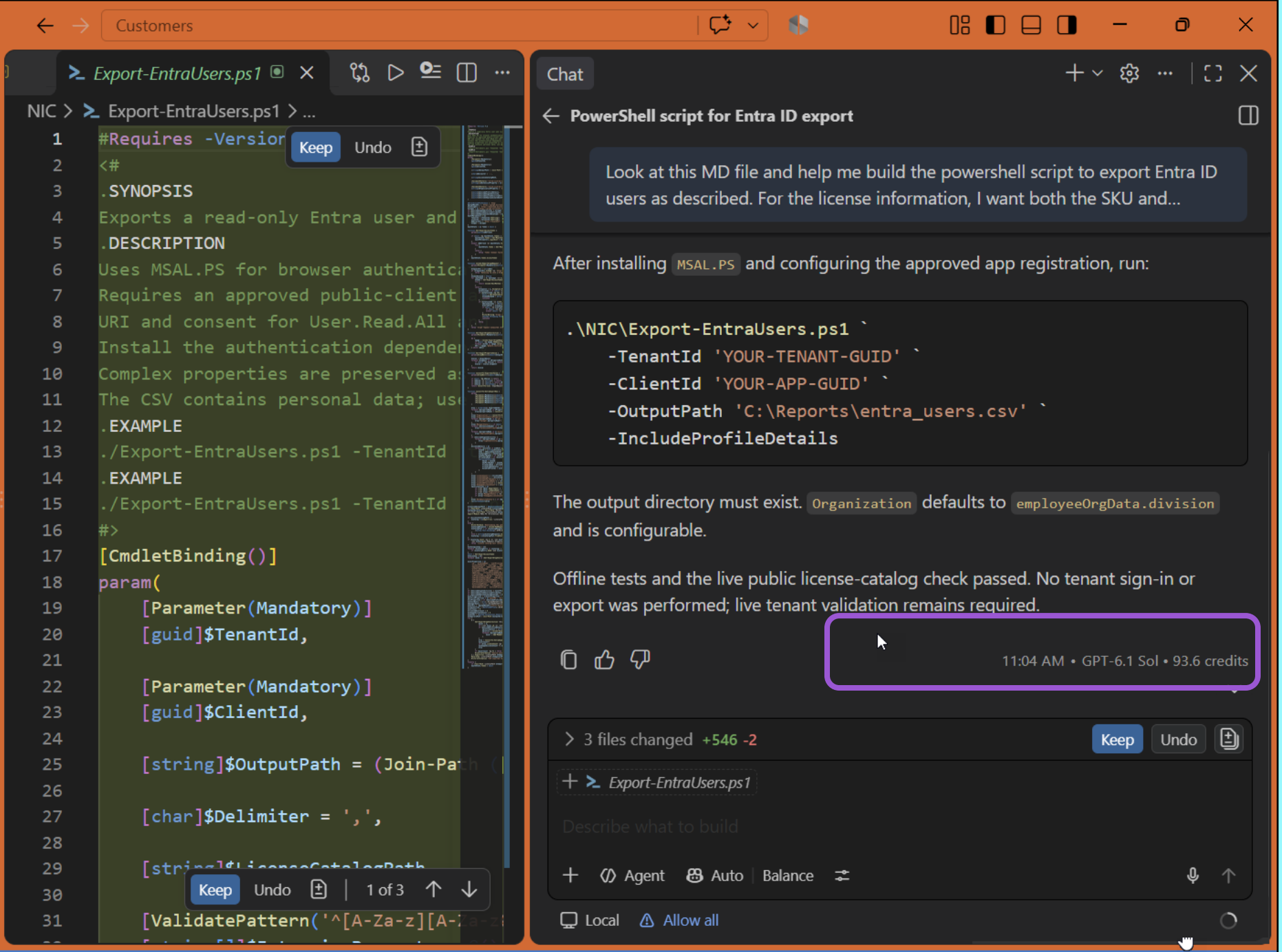
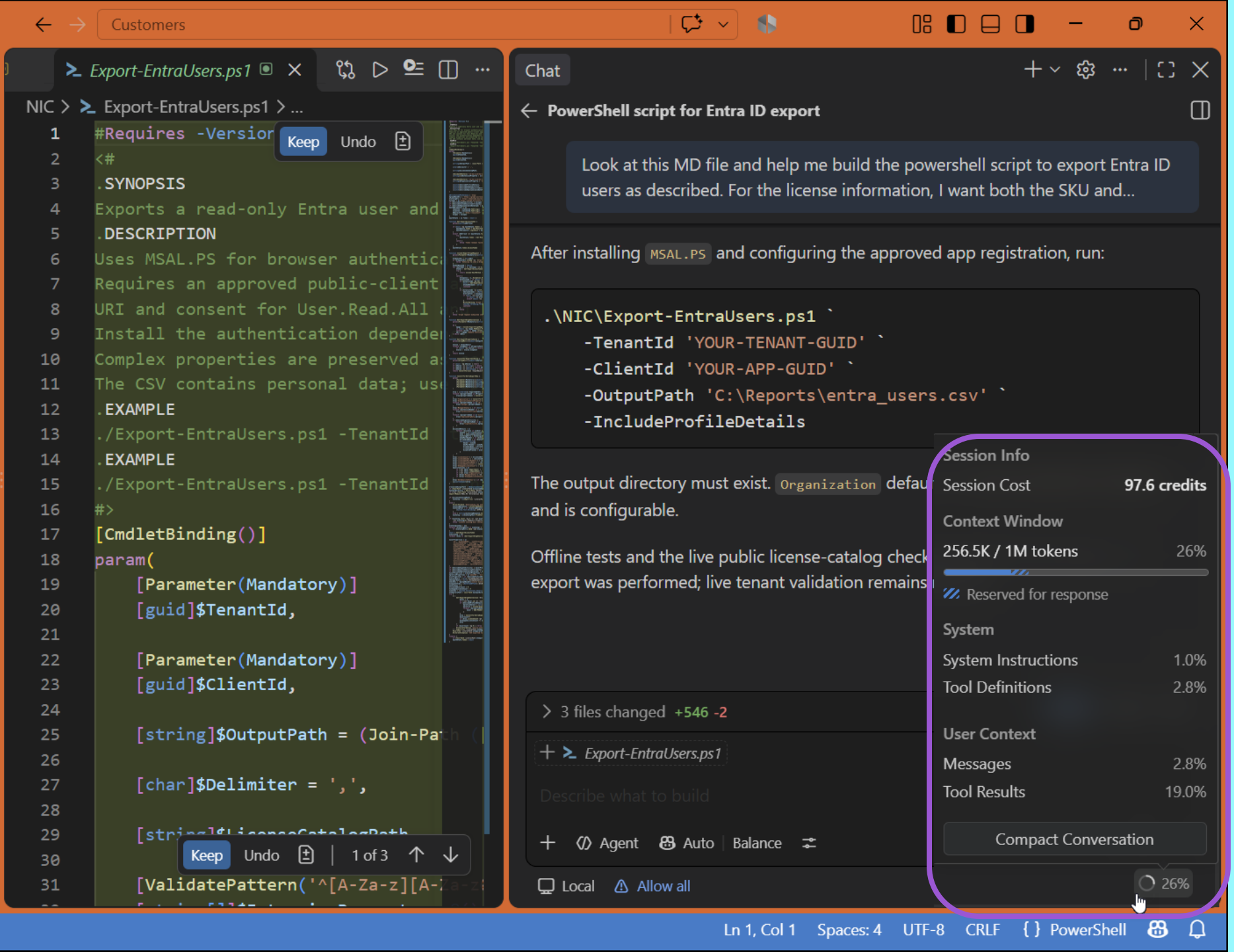


Explore the output

- First iteration completed.
- Hover to see info and cost
- Now we are in the Zone – in the Vibe!



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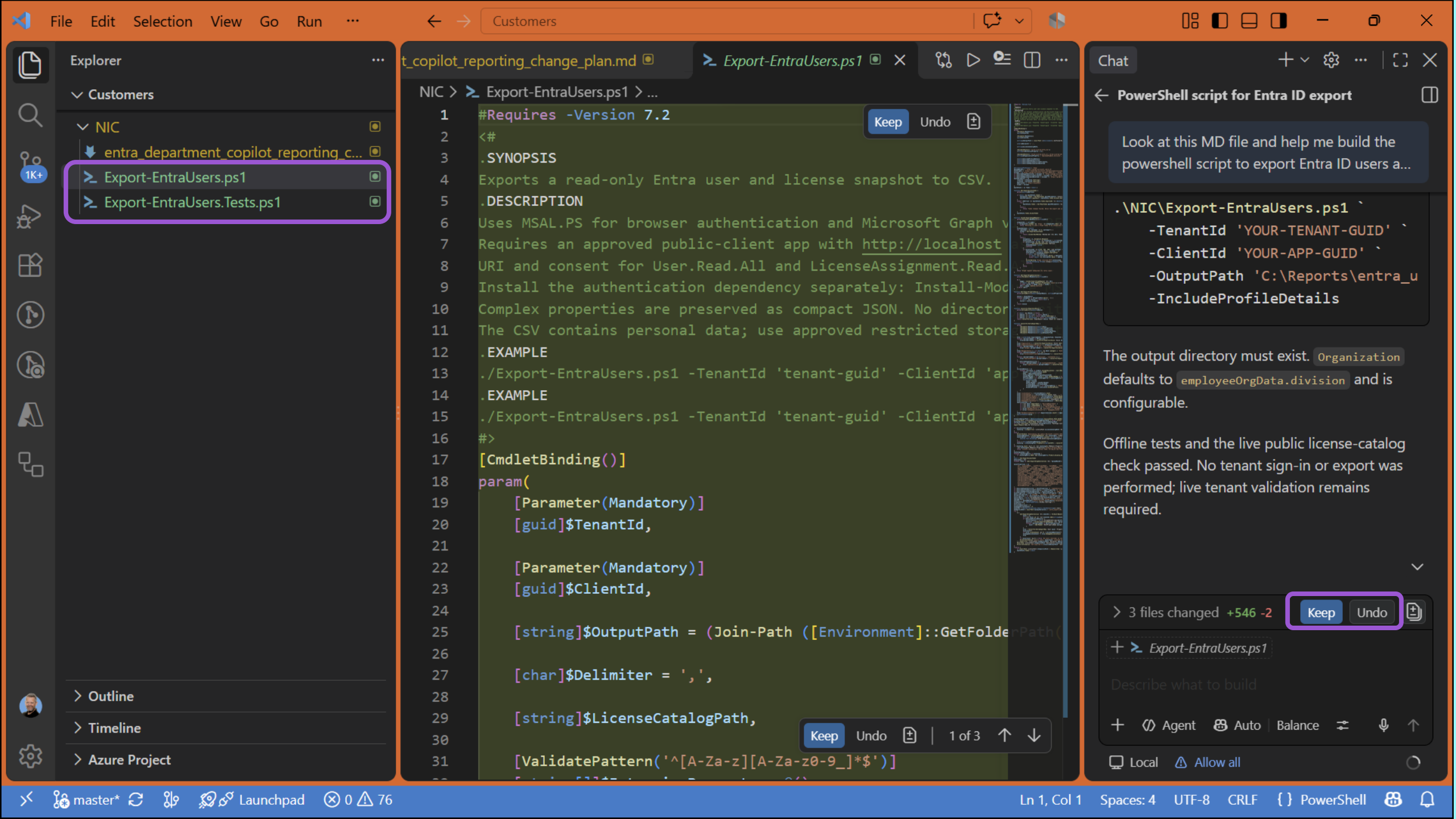

File Edit Selection View Go Run
Customers
Explorer
Customers
NIC
entra_department_copilot_reporting_c...
Export-EntraUsers.ps1
Export-EntraUsers.Tests.ps1
Outline
Timeline
Azure Project
t_copilot_reporting_change_plan.md
Export-EntraUsers.ps1
NIC > Export-EntraUsers.ps1 > ...
Keep Undo
#Requires -Version 7.2
<#
.SYNOPSIS
Exports a read-only Entra user and license snapshot to CSV.
.DESCRIPTION
Uses MSAL.PS for browser authentication and Microsoft Graph
Requires an approved public-client app with http://localhost
URI and consent for User.Read.All and LicenseAssignment.Read.
Install the authentication dependency separately: Install-Mod
Complex properties are preserved as compact JSON. No director
The CSV contains personal data; use approved restricted stora
.EXAMPLE
./Export-EntraUsers.ps1 -TenantId 'tenant-guid' -ClientId 'ap
.EXAMPLE
./Export-EntraUsers.ps1 -TenantId 'tenant-guid' -ClientId 'ap
#>
[CmdletBinding()]
param(
[Parameter(Mandatory)]
[guid]$TenantId,
[Parameter(Mandatory)]
[guid]$ClientId,
[string]$OutputPath = (Join-Path ([Environment]::GetFolderPath(
[char]$Delimiter = ',',
[string]$LicenseCatalogPath,
Keep Undo 1 of 3
[ValidatePattern('^[A-Za-z][A-Za-z0-9_]*$')]
Chat
PowerShell script for Entra ID export
Look at this MD file and help me build the powershell script to export Entra ID users a...
.\NIC\Export-EntraUsers.ps1 `
-TenantId 'YOUR-TENANT-GUID' `
-ClientId 'YOUR-APP-GUID' `
-OutputPath 'C:\Reports\entra_u
-IncludeProfileDetails
The output directory must exist. Organization defaults to employeeOrgData.division and is configurable.
Offline tests and the live public license-catalog check passed. No tenant sign-in or export was performed; live tenant validation remains required.
3 files changed +546 -2
Keep Undo
Export-EntraUsers.ps1
Describe what to build
Agent Auto Balance
Local Allow all
master* Launchpad 0 76
Ln 1, Col 1 Spaces: 4 UTF-8 CRLF { } PowerShell

The Vibe Coding Workflow

The Vibe Coding Workflow

4. Iterate
Refine through conversation until it works perfectly

entra_department_copilot_reporting_change_plan.md

I will in a while create a script to update the Entra Users with variables from the CSV file. Meanwhile I want you to update the MD file with information about the powershell script I have now, the risk of running it, and also rollback strategies.

Agent | Auto | Balance

3. Review
Run the code, check results, identify issues

The Vibe Coding Workflow

4. Iterate
Refine through conversation
until it works perfectly

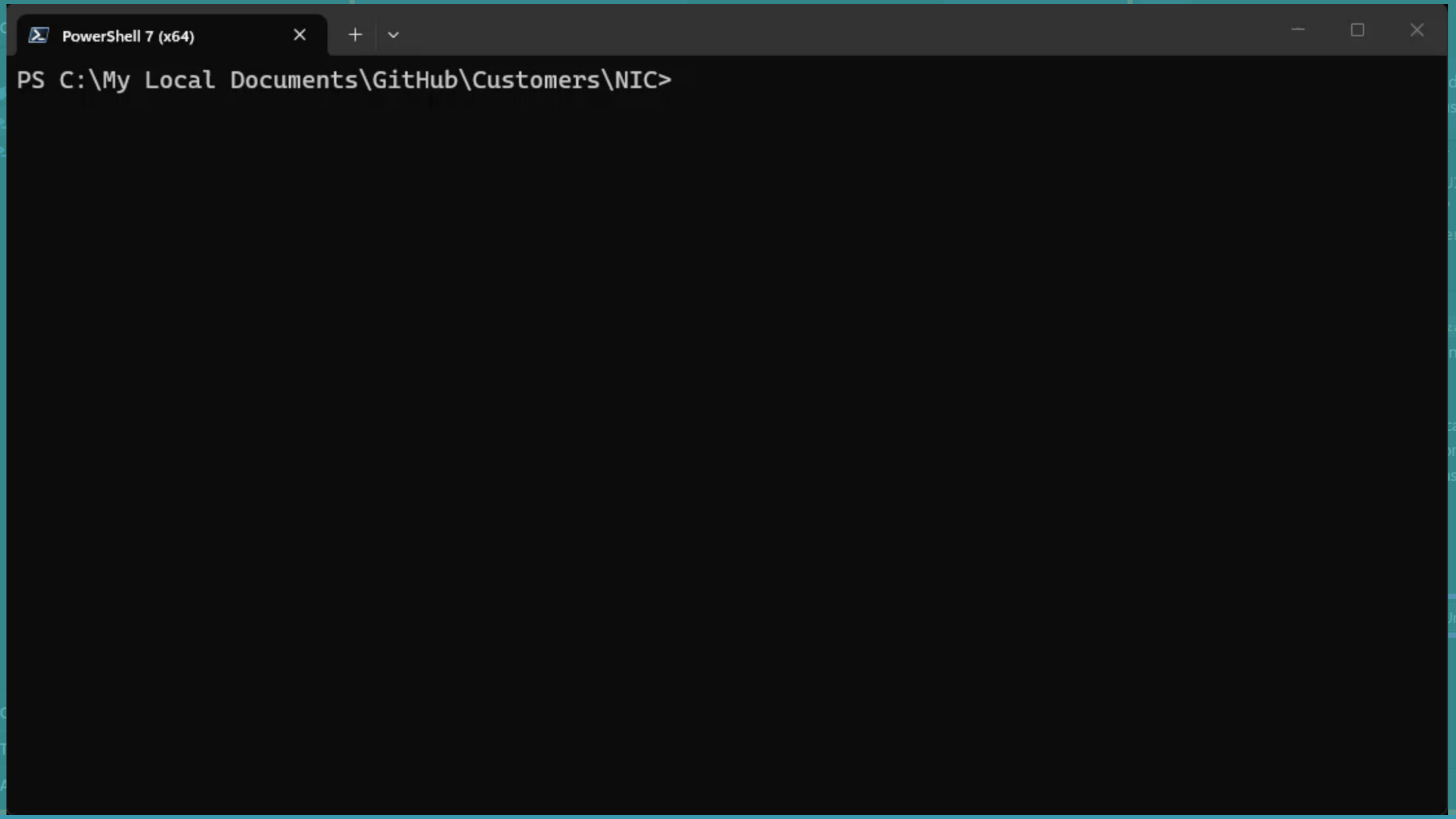
1 file changed +58 -0

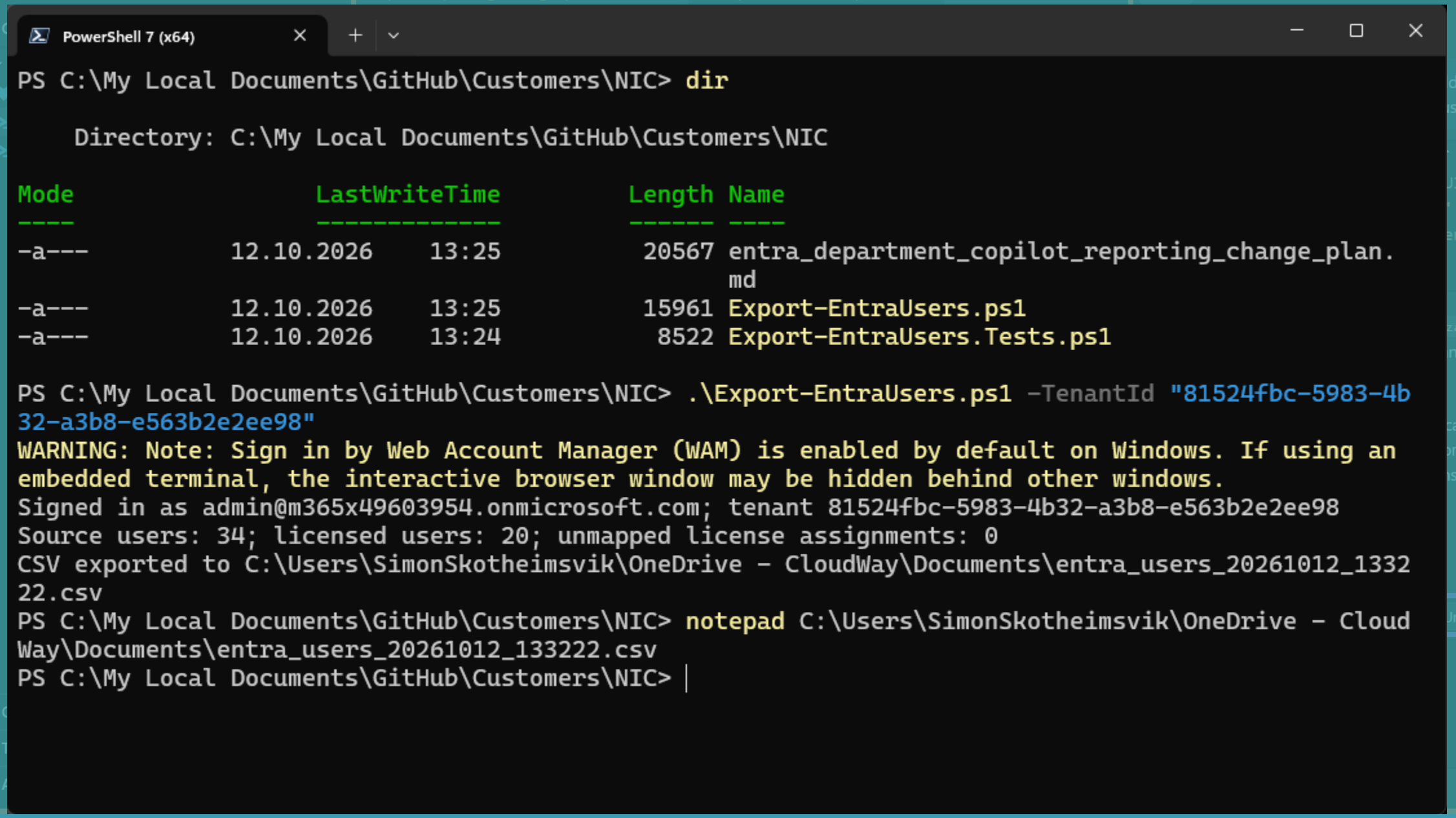
Keep Undo

entra_department_copilot_reporting_change_plan.md

Create the script to update users in Entra ID based on the updated data in the CSV file. Follow the same principels as allready used for the export script. I want the script to have a dry-run option.
Also update the MD file with information of this script, the risks of running it, and also rollback options.

Agent Auto | Balance


PowerShell 7 (x64)
PS C:\My Local Documents\GitHub\Customers\NIC>


PowerShell 7 (x64)
PS C:\My Local Documents\GitHub\Customers\NIC> dir
Directory: C:\My Local Documents\GitHub\Customers\NIC
Mode LastWriteTime Length Name
-a--- 12.10.2026 13:25 20567 entra_department_copilot_reporting_change_plan.md
-a--- 12.10.2026 13:25 15961 Export-EntraUsers.ps1
-a--- 12.10.2026 13:24 8522 Export-EntraUsers.Tests.ps1
PS C:\My Local Documents\GitHub\Customers\NIC> .\Export-EntraUsers.ps1 -TenantId "81524fbc-5983-4b32-a3b8-e563b2e2ee98"
WARNING: Note: Sign in by Web Account Manager (WAM) is enabled by default on Windows. If using an embedded terminal, the interactive browser window may be hidden behind other windows.
Signed in as admin@m365x49603954.onmicrosoft.com; tenant 81524fbc-5983-4b32-a3b8-e563b2e2ee98
Source users: 34; licensed users: 20; unmapped license assignments: 0
CSV exported to C:\Users\SimonSkotheimsvik\OneDrive - CloudWay\Documents\entra_users_20261012_133222.csv
PS C:\My Local Documents\GitHub\Customers\NIC> notepad C:\Users\SimonSkotheimsvik\OneDrive - CloudWay\Documents\entra_users_20261012_133222.csv
PS C:\My Local Documents\GitHub\Customers\NIC>

"ExportedAtUtc","TenantId","id","displayName","userPrincipalName","givenName","surname","mail","mailNickname","accountEnabled","userType","department","company","jobTitle","employeeId","employeeType","employeeHireDate","employeeOrgData","officeLocation","usageLocation","preferredLanguage","preferredDataLocation","businessPhones","mobilePhone","otherMails","proxyAddresses","imAddresses","faxNumber","streetAddress","city","state","postalCode","country","ageGroup","consentProvidedForMinor","legalAgeGroupClassification","createdDateTime","creationType","externalUserState","externalUserStateChangeDateTime","identities","onPremisesSyncEnabled","onPremisesLastSyncDateTime","onPremisesImmutableId","onPremisesDomainName","onPremisesSamAccountName","onPremisesUserPrincipalName","onPremisesDistinguishedName","onPremisesSecurityIdentifier","onPremisesExtensionAttributes","onPremisesProvisioningErrors","assignedLicenses","assignedPlans","licenseAssignmentStates","provisionedPlans","passwordPolicies","lastPasswordChangeDateTime","isManagementRestricted","securityIdentifier","signInSessionsValidFromDateTime","refreshTokensValidFromDateTime","deletedDateTime","serviceProvisioningErrors","Organization","OrganizationSourceProperty","manager_id","manager_displayName","manager_userPrincipalName","manager_mail","manager_department","manager_companyName","manager_jobTitle","ManagerStatus","extensionAttribute1","extensionAttribute2","extensionAttribute3","extensionAttribute4","extensionAttribute5","extensionAttribute6","extensionAttribute7","extensionAttribute8","extensionAttribute9","extensionAttribute10","extensionAttribute11","extensionAttribute12","extensionAttribute13","extensionAttribute14","extensionAttribute15","employeeOrgData_division","employeeOrgData_costCenter","LicenseCount","LicenseSkuIds","LicenseSkuPartNumbers","LicenseFriendlyNames","LicenseDetailsJson","UnmappedLicenseCount","DataQualityStatus"
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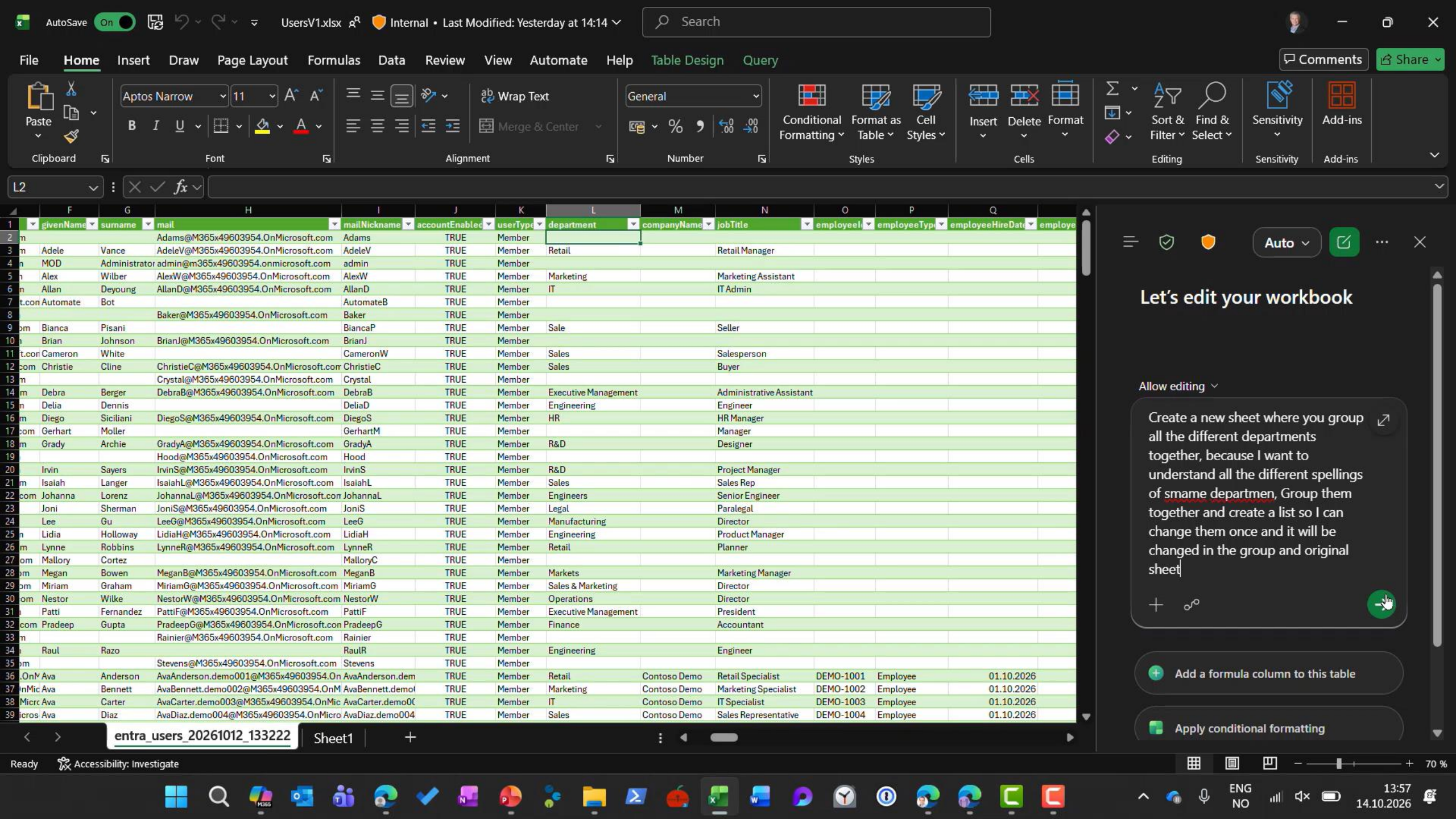
If you are using code
to figure out what to change
you are doing wrong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Internal • Last Modified: Yesterday at 14:14
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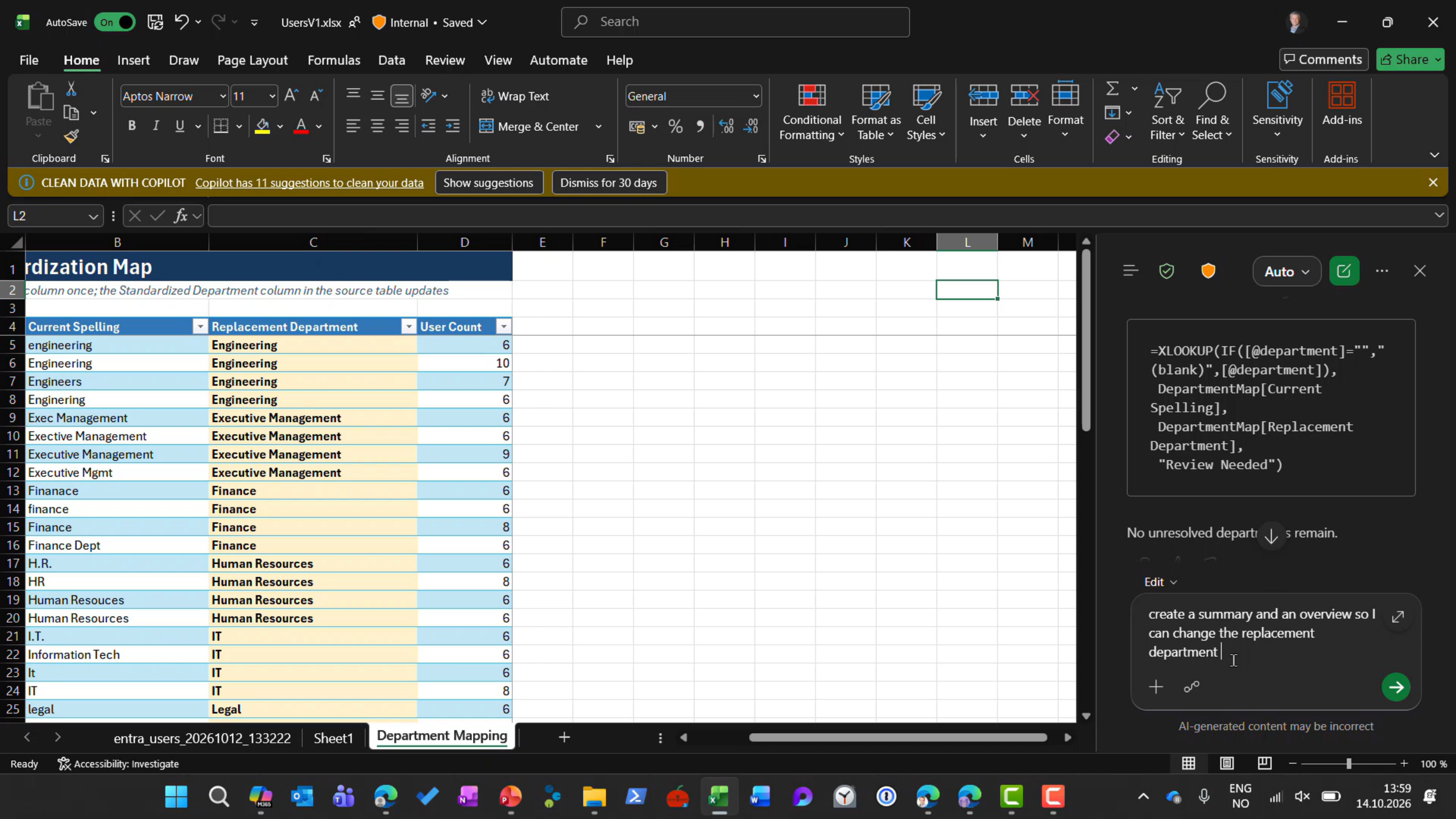
File Home Insert Draw Page Layout Formulas Data Review View Automate Help Table Design Query
Comments Share
Paste Clipboard
Aptos Narrow 11 Font
Wrap Text Merge & Center Alignment
General Number
Conditional Formatting Format as Table Cell Styles Styles
Insert Delete Format Cells
Sort & Filter Find & Select Editing
Sensitivity Sensitivity
Add-ins Add-ins
L2
givenName surname mail mailNickname accountEnabled userType department companyName jobTitle employeeId employeeType employeeHireDate
Adams@M365x49603954.OnMicrosoft.com Adams TRUE Member
Adele Vance AdeleV@M365x49603954.OnMicrosoft.com AdeleV TRUE Member Retail Retail Manager
MOD Administrator admin@m365x49603954.onmicrosoft.com admin TRUE Member
Alex Wilber AlexW@M365x49603954.OnMicrosoft.com AlexW TRUE Member Marketing Marketing Assistant
Allan Deyoung AllanD@M365x49603954.OnMicrosoft.com AllanD TRUE Member IT IT Admin
Automate Bot AutomateB TRUE Member
Baker@M365x49603954.OnMicrosoft.com Baker TRUE Member
Bianca Pisani BiancaP TRUE Member Sale Seller
Brian Johnson BrianJ@M365x49603954.OnMicrosoft.com BrianJ TRUE Member
Cameron White CameronW TRUE Member Sales Salesperson
Christie Cline ChristieC@M365x49603954.OnMicrosoft.com ChristieC TRUE Member Sales Buyer
Crystal@M365x49603954.OnMicrosoft.com Crystal TRUE Member
Debra Berger DebraB@M365x49603954.OnMicrosoft.com DebraB TRUE Member Executive Management Administrative Assistant
Delia Dennis DeliaD TRUE Member Engineering Engineer
Diego Siciliani DiegoS@M365x49603954.OnMicrosoft.com DiegoS TRUE Member HR HR Manager
Gerhart Moller GerhartM TRUE Member Manager
Grady Archie GradyA@M365x49603954.OnMicrosoft.com GradyA TRUE Member R&D Designer
Hood@M365x49603954.OnMicrosoft.com Hood TRUE Member
Irvin Sayers IrvinS@M365x49603954.OnMicrosoft.com IrvinS TRUE Member R&D Project Manager
Isaiah Langer IsaiahL@M365x49603954.OnMicrosoft.com IsaiahL TRUE Member Sales Sales Rep
Johanna Lorenz JohannaL@M365x49603954.OnMicrosoft.com JohannaL TRUE Member Engineers Senior Engineer
Joni Sherman JoniS@M365x49603954.OnMicrosoft.com JoniS TRUE Member Legal Paralegal
Lee Gu LeeG@M365x49603954.OnMicrosoft.com LeeG TRUE Member Manufacturing Director
Lidia Holloway LidiaH@M365x49603954.OnMicrosoft.com LidiaH TRUE Member Engineering Product Manager
Lynne Robbins LynneR@M365x49603954.OnMicrosoft.com LynneR TRUE Member Retail Planner
Mallory Cortez MalloryC TRUE Member
Megan Bowen MeganB@M365x49603954.OnMicrosoft.com MeganB TRUE Member Markets Marketing Manager
Miriam Graham MiriamG@M365x49603954.OnMicrosoft.com MiriamG TRUE Member Sales & Marketing Director
Nestor Wilke NestorW@M365x49603954.OnMicrosoft.com NestorW TRUE Member Operations Director
Patti Fernandez PattiF@M365x49603954.OnMicrosoft.com PattiF TRUE Member Executive Management President
Pradeep Gupta PradeepG@M365x49603954.OnMicrosoft.com PradeepG TRUE Member Finance Accountant
Rainier@M365x49603954.OnMicrosoft.com Rainier TRUE Member
Raul Razo RaulR TRUE Member Engineering Engineer
Stevens@M365x49603954.OnMicrosoft.com Stevens TRUE Member
Ava Anderson AvaAnderson.demo001@M365x49603954.On AvaAnderson.dem TRUE Member Retail Contoso Demo Retail Specialist DEMO-1001 Employee 01.10.2026
Ava Bennett AvaBennett.demo002@M365x49603954.OnM AvaBennett.demo TRUE Member Marketing Contoso Demo Marketing Specialist DEMO-1002 Employee 01.10.2026
Ava Carter AvaCarter.demo003@M365x49603954.OnMic AvaCarter.demo00 TRUE Member IT Contoso Demo IT Specialist DEMO-1003 Employee 01.10.2026
Ava Diaz AvaDiaz.demo004@M365x49603954.OnMicro AvaDiaz.demo004 TRUE Member Sales Contoso Demo Sales Representative DEMO-1004 Employee 01.10.2026
entra_users_20261012_133222
Sheet1
Auto
Let's edit your workbook
Allow editing
Create a new sheet where you group all the different departments together, because I want to understand all the different spellings of smame departmen, Group them together and create a list so I can change them once and it will be changed in the group and original sheet
Add a formula column to this table
Apply conditional formatting
Ready
Accessibility: Investigate
70 %
ENG NO
13:57 14.10.2026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Search
File Home Insert Draw Page Layout Formulas Data Review View Automate Help
Comments Share
Paste Clipboard
Aptos Narrow 11 Font
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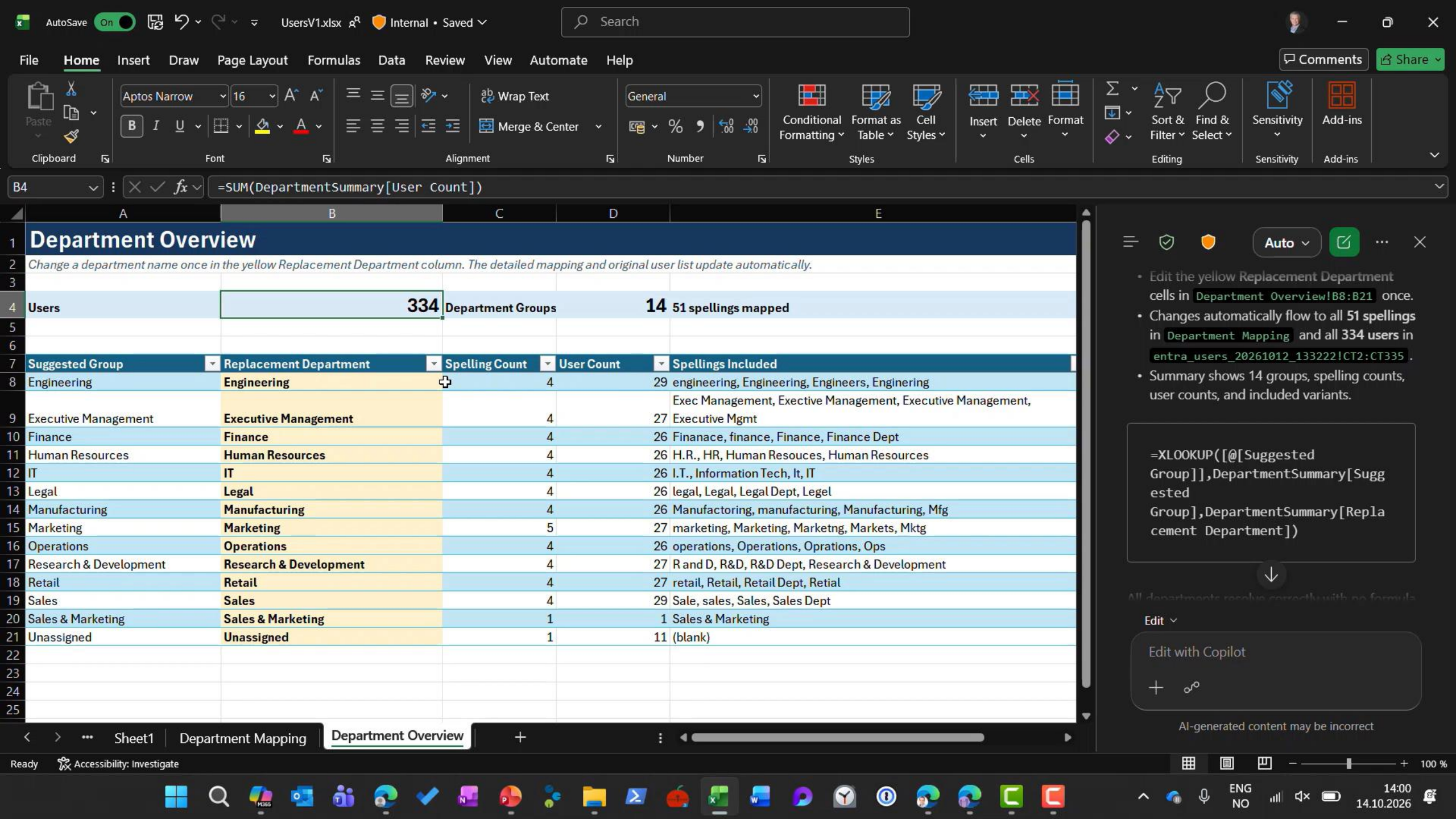
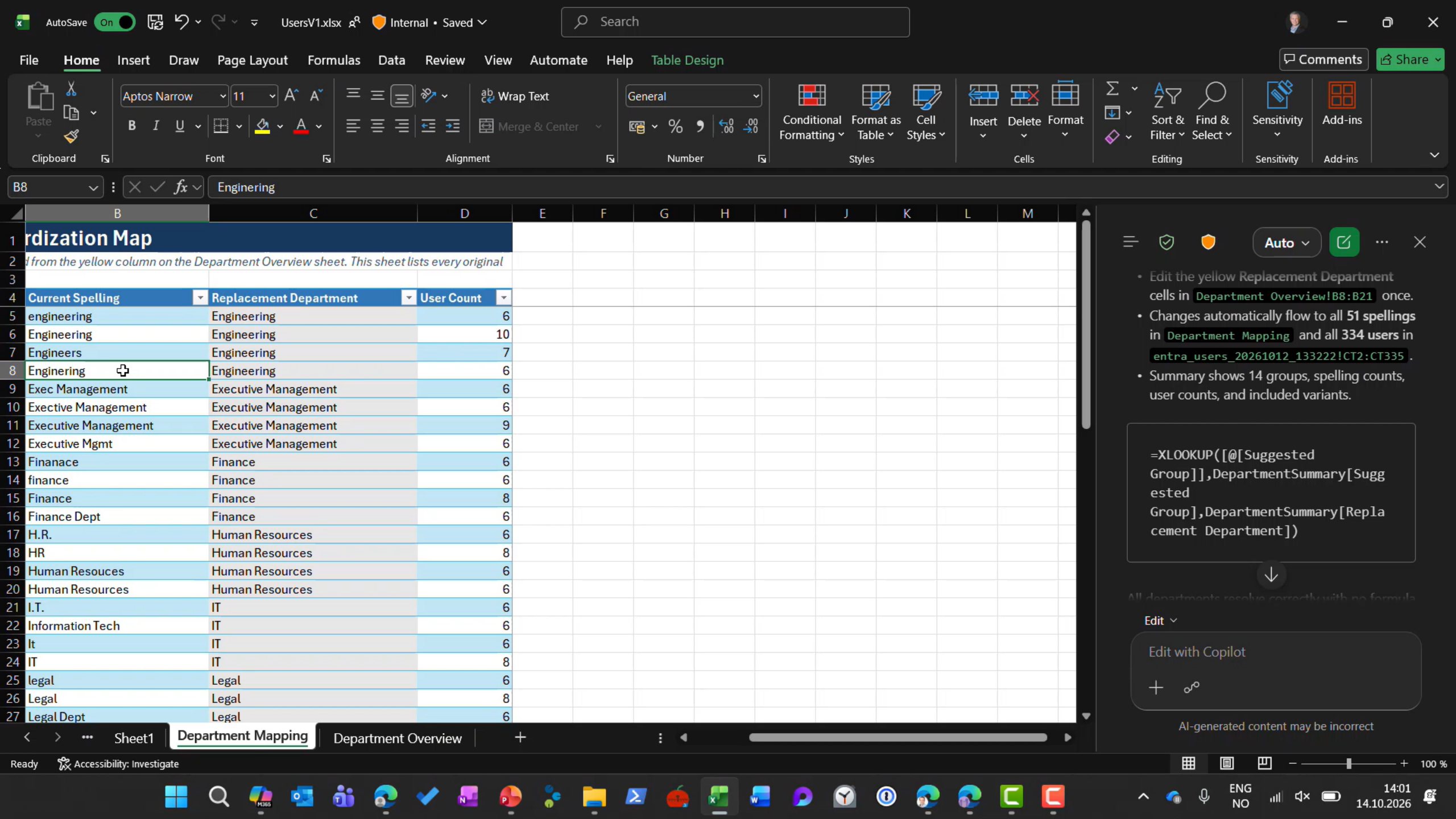
General Number
Conditional Formatting Format as Table Cell Styles Styles
Insert Delete Format Cells
Sort & Filter Find & Select Editing
Sensitivity Sensitivity
Add-ins Add-ins
CLEAN DATA WITH COPILOT Copilot has 11 suggestions to clean your data Show suggestions Dismiss for 30 days
L2
rdization Map
column once; the Standardized Department column in the source table updates
Current Spelling | Replacement Department | User Count
engineering | Engineering | 6
Engineering | Engineering | 10
Engineers | Engineering | 7
Enginering | Engineering | 6
Exec Management | Executive Management | 6
Exective Management | Executive Management | 6
Executive Management | Executive Management | 9
Executive Mgmt | Executive Management | 6
Finanace | Finance | 6
finance | Finance | 6
Finance | Finance | 8
Finance Dept | Finance | 6
H.R. | Human Resources | 6
HR | Human Resources | 8
Human Resouces | Human Resources | 6
Human Resources | Human Resources | 6
I.T. | IT | 6
Information Tech | IT | 6
It | IT | 6
IT | IT | 8
legal | Legal | 6
=XLOOKUP(IF([@department]="","(blank)",[@department]), DepartmentMap[Current Spelling], DepartmentMap[Replacement Department], "Review Needed")
No unresolved departi s remain.
Edit
create a summary and an overview so I can change the replacement department
AI-generated content may be incorrect
entra_users_20261012_133222 Sheet1 Department Mapping
Ready Accessibility: Investigate
100 %
ENG NO
13:59 14.10.2026


AutoSave On
UsersV1.xlsx
Internal • Saved
Search
File Home Insert Draw Page Layout Formulas Data Review View Automate Help
Comments Share
Paste Clipboard
Aptos Narrow 16 Font
Wrap Text Merge & Center Alignment
General Number
Conditional Formatting Format as Table Cell Styles Styles
Insert Delete Format Cells
Sort & Filter Find & Select Editing
Sensitivity Sensitivity
Add-ins Add-ins
B4
=SUM(DepartmentSummary[User Count])
Department Overview
Change a department name once in the yellow Replacement Department column. The detailed mapping and original user list update automatically.
Users 334 Department Groups 14 51 spellings mapped
Suggested Group | Replacement Department | Spelling Count | User Count | Spellings Included
Engineering | Engineering | 4 | 29 | engineering, Engineering, Engineers, Enginering
Executive Management | Executive Management | 4 | 27 | Exec Management, Exective Management, Executive Management, Executive Mgmt
Finance | Finance | 4 | 26 | Finanace, finance, Finance, Finance Dept
Human Resources | Human Resources | 4 | 26 | H.R., HR, Human Resouces, Human Resources
IT | IT | 4 | 26 | I.T., Information Tech, It, IT
Legal | Legal | 4 | 26 | legal, Legal, Legal Dept, Legel
Manufacturing | Manufacturing | 4 | 26 | Manufactoring, manufacturing, Manufacturing, Mfg
Marketing | Marketing | 5 | 27 | marketing, Marketing, Marketng, Markets, Mktg
Operations | Operations | 4 | 26 | operations, Operations, Oprations, Ops
Research & Development | Research & Development | 4 | 27 | R and D, R&D, R&D Dept, Research & Development
Retail | Retail | 4 | 27 | retail, Retail, Retail Dept, Retial
Sales | Sales | 4 | 29 | Sale, sales, Sales, Sales Dept
Sales & Marketing | Sales & Marketing | 1 | 1 | Sales & Marketing
Unassigned | Unassigned | 1 | 11 | (blank)
Sheet1 Department Mapping Department Overview
Auto
Edit the yellow Replacement Department cells in Department Overview!B8:B21 once.
Changes automatically flow to all 51 spellings in Department Mapping and all 334 users in entra_users_20261012_133222!CT2:CT335.
Summary shows 14 groups, spelling counts, user counts, and included variants.
=XLOOKUP([@[Suggested Group]],DepartmentSummary[Suggested Group],DepartmentSummary[Replacement Department])
All departments resolve correctly with no formula
Edit
Edit with Copilot
AI-generated content may be incorrect
Ready Accessibility: Investigate
100 %
ENG NO
14:00 14.10.2026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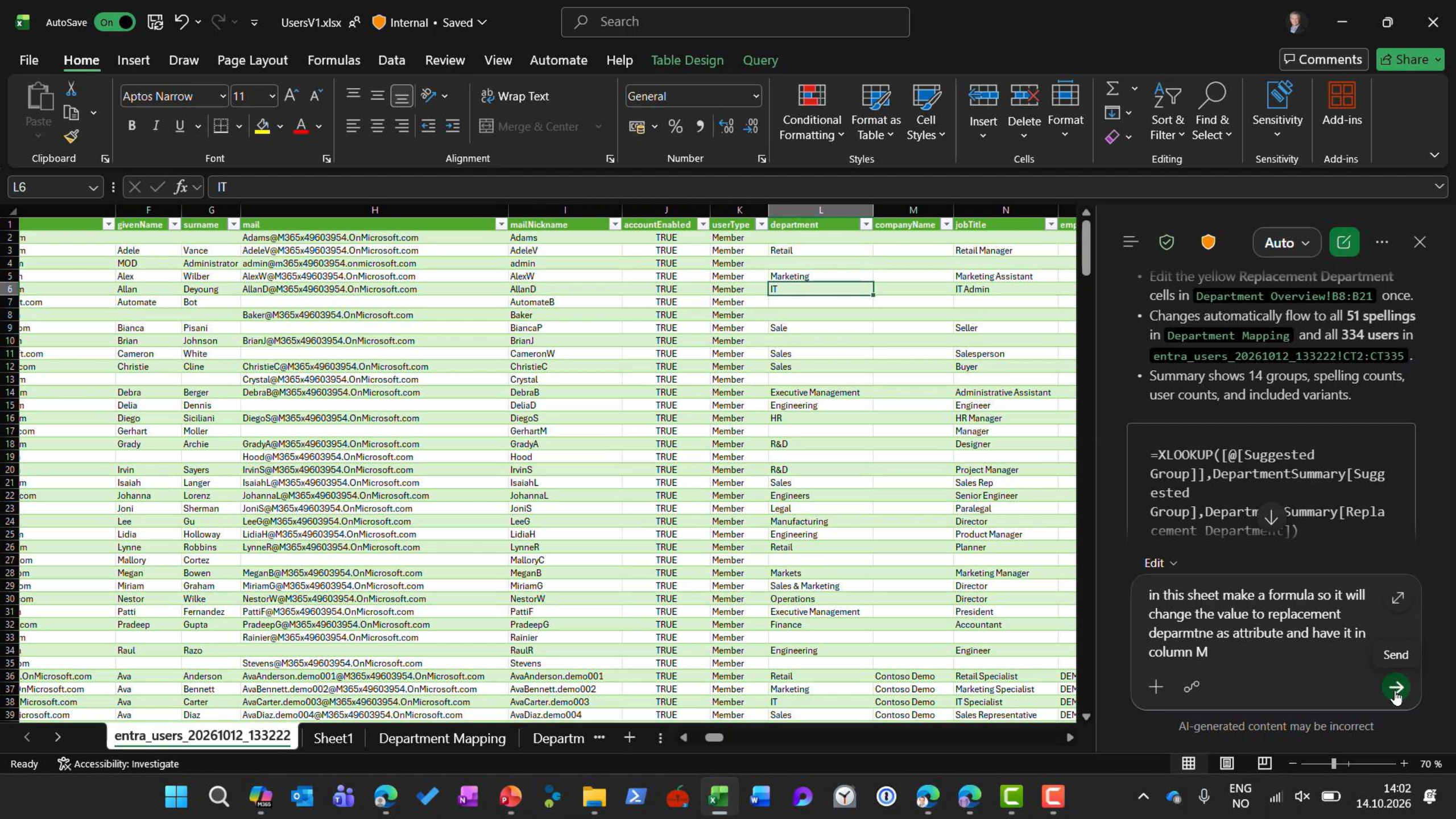
File Home Insert Draw Page Layout Formulas Data Review View Automate Help Table Design
Comments Share
Paste Clipboard
Aptos Narrow 11 Font
Wrap Text Merge & Center Alignment
General Number
Conditional Formatting Format as Table Cell Styles Styles
Insert Delete Format Cells
Sort & Filter Find & Select Editing
Sensitivity Sensitivity
Add-ins Add-ins
B8 Enginering
rdization Map
l from the yellow column on the Department Overview sheet. This sheet lists every original
Current Spelling | Replacement Department | User Count
engineering | Engineering | 6
Engineering | Engineering | 10
Engineers | Engineering | 7
Enginering | Engineering | 6
Exec Management | Executive Management | 6
Exective Management | Executive Management | 6
Executive Management | Executive Management | 9
Executive Mgmt | Executive Management | 6
Finanace | Finance | 6
finance | Finance | 6
Finance | Finance | 8
Finance Dept | Finance | 6
H.R. | Human Resources | 6
HR | Human Resources | 8
Human Resouces | Human Resources | 6
Human Resources | Human Resources | 6
I.T. | IT | 6
Information Tech | IT | 6
It | IT | 6
IT | IT | 8
legal | Legal | 6
Legal | Legal | 8
Legal Dept | Legal | 6
Auto
• Edit the yellow Replacement Department cells in Department Overview!B8:B21 once.
• Changes automatically flow to all 51 spellings in Department Mapping and all 334 users in entra_users_20261012_133222!CT2:CT335.
• Summary shows 14 groups, spelling counts, user counts, and included variants.
=XLOOKUP([@[Suggested Group]],DepartmentSummary[Suggested Group],DepartmentSummary[Replacement Department])
All departments resolve correctly with no formula
Edit
Edit with Copilot
AI-generated content may be incorrect
Sheet1 Department Mapping Department Overview
Ready Accessibility: Investigate
100 %
ENG NO
14:01 14.10.2026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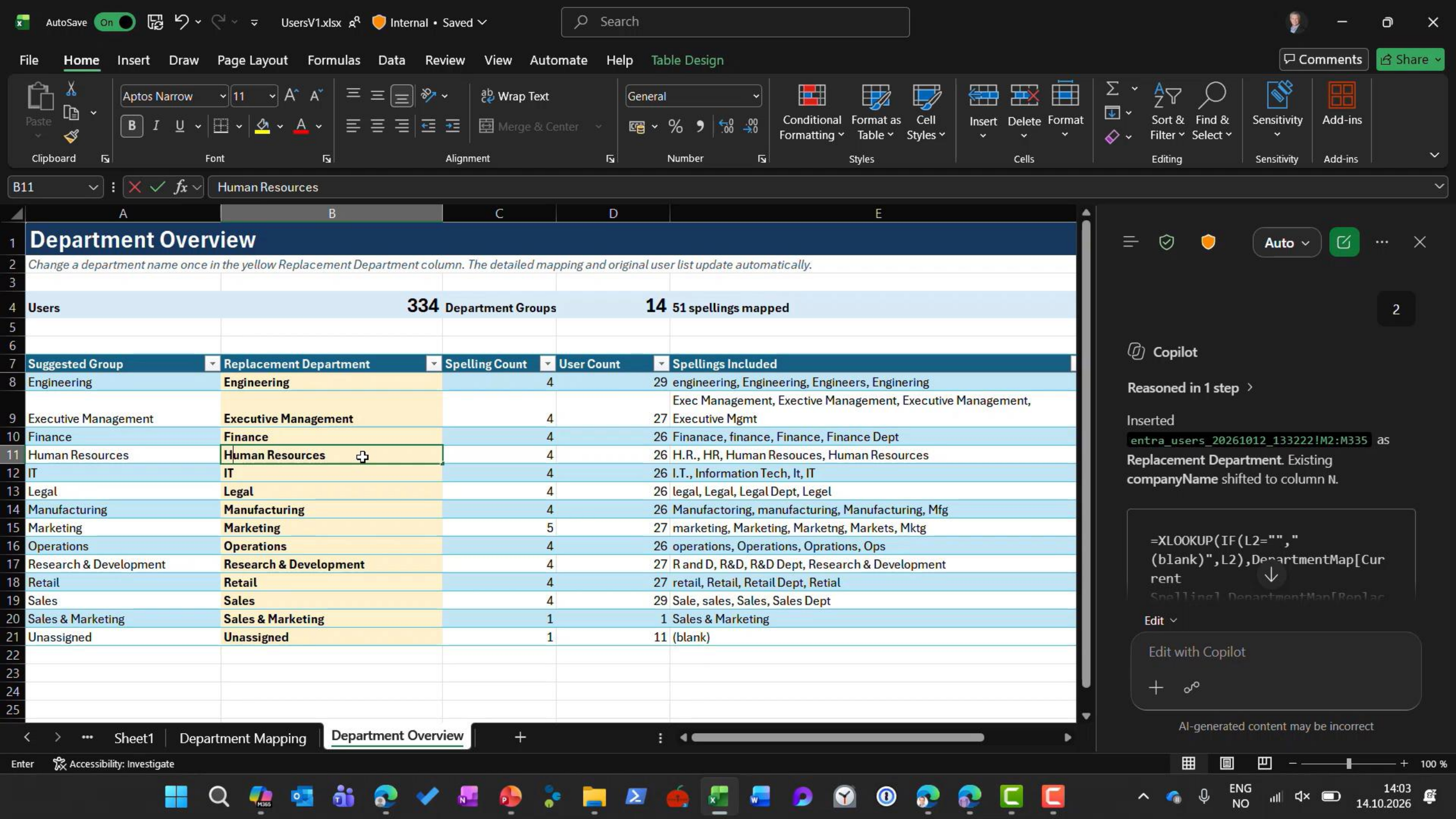
File Home Insert Draw Page Layout Formulas Data Review View Automate Help Table Design Query
Comments Share
Aptos Narrow 11
Wrap Text
General
Conditional Formatting
Format as Table
Cell Styles
Insert Delete Format
Sort & Filter
Find & Select
Sensitivity
Add-ins
Clipboard Font Alignment Number Styles Cells Editing Sensitivity Add-ins
L6
IT
givenName surname mail mailNickname accountEnabled userType department companyName jobTitle
Adams@M365x49603954.OnMicrosoft.com Adams TRUE Member
Adele Vance AdeleV@M365x49603954.OnMicrosoft.com AdeleV TRUE Member Retail Retail Manager
MOD Administrator admin@m365x49603954.onmicrosoft.com admin TRUE Member
Alex Wilber AlexW@M365x49603954.OnMicrosoft.com AlexW TRUE Member Marketing Marketing Assistant
Allan Deyoung AllanD@M365x49603954.OnMicrosoft.com AllanD TRUE Member IT IT Admin
Automate Bot AutomateB TRUE Member
Baker@M365x49603954.OnMicrosoft.com Baker TRUE Member
Bianca Pisani BiancaP TRUE Member Sale Seller
Brian Johnson BrianJ@M365x49603954.OnMicrosoft.com BrianJ TRUE Member
Cameron White CameronW TRUE Member Sales Salesperson
Christie Cline ChristieC@M365x49603954.OnMicrosoft.com ChristieC TRUE Member Sales Buyer
Crystal@M365x49603954.OnMicrosoft.com Crystal TRUE Member
Debra Berger DebraB@M365x49603954.OnMicrosoft.com DebraB TRUE Member Executive Management Administrative Assistant
Delia Dennis DeliaD TRUE Member Engineering Engineer
Diego Siciliani DiegoS@M365x49603954.OnMicrosoft.com DiegoS TRUE Member HR HR Manager
Gerhart Moller GerhartM TRUE Member Manager
Grady Archie GradyA@M365x49603954.OnMicrosoft.com GradyA TRUE Member R&D Designer
Hood@M365x49603954.OnMicrosoft.com Hood TRUE Member
Irvin Sayers IrvinS@M365x49603954.OnMicrosoft.com IrvinS TRUE Member R&D Project Manager
Isaiah Langer IsaiahL@M365x49603954.OnMicrosoft.com IsaiahL TRUE Member Sales Sales Rep
Johanna Lorenz JohannaL@M365x49603954.OnMicrosoft.com JohannaL TRUE Member Engineers Senior Engineer
Joni Sherman JoniS@M365x49603954.OnMicrosoft.com JoniS TRUE Member Legal Paralegal
Lee Gu LeeG@M365x49603954.OnMicrosoft.com LeeG TRUE Member Manufacturing Director
Lidia Holloway LidiaH@M365x49603954.OnMicrosoft.com LidiaH TRUE Member Engineering Product Manager
Lynne Robbins LynneR@M365x49603954.OnMicrosoft.com LynneR TRUE Member Retail Planner
Mallory Cortez MalloryC TRUE Member
Megan Bowen MeganB@M365x49603954.OnMicrosoft.com MeganB TRUE Member Markets Marketing Manager
Miriam Graham MiriamG@M365x49603954.OnMicrosoft.com MiriamG TRUE Member Sales & Marketing Director
Nestor Wilke NestorW@M365x49603954.OnMicrosoft.com NestorW TRUE Member Operations Director
Patti Fernandez PattiF@M365x49603954.OnMicrosoft.com PattiF TRUE Member Executive Management President
Pradeep Gupta PradeepG@M365x49603954.OnMicrosoft.com PradeepG TRUE Member Finance Accountant
Rainier@M365x49603954.OnMicrosoft.com Rainier TRUE Member
Raul Razo RaulR TRUE Member Engineering Engineer
Stevens@M365x49603954.OnMicrosoft.com Stevens TRUE Member
AvaAnderson.demo001@M365x49603954.OnMicrosoft.com Ava Anderson AvaAnderson.demo001 TRUE Member Retail Contoso Demo Retail Specialist
AvaBennett.demo002@M365x49603954.OnMicrosoft.com Ava Bennett AvaBennett.demo002 TRUE Member Marketing Contoso Demo Marketing Specialist
AvaCarter.demo003@M365x49603954.OnMicrosoft.com Ava Carter AvaCarter.demo003 TRUE Member IT Contoso Demo IT Specialist
AvaDiaz.demo004@M365x49603954.OnMicrosoft.com Ava Diaz AvaDiaz.demo004 TRUE Member Sales Contoso Demo Sales Representative
entra_users_20261012_133222 Sheet1 Department Mapping Departm
Auto
Edit the yellow Replacement Department cells in Department Overview!B8:B21 once.
Changes automatically flow to all 51 spellings in Department Mapping and all 334 users in entra_users_20261012_133222!CT2:CT335.
Summary shows 14 groups, spelling counts, user counts, and included variants.
=XLOOKUP([@[Suggested Group]],DepartmentSummary[Suggested Group],Departm Summary[Replacement Departme])
Edit
in this sheet make a formula so it will change the value to replacement deparmtne as attribute and have it in column M
Send
AI-generated content may be incorrect
Ready
Accessibility: Investigate
70 %
ENG NO
14:02 14.10.2026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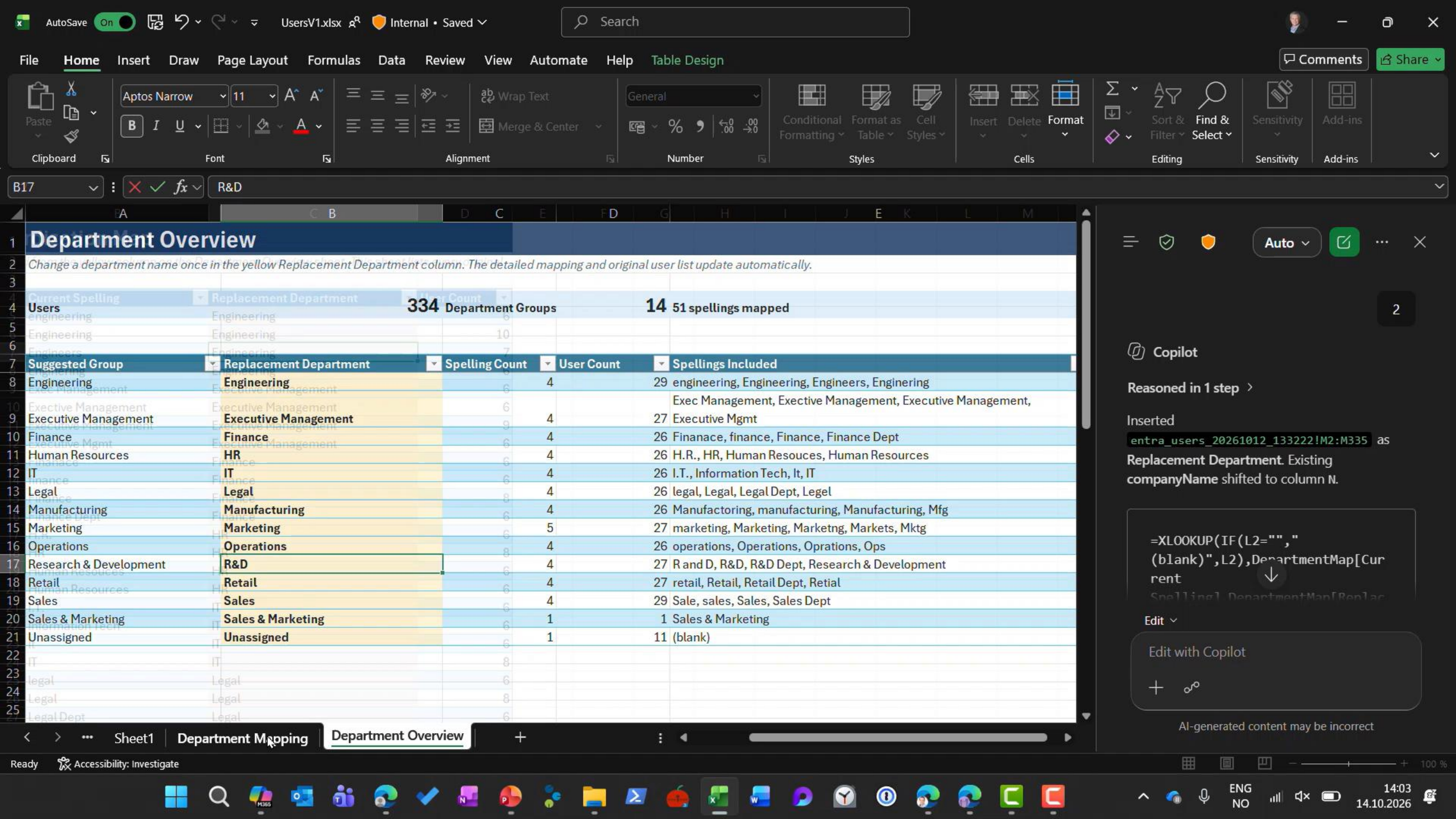
File Home Insert Draw Page Layout Formulas Data Review View Automate Help Table Design
Comments Share
Paste Clipboard
Aptos Narrow 11 Font
Wrap Text Merge & Center Alignment
General Number
Conditional Formatting Format as Table Cell Styles Styles
Insert Delete Format Cells
Sort & Filter Find & Select Editing
Sensitivity Sensitivity
Add-ins Add-ins
B11
Human Resources
Department Overview
Change a department name once in the yellow Replacement Department column. The detailed mapping and original user list update automatically.
Users 334 Department Groups 14 51 spellings mapped
Suggested Group | Replacement Department | Spelling Count | User Count | Spellings Included
Engineering | Engineering | 4 | 29 | engineering, Engineering, Engineers, Enginering
Executive Management | Executive Management | 4 | 27 | Exec Management, Exective Management, Executive Management, Executive Mgmt
Finance | Finance | 4 | 26 | Finanace, finance, Finance, Finance Dept
Human Resources | Human Resources | 4 | 26 | H.R., HR, Human Resouces, Human Resources
IT | IT | 4 | 26 | I.T., Information Tech, It, IT
Legal | Legal | 4 | 26 | legal, Legal, Legal Dept, Legel
Manufacturing | Manufacturing | 4 | 26 | Manufactoring, manufacturing, Manufacturing, Mfg
Marketing | Marketing | 5 | 27 | marketing, Marketing, Marketng, Markets, Mktg
Operations | Operations | 4 | 26 | operations, Operations, Oprations, Ops
Research & Development | Research & Development | 4 | 27 | R and D, R&D, R&D Dept, Research & Development
Retail | Retail | 4 | 27 | retail, Retail, Retail Dept, Retial
Sales | Sales | 4 | 29 | Sale, sales, Sales, Sales Dept
Sales & Marketing | Sales & Marketing | 1 | 1 | Sales & Marketing
Unassigned | Unassigned | 1 | 11 | (blank)
Sheet1 Department Mapping Department Overview
Auto
2
Copilot
Reasoned in 1 step
Inserted entra_users_20261012_133222!M2:M335 as Replacement Department. Existing companyName shifted to column N.
=XLOOKUP(IF(L2="","(blank)",L2),DepartmentMap[Current
Edit
Edit with Copilot
AI-generated content may be incorrect
Enter Accessibility: Investigate
100 %
ENG NO
14:03 14.10.2026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Aptos Narrow 11
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Format as Table
Cell Styles
Insert Delete Format
Sort & Filter
Find & Select
Sensitivity
Add-ins
Clipboard Font Alignment Number Styles Cells Editing Sensitivity Add-ins
B17
R&D
Department Overview
Change a department name once in the yellow Replacement Department column. The detailed mapping and original user list update automatically.
Users 334 Department Groups 14 51 spellings mapped
Suggested Group | Replacement Department | Spelling Count | User Count | Spellings Included
Engineering | Engineering | 4 | 29 | engineering, Engineering, Engineers, Enginering
Executive Management | Executive Management | 4 | 27 | Exec Management, Exective Management, Executive Management, Executive Mgmt
Finance | Finance | 4 | 26 | Finanace, finance, Finance, Finance Dept
Human Resources | HR | 4 | 26 | H.R., HR, Human Resouces, Human Resources
IT | IT | 4 | 26 | I.T., Information Tech, It, IT
Legal | Legal | 4 | 26 | legal, Legal, Legal Dept, Legel
Manufacturing | Manufacturing | 4 | 26 | Manufactoring, manufacturing, Manufacturing, Mfg
Marketing | Marketing | 5 | 27 | marketing, Marketing, Marketng, Markets, Mktg
Operations | Operations | 4 | 26 | operations, Operations, Oprations, Ops
Research & Development | R&D | 4 | 27 | R and D, R&D, R&D Dept, Research & Development
Retail | Retail | 4 | 27 | retail, Retail, Retail Dept, Retial
Sales | Sales | 4 | 29 | Sale, sales, Sales, Sales Dept
Sales & Marketing | Sales & Marketing | 1 | 1 | Sales & Marketing
Unassigned | Unassigned | 1 | 11 | (blank)
Sheet1 Department Mapping Department Overview
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2
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Reasoned in 1 step
Inserted entra_users_20261012_133222!M2:M335 as Replacement Department. Existing companyName shifted to column N.
=XLOOKUP(IF(L2="","(blank)",L2),DepartmentMap[Current Spelling],DepartmentMap[Replac
Edit
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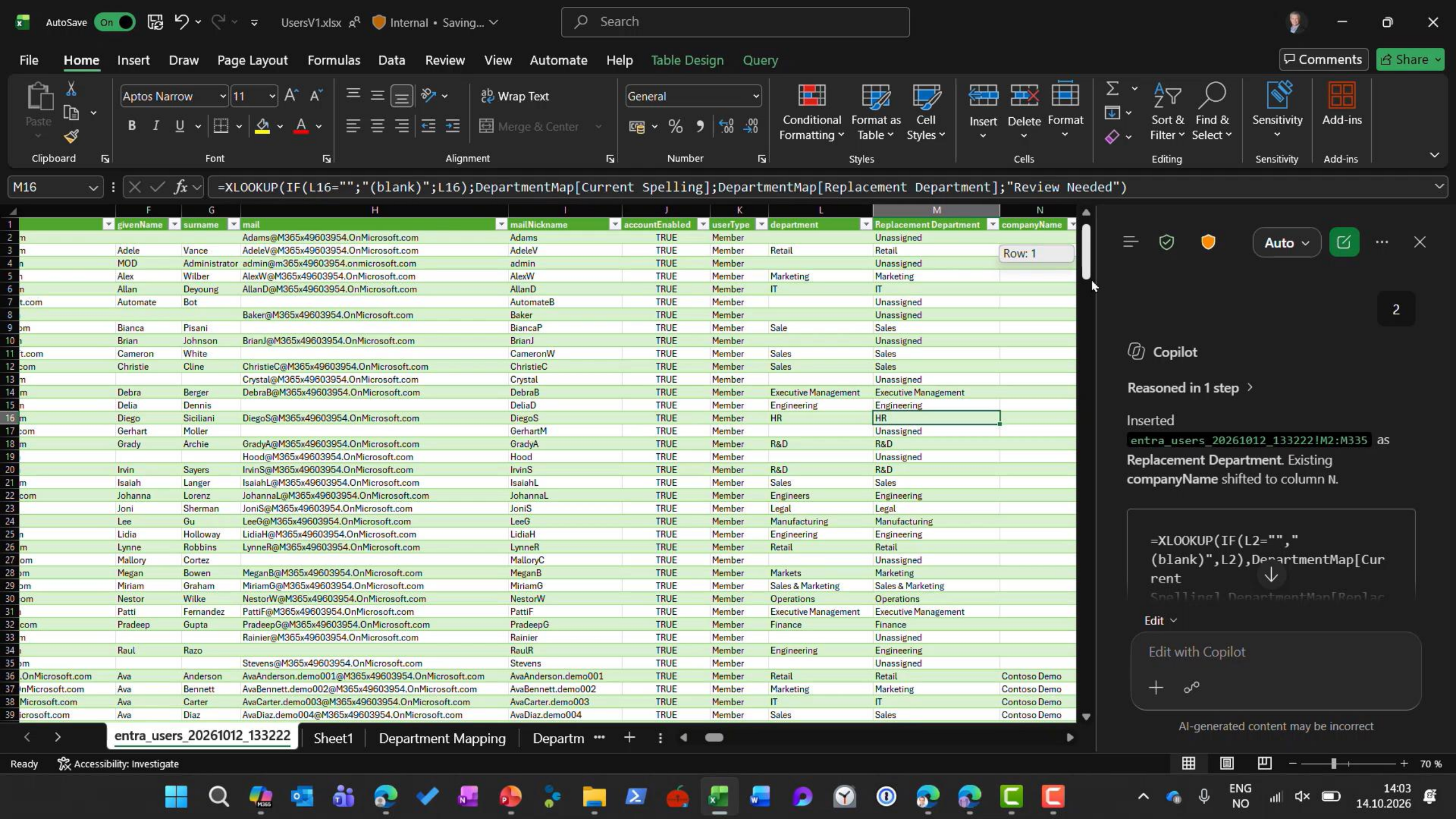
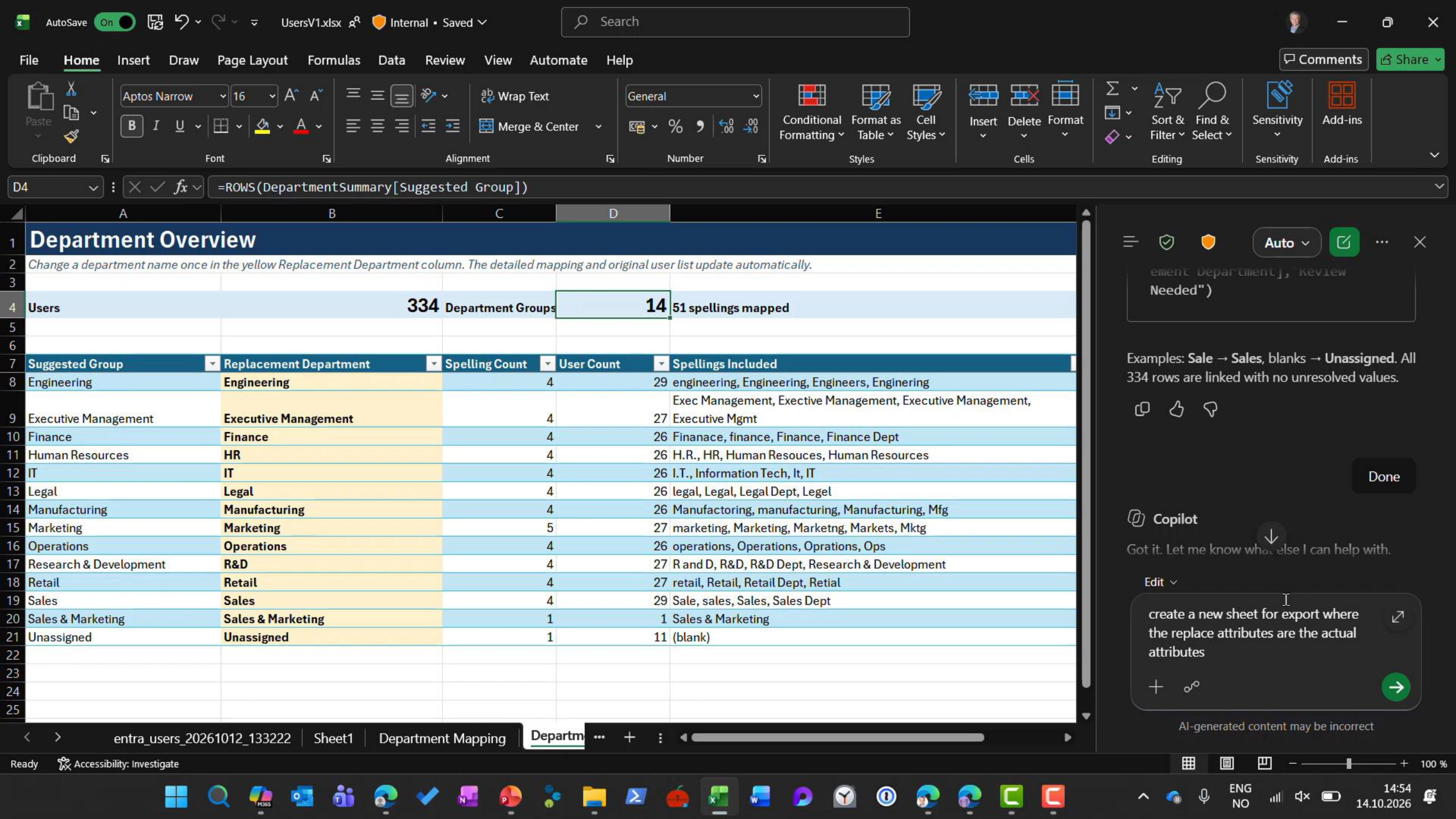


	F	G	H	I	J	K	L	M	N
1	givenName	surname	mail	mailNickname	accountEnabled	userType	department	Replacement Department	companyName
2			Adams@M365x49603954.OnMicrosoft.com	Adams	TRUE	Member		Unassigned	
3	Adele	Vance	AdeleV@M365x49603954.OnMicrosoft.com	AdeleV	TRUE	Member	Retail	Retail	
4	MOD	Administrator	admin@m365x49603954.onmicrosoft.com	admin	TRUE	Member		Unassigned	
5	Alex	Wilber	AlexW@M365x49603954.OnMicrosoft.com	AlexW	TRUE	Member	Marketing	Marketing	
6	Allan	Deyoung	AllanD@M365x49603954.OnMicrosoft.com	AllanD	TRUE	Member	IT	IT	
7	Automate	Bot		AutomateB	TRUE	Member		Unassigned	
8			Baker@M365x49603954.OnMicrosoft.com	Baker	TRUE	Member		Unassigned	
9	Bianca	Pisani		BiancaP	TRUE	Member	Sale	Sales	
10	Brian	Johnson	BrianJ@M365x49603954.OnMicrosoft.com	BrianJ	TRUE	Member		Unassigned	
11	Cameron	White		CameronW	TRUE	Member	Sales	Sales	
12	Christie	Cline	ChristieC@M365x49603954.OnMicrosoft.com	ChristieC	TRUE	Member	Sales	Sales	
13			Crystal@M365x49603954.OnMicrosoft.com	Crystal	TRUE	Member		Unassigned	
14	Debra	Berger	DebraB@M365x49603954.OnMicrosoft.com	DebraB	TRUE	Member	Executive Management	Executive Management	
15	Delia	Dennis		DeliaD	TRUE	Member	Engineering	Engineering	
16	Diego	Siciliani	DiegoS@M365x49603954.OnMicrosoft.com	DiegoS	TRUE	Member	HR	HR	
17	Gerhart	Moller		GerhartM	TRUE	Member		Unassigned	
18	Grady	Archie	GradyA@M365x49603954.OnMicrosoft.com	GradyA	TRUE	Member	R&D	R&D	
19			Hood@M365x49603954.OnMicrosoft.com	Hood	TRUE	Member		Unassigned	
20	Irvin	Sayers	IrvinS@M365x49603954.OnMicrosoft.com	IrvinS	TRUE	Member	R&D	R&D	
21	Isaiah	Langer	IsaiahL@M365x49603954.OnMicrosoft.com	IsaiahL	TRUE	Member	Sales	Sales	
22	Johanna	Lorenz	JohannaL@M365x49603954.OnMicrosoft.com	JohannaL	TRUE	Member	Engineers	Engineering	
23	Joni	Sherman	JoniS@M365x49603954.OnMicrosoft.com	JoniS	TRUE	Member	Legal	Legal	
24	Lee	Gu	LeeG@M365x49603954.OnMicrosoft.com	LeeG	TRUE	Member	Manufacturing	Manufacturing	
25	Lidia	Holloway	LidiaH@M365x49603954.OnMicrosoft.com	LidiaH	TRUE	Member	Engineering	Engineering	
26	Lynne	Robbins	LynneR@M365x49603954.OnMicrosoft.com	LynneR	TRUE	Member	Retail	Retail	
27	Mallory	Cortez		MalloryC	TRUE	Member		Unassigned	
28	Megan	Bowen	MeganB@M365x49603954.OnMicrosoft.com	MeganB	TRUE	Member	Markets	Marketing	
29	Miriam	Graham	MiriamG@M365x49603954.OnMicrosoft.com	MiriamG	TRUE	Member	Sales & Marketing	Sales & Marketing	
30	Nestor	Wilke	NestorW@M365x49603954.OnMicrosoft.com	NestorW	TRUE	Member	Operations	Operations	
31	Patti	Fernandez	PattiF@M365x49603954.OnMicrosoft.com	PattiF	TRUE	Member	Executive Management	Executive Management	
32	Pradeep	Gupta	PradeepG@M365x49603954.OnMicrosoft.com	PradeepG	TRUE	Member	Finance	Finance	
33			Rainier@M365x49603954.OnMicrosoft.com	Rainier	TRUE	Member		Unassigned	
34	Raul	Razo		RaulR	TRUE	Member	Engineering	Engineering	
35			Stevens@M365x49603954.OnMicrosoft.com	Stevens	TRUE	Member		Unassigned	
36	Ava	Anderson	AvaAnderson.demo001@M365x49603954.OnMicrosoft.com	AvaAnderson.demo001	TRUE	Member	Retail	Retail	Contoso Demo
37	Ava	Bennett	AvaBennett.demo002@M365x49603954.OnMicrosoft.com	AvaBennett.demo002	TRUE	Member	Marketing	Marketing	Contoso Demo
38	Ava	Carter	AvaCarter.demo003@M365x49603954.OnMicrosoft.com	AvaCarter.demo003	TRUE	Member	IT	IT	Contoso Demo
39	Ava	Diaz	AvaDiaz.demo004@M365x49603954.OnMicrosoft.com	AvaDiaz.demo004	TRUE	Member	Sales	Sales	Contoso Demo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Add-ins
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D4
=ROWS(DepartmentSummary[Suggested Group])
Department Overview
Change a department name once in the yellow Replacement Department column. The detailed mapping and original user list update automatically.
Users 334 Department Groups 14 51 spellings mapped
Suggested Group | Replacement Department | Spelling Count | User Count | Spellings Included
Engineering | Engineering | 4 | 29 | engineering, Engineering, Engineers, Enginering
Executive Management | Executive Management | 4 | 27 | Exec Management, Exective Management, Executive Management, Executive Mgmt
Finance | Finance | 4 | 26 | Finanace, finance, Finance, Finance Dept
Human Resources | HR | 4 | 26 | H.R., HR, Human Resouces, Human Resources
IT | IT | 4 | 26 | I.T., Information Tech, It, IT
Legal | Legal | 4 | 26 | legal, Legal, Legal Dept, Legel
Manufacturing | Manufacturing | 4 | 26 | Manufactoring, manufacturing, Manufacturing, Mfg
Marketing | Marketing | 5 | 27 | marketing, Marketing, Marketng, Markets, Mktg
Operations | Operations | 4 | 26 | operations, Operations, Oprations, Ops
Research & Development | R&D | 4 | 27 | R and D, R&D, R&D Dept, Research & Development
Retail | Retail | 4 | 27 | retail, Retail, Retail Dept, Retial
Sales | Sales | 4 | 29 | Sale, sales, Sales, Sales Dept
Sales & Marketing | Sales & Marketing | 1 | 1 | Sales & Marketing
Unassigned | Unassigned | 1 | 11 | (blank)
entra_users_20261012_133222 Sheet1 Department Mapping Departm
Auto
Needed")
Examples: Sale → Sales, blanks → Unassigned. All 334 rows are linked with no unresolved values.
Done
Copilot
Got it. Let me know what else I can help with.
Edit
create a new sheet for export where the replace attributes are the actual attributes
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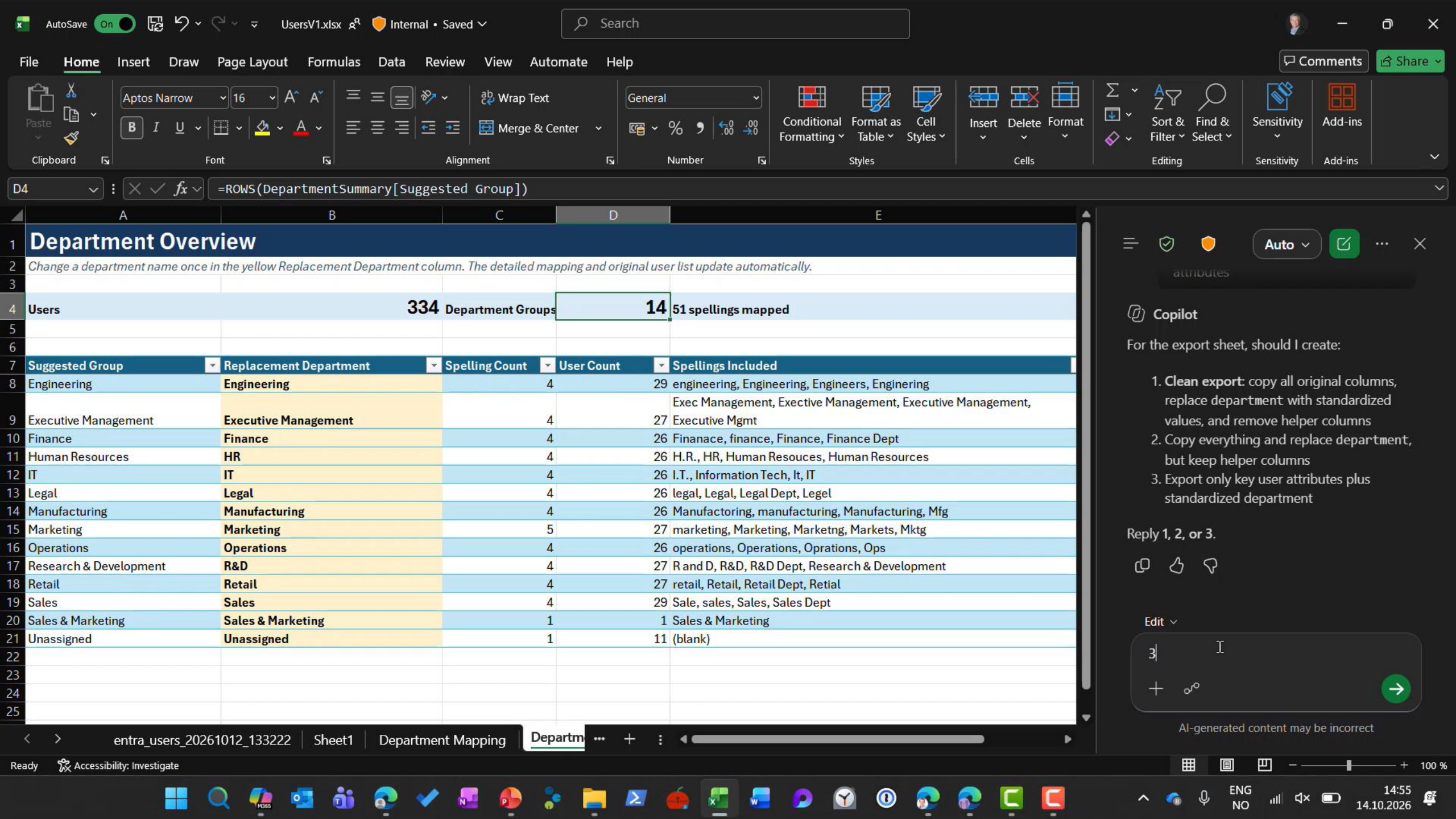
Ready
Accessibility: Investigate
100 %
ENG NO
14:54 14.10.2026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Conditional Formatting Format as Table Cell Styles Styles
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Add-ins Add-ins
D4 =ROWS(DepartmentSummary[Suggested Group])
Department Overview
Change a department name once in the yellow Replacement Department column. The detailed mapping and original user list update automatically.
Users 334 Department Groups 14 51 spellings mapped
Suggested Group | Replacement Department | Spelling Count | User Count | Spellings Included
Engineering | Engineering | 4 | 29 | engineering, Engineering, Engineers, Enginering
Executive Management | Executive Management | 4 | 27 | Exec Management, Exective Management, Executive Management, Executive Mgmt
Finance | Finance | 4 | 26 | Finanace, finance, Finance, Finance Dept
Human Resources | HR | 4 | 26 | H.R., HR, Human Resouces, Human Resources
IT | IT | 4 | 26 | I.T., Information Tech, It, IT
Legal | Legal | 4 | 26 | legal, Legal, Legal Dept, Legel
Manufacturing | Manufacturing | 4 | 26 | Manufactoring, manufacturing, Manufacturing, Mfg
Marketing | Marketing | 5 | 27 | marketing, Marketing, Marketng, Markets, Mktg
Operations | Operations | 4 | 26 | operations, Operations, Oprations, Ops
Research & Development | R&D | 4 | 27 | R and D, R&D, R&D Dept, Research & Development
Retail | Retail | 4 | 27 | retail, Retail, Retail Dept, Retial
Sales | Sales | 4 | 29 | Sale, sales, Sales, Sales Dept
Sales & Marketing | Sales & Marketing | 1 | 1 | Sales & Marketing
Unassigned | Unassigned | 1 | 11 | (blank)
entra_users_20261012_133222 Sheet1 Department Mapping Departm
Auto
Copilot
For the export sheet, should I create:
1. Clean export: copy all original columns, replace department with standardized values, and remove helper columns
2. Copy everything and replace department, but keep helper columns
3. Export only key user attributes plus standardized department
Reply 1, 2, or 3.
Edit
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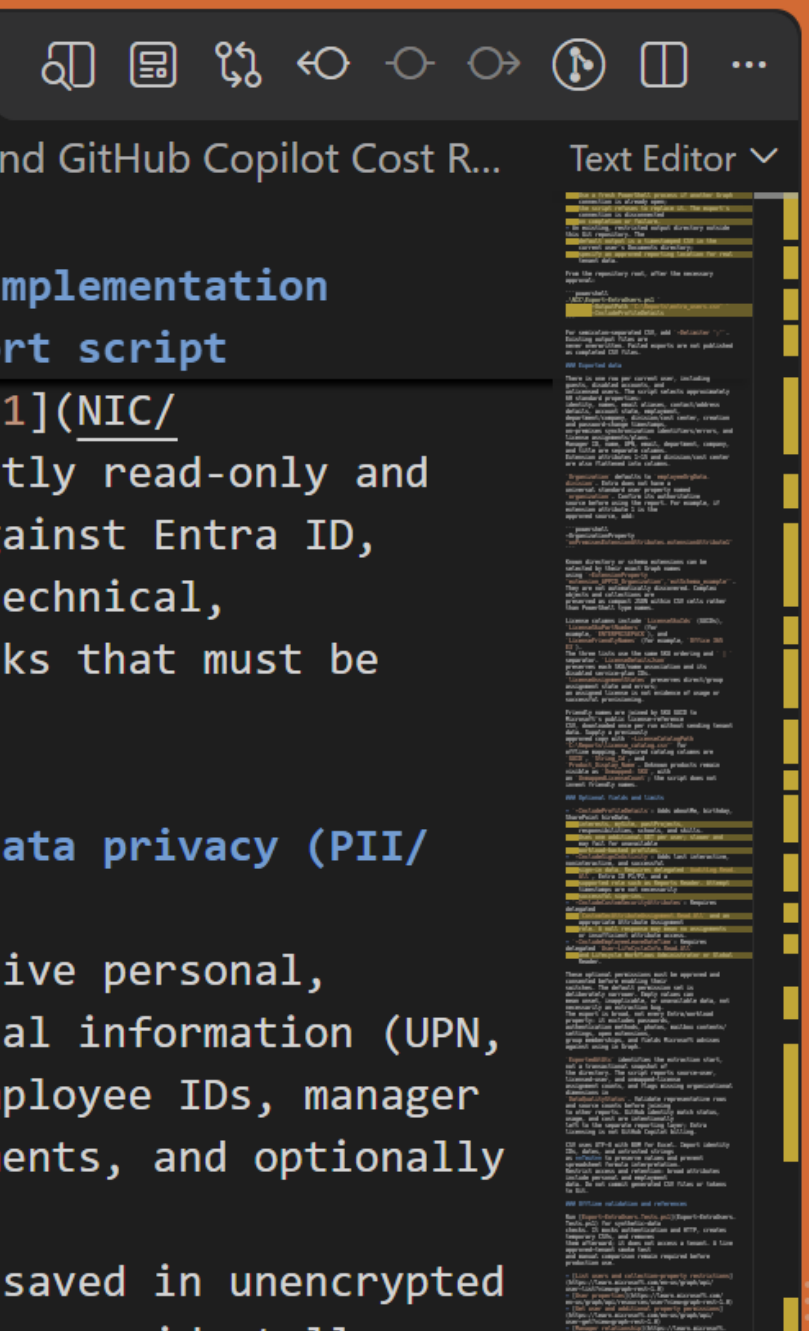
CSV is ready
Please give me
Risk and rollback strategy



Chat

PowerShell script for Entra ID export

Create the script to update users in Entra ID based on the updated data in the CSV file. Follow the same principels as allready used for the export script. I want the script to have a dry-run option. Also update the MD file with information of this script, the risks of running it, and also rollback options.

Completed 22 steps in 18m 3s

Created `Update-EntraUsers.ps1`, along with automated offline tests in `Update-EntraUsers.Tests.ps1`, and updated the plan documentation in `entra_department_copilot_reporting_change_plan.md`.

---

1. Script Architecture & Features (`Update-EntraUsers.ps1`)

- **Authentication & Consistency**: Uses `Microsoft.Graph.Authentication` with interactive WAM/account-picker sign-in (scoped to `Process` and cleanly disconnected on exit), matching the export script.
- **Identity Matching**: Matches on `id` (object GUID) or `userPrincipalName`.
- **Live Diffing**: Queries live state before modifying anything; only fields whose values actually differ are added to the update payload (`PATCH /users/{id}`).
- **Supported Attributes**:
  - Organizational: `department`, `companyName`, `jobTitle`, `officeLocation`

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Chat

PowerShell script for Entra ID export

- **Supported Attributes**:
  - Organizational: `department`, `companyName`, `jobTitle`, `officeLocation`
  - Contact/Location: `city`, `state`, `country`, `postalCode`, `streetAddress`, `usageLocation`, `preferredLanguage`
  - Worker Metadata: `employeeId`, `employeeType`, `employeeHireDate`
  - Organization Data: `division`, `costCenter` (`employeeOrgData`)
  - Extension Attributes: `extensionAttribute1`..`15` (for cloud-only users; skipped automatically if `onPremisesSyncEnabled = $true` to prevent API errors)
  - Manager: `manager_id` (via `PUT /users/{id}/manager/$ref` when `-UpdateManager` is specified)
  - Specific fields can be targeted with `-UpdateProperties @('department', 'companyName')`.
- **Dry-Run Mode** (`-DryRun` or `-WhatIf`):
  - Compares the CSV against live Entra ID objects.
  - Outputs console diffs: `[DIFF] User 'user@contoso.com' (id): department -> 'New' (was 'Old')`.
  - Generates `update_audit_<timestamp>.csv` with `Action = 'DryRun'` and `Status = 'Planned'`.
  - Executes zero `PATCH`, `PUT`, or `DELETE` requests.
- **Pre-Change Rollback Snapshot**:
  - On live runs, writes `rollback_users_<timestamp>.csv` containing the exact pre-change attribute values before modifying any user.

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Chat

PowerShell script for Entra ID export

2. Execution Examples

```
# 1. Preview changes in dry-run mode (no directory modifications)
.\NIC\Update-EntraUsers.ps1 -InputPath 'C:\Reports\entra_users_updated.csv' -DryRun

# 2. Preview specific properties with manager updates
.\NIC\Update-EntraUsers.ps1 -InputPath 'C:\Reports\entra_users_updated.csv' -Update

# 3. Live run (creates pre-change rollback snapshot and prompts for confirmation)
.\NIC\Update-EntraUsers.ps1 -InputPath 'C:\Reports\entra_users_updated.csv' -Update

# 4. Instant rollback using the auto-generated pre-change snapshot
.\NIC\Update-EntraUsers.ps1 -InputPath 'C:\Reports\rollback_users_20261013_120000.c
```

3. Change Plan Updates (entra_department_copilot_reporting_change_plan.md)

The change plan has been updated with:

- **Operational Details**: Prerequisites (`User.ReadWrite.All`), invocation flags, and dry-run reporting.
- **Execution Risks**:

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Chat

PowerShell script for Entra ID export

3. Change Plan Updates (entra_department_copilot_reporting_change_plan.md **)**

The change plan has been updated with:

- **Operational Details**: Prerequisites (`User.ReadWrite.All`), invocation flags, and dry-run reporting.
- **Execution Risks**:
  1. *Mass attribute overwrite*: Overwriting valid data if the CSV contains errors or unintended blanks.
  2. *Dynamic group & license side effects*: Changing `department`, `jobTitle`, or `companyName` triggers membership re-evaluation, altering app/license access.
  3. *Conditional Access / Access Package impact*: Changing department/worker attributes may alter sign-in risk or revoke access.
  4. *Manager loops*: Invalid manager IDs breaking org charts or approval chains.
  5. *Hybrid sync conflicts*: Modifying attributes in the cloud that are mastered on-premises.
  6. *Partial execution*: Mid-batch failures leaving the directory in a mixed state.
- **Rollback Options**:
  1. *Automated snapshot restoration*: Passing `rollback_users_<timestamp>.csv` back into the update script.
  2. *Targeted audit recovery*: Using `update_audit_<timestamp>.csv` to isolate and revert only successful rows from partial runs.
  3. *Disaster recovery*: Re-applying the original baseline export from Export-EntraUsers.ps1.

Script and MD file updated

AutoSave Off | Document1 - Word | Internal* | Search

File Home Insert Draw Design Layout References Mailings Review View Help | Comments | Editing | Share

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Allow editing

https://teamclou... UsersV1.xlsx

I am preparing a change for CAB, please make a documented writeup of the process of the impact, risk and rollback for updtating and consolidating department, also document which departments are consolidated from a typing perspective and how many users are imacted https://teamcloudway.sharepoint.com/:t:/r/sites/CloudWay53/Shared%20Documents/PPT%20Materiale/CloudWayTalks/NICCONF2026/VibeFiles/entra_department_copilot_reporting_change_plan%20-%20from%20Simon.md?d=w00b56da8393544a7a1db1cc8c6f5bcd0&csf=1&web=1&e=WfVtoh UsersV1.xlsx

Send

Get me started writing

Create an outline for...

Page 1 of 1 | 0 words | English (United States) | Text Predictions: Off | Accessibility: Good to go | Focus | 100 %

ENG NO 17:01 14.10.2026

SENSITIVITY Due to content created by Copilot, your organization automatically applied the sensitivity label: Internal

4. Department consolidation and affected-user evidence

The source workbook contains the relevant fields **department**, **Replacement Department**, and **Standardized Department**. The CAB evidence must be produced by grouping rows where the current department differs from the approved target. Typing-only consolidation includes differences in capitalization, spacing, punctuation, abbreviations, and spelling that do not represent a genuine organizational transfer. Rows whose target represents a real move between organizational units must be reviewed separately and must not be approved as typing cleanup without the relevant data owner.

Required CAB metric	Definition	Status
Total source users	All data rows in UsersV1.xlsx, excluding the header.	Not verified from the available workbook extract.
Users with a department change	Rows where normalized current department is not equal to normalized approved target department.	Not verified from the available workbook extract.
Typing-only consolidations	Changed rows where the old and target values represent the same business department.	Not verified from the available workbook extract.
Actual organizational moves	Changed rows where the target represents a different business department.	Must be separately reviewed and approved.
Unassigned or	Rows with an empty, Unassigned, or	Must be excluded

Page: 2
4. Department consoli...

Reasoning completed in 4 steps

I created a **six-page CAB-ready change plan** covering:

- Change scope and business impact
- Controlled implementation process
- Dependency checks for dynamic groups, licensing, access, and workflows
- Detailed risk assessment and mitigations
- Step-by-step rollback procedure
- Validation and reconciliation criteria
- Go/no-go conditions
- Required department-mapping appendix

I also documented how to distinguish **typing-only department consolidation** from genuine organizational moves.

Important limitation: I could verify that the workbook contains the `department`, `Replacement Department`, and `Standardized Department` columns, but the available workbook extract did not expose the complete dataset for

Edit

Describe what you'd like to edit

AI-generated content may be incorrect

CAB Change Plan: Entra ID Department Standardization and Consolidation

Controlled update of department values to improve Microsoft 365 and Copilot reporting

1. Change summary

This change will standardize and consolidate the Microsoft Entra ID **department** attribute for an approved set of user accounts. The purpose is to replace inconsistent typing, spelling, capitalization, abbreviations, and duplicate department labels with an agreed controlled vocabulary. The resulting values will provide a more reliable organizational dimension for Microsoft 365 and Copilot usage reporting. The change does not alter user identities, passwords, licenses, manager relationships, job titles, group ownership, or account state.

2. Business justification

- Reduce fragmented departmental reporting caused by equivalent values entered differently.
- Improve grouping and filtering in Copilot Analytics and other Microsoft 365 reports

Reasoning completed in 4 steps

I created a **six-page CAB-ready change plan** covering:

- Change scope and business impact
- Controlled implementation process
- Dependency checks for dynamic groups, licensing, access, and workflows
- Detailed risk assessment and mitigations
- Step-by-step rollback procedure
- Validation and reconciliation criteria
- Go/no-go conditions
- Required department-mapping appendix

I also documented how to distinguish **typing-only department consolidation** from genuine organizational moves.

Important limitation: I could verify that the workbook contains the `department`, `Replacement Department`, and `Standardized Department` columns, but the available workbook extract did not expose the complete dataset for

Edit

the summary is a bit hard to read, can you make into bulletpoints

AI-generated content may be incorrect

CAB Change Plan: Entra ID Department Standardization and Consolidation

Controlled update of department values to improve Microsoft 365 and Copilot reporting

1. Change summary

- **Change:** Standardize and consolidate the Microsoft Entra ID **department** attribute for an approved set of user accounts.
- **Purpose:** Replace inconsistent typing, spelling, capitalization, abbreviations, and duplicate department labels with an agreed controlled vocabulary.
- **Expected outcome:** Provide a more reliable organizational dimension for Microsoft 365 and Copilot usage reporting.
- **Not affected:** User identities, passwords, licenses, manager relationships, job titles, group ownership, and account state.

2. Business justification

- Reduce fragmented departmental reporting caused by equivalent values entered differently.

the summary is a bit hard to read, can you make into bulletpoints

Copilot

Reasoning completed in 2 steps

I converted the selected **Change summary** into four concise bullet points:

- Change
- Purpose
- Expected outcome
- Items not affected

The meaning is unchanged, and the rest of the CAB document was left untouched.

Edit

Describe what you'd like to edit

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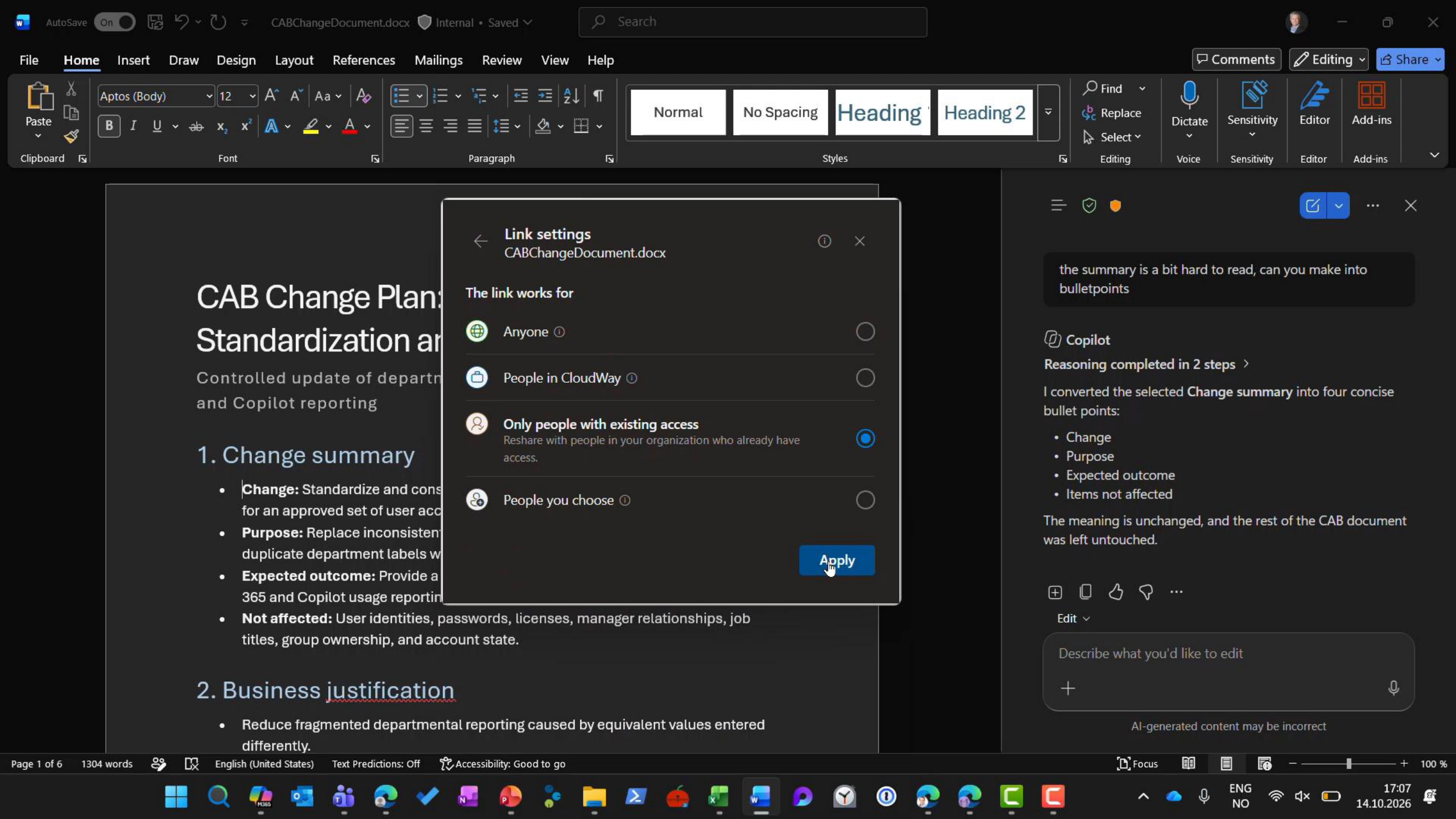
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1. Change summary
Change: Standardize and cons
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Purpose: Replace inconsisten
duplicate department labels w
Expected outcome: Provide a
365 and Copilot usage reportin
Not affected: User identities, passwords, licenses, manager relationships, job titles, group ownership, and account state.
2. Business justification
Reduce fragmented departmental reporting caused by equivalent values entered differently.
Link settings
CABChangeDocument.docx
The link works for
Anyone
People in CloudWay
Only people with existing access
Reshare with people in your organization who already have access.
People you choose
Apply
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Copilot
Reasoning completed in 2 steps
I converted the selected Change summary into four concise bullet points:
Change
Purpose
Expected outcome
Items not affected
The meaning is unchanged, and the rest of the CAB document was left untouched.
Edit
Describe what you'd like to edit
AI-generated content may be incorrect
Page 1 of 6
1304 words
English (United States)
Text Predictions: Off
Accessibility: Good to go
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Click to add subtitle

Click to add title

Name, Title
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https://cloudway.com

CloudWay

@CloudWay.com

Click to add notes

Auto

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Allow editing

CABChangeDocument.docx

Please create a three slide cab change presentation, 2 slides background and 1 slide detailed onepager
CABChangeDocument.docx

Send

Create a presentation about topic

Add a slide about description

Create a branded presentation

See more

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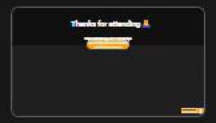

Click to add subtitle

Click to add title

Name, Title
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CloudWay

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Click to add notes

Auto

Reasoned in 3 steps

Should I keep the current CloudWay style for the three-slide CAB presentation, or use a new style?

- **Keep current style** Recommended
  Use the current deck's CloudWay theme for two background slides and one detailed CAB one-pager.
- **Use a new style**
  Choose a different visual style for the three-slide presentation.
- Enter another option

Confirm | Skip all

Edit

Describe what you'd like to create or edit

AI-generated responses may be incorrect

Slide 1 of 4 | Norwegian (Bokmål) | Accessibility: Investigate | Notes | 74 %

ENG NO | 17:09 14.10.2026

Entra ID department standardization

Background: consistent department data for Microsoft 365 and Copilot reporting

Why change	Spelling, capitalization, abbreviations and duplicate labels fragment departmental reporting.
Proposed change	Map each current department value to one approved target in a controlled vocabulary.
Expected outcome	More reliable grouping and filtering, with an auditable record of every consolidation.

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[Sources]
CABChangeDocument.docx, supplied by user, Sections 1–2: change summary and business justification

Auto

I'm using slide 3 as a style template, removing its logos, and replacing them with the requested title and body while checking content and quality requirements

I'm removing unrelated branding from the source material while preserving the master content and existing browser logo

Mapping it out

I'm clarifying requirements and considering unique labels for map elements to support more precise handling than the built-in options

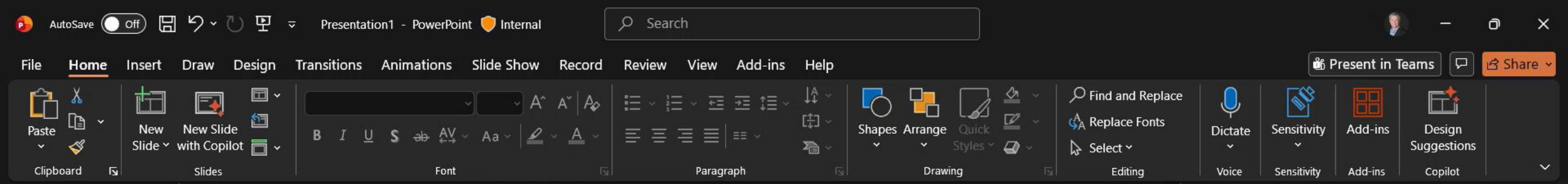
Thinking it through

Exploring approaches

Edit

Add a follow-up to queue

AI-generated responses may be incorrect



CAB change one-pager

Entra ID department consolidation / Approval evidence remains outstanding

Scope and evidence

Department only, approved accounts, no planned outage. User totals and typing-only counts remain unverified. Attach the reconciled current-to-target mapping, classification, counts and owner approvals.

Execution and safeguards

Approve vocabulary and mappings. Freeze input, export baseline, match object IDs, dry run, pilot, then update in logged batches. Exclude ambiguous accounts and blank, Unassigned or unapproved targets.

Risks and validation

Review department-dependent access and licensing. Confirm the authoritative source. Reconcile successes, failures, skips and unchanged rows to the dry run. Verify every successful write and downstream reporting after refresh.

Rollback

Stop on wrong mappings, access or licensing effects, excess failures or reconciliation mismatch. Restore exact prior values by object ID for successful writes only. Re-export and verify values plus dependency behavior.

GO only after reconciled counts, owner approval, dependency review, secured baseline, clean dry run and passed pilot. Otherwise NO-GO.

To confirm: change owner, execution window, failure threshold and validation window.

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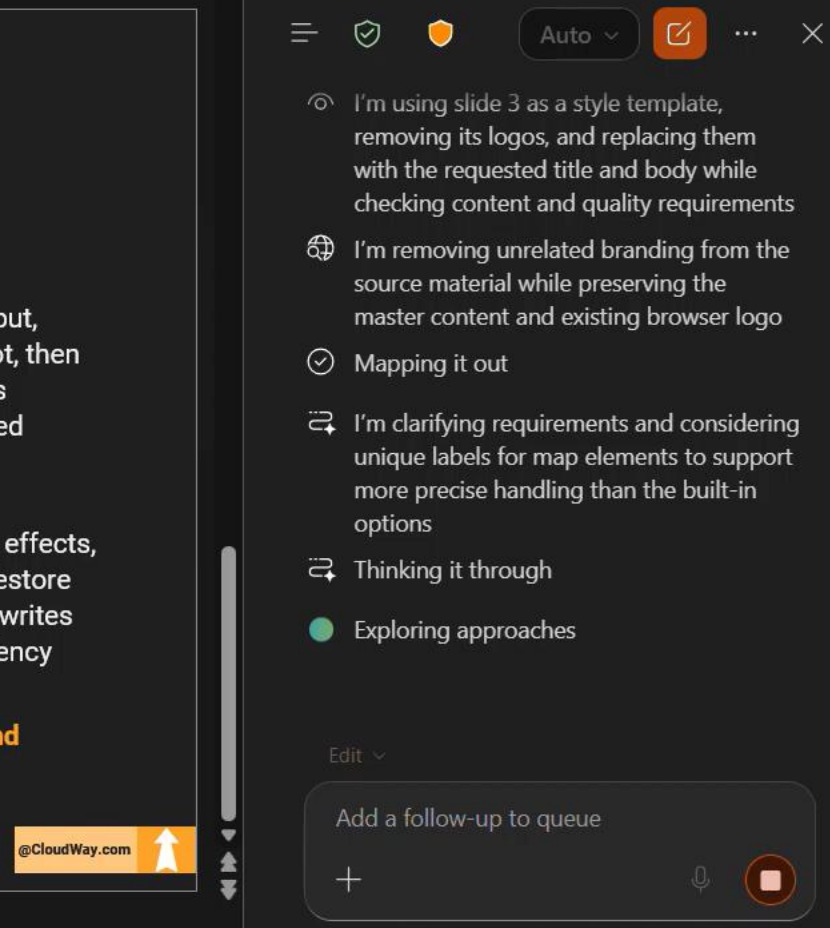
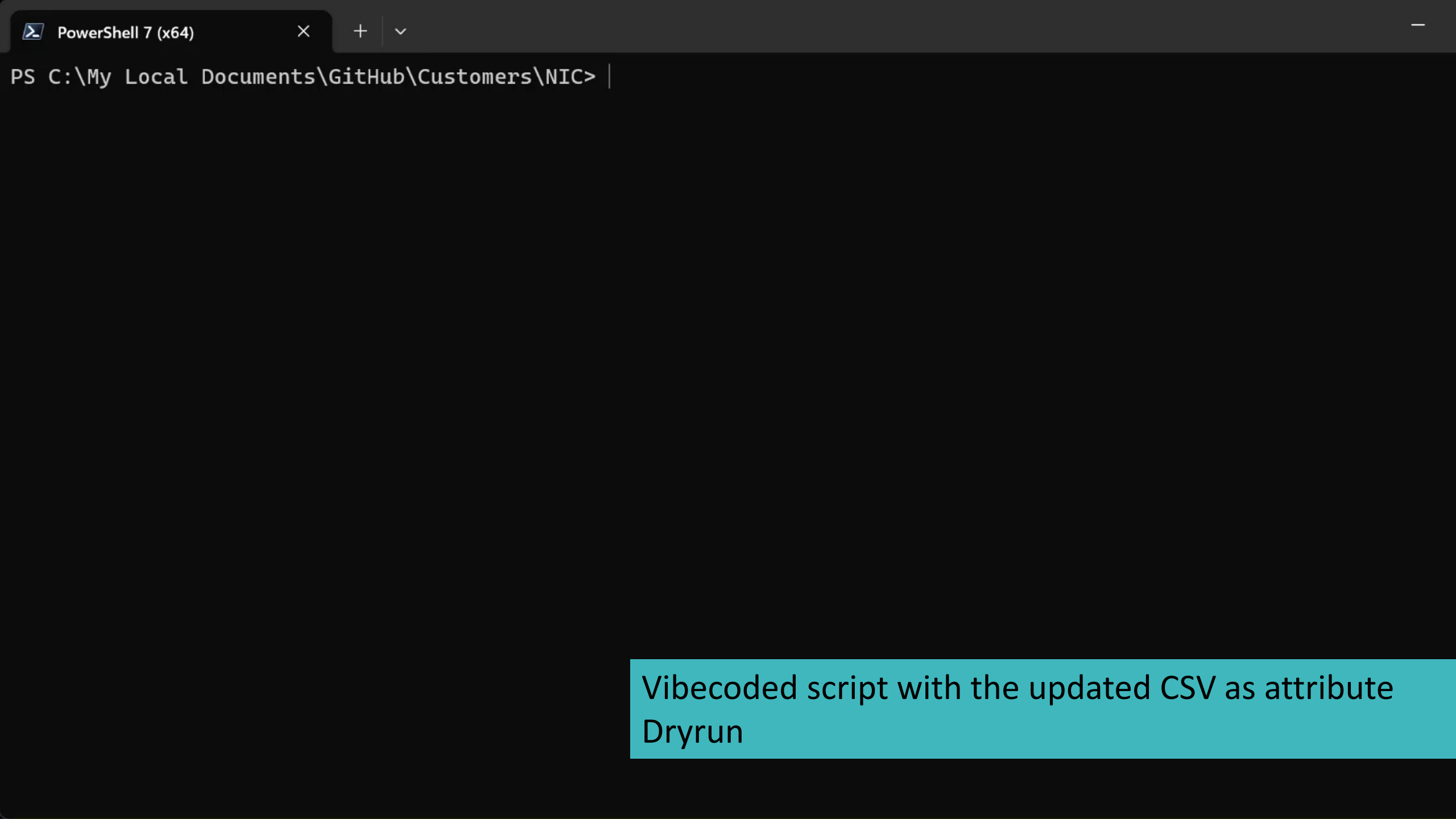
[Sources]
CABChangeDocument.docx, supplied by user, Sections 4–9: evidence, implementation, risks, rollback, success criteria and go/no-go. The source

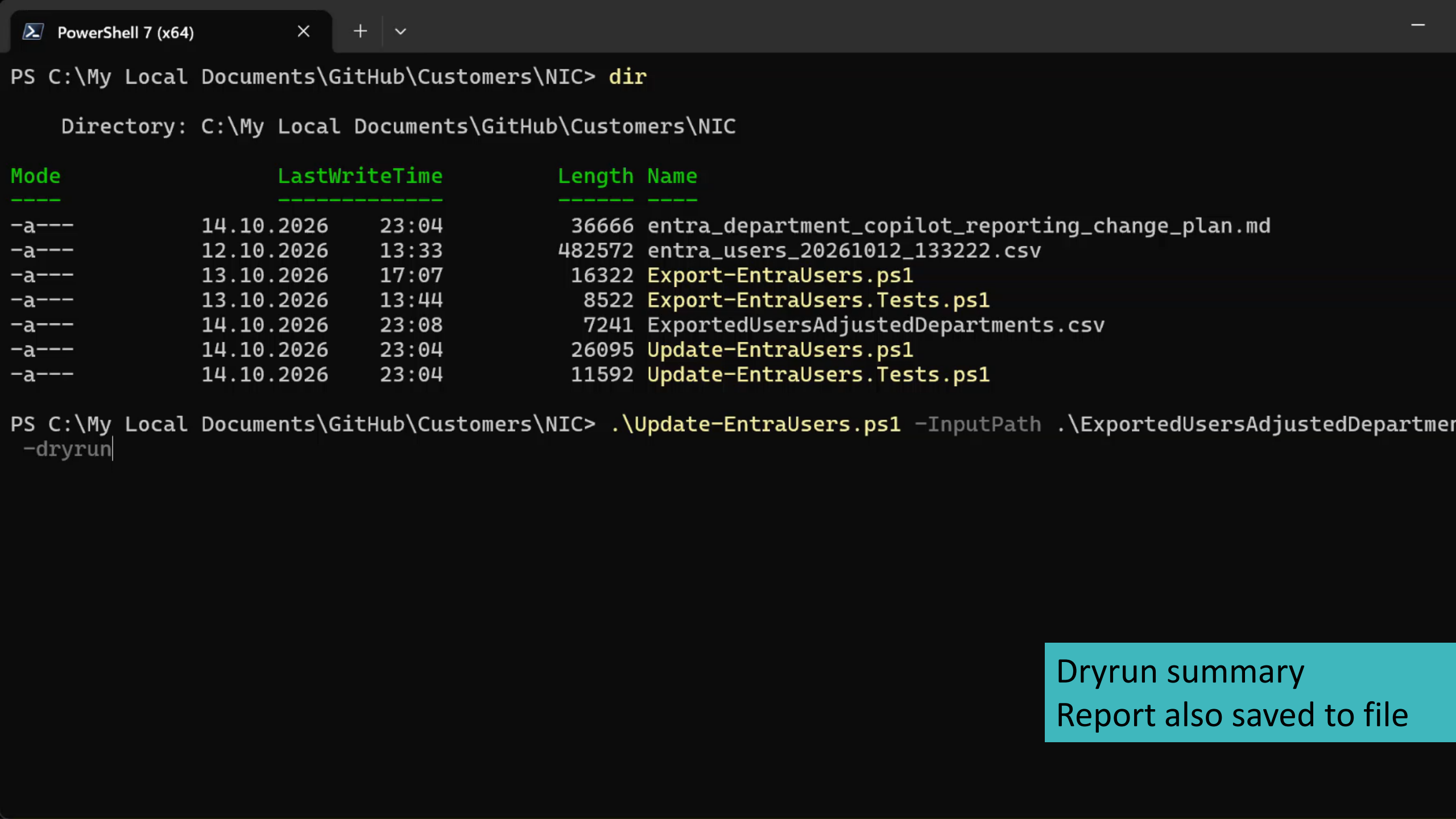


ENG NO 17:12 14.10.2026

Change approved
Go do!



Vibecoded script with the updated CSV as attribute Dryrun


PowerShell 7 (x64)
PS C:\My Local Documents\GitHub\Customers\NIC> dir
Directory: C:\My Local Documents\GitHub\Customers\NIC
Mode LastWriteTime Length Name
-a--- 14.10.2026 23:04 36666 entra_department_copilot_reporting_change_plan.md
-a--- 12.10.2026 13:33 482572 entra_users_20261012_133222.csv
-a--- 13.10.2026 17:07 16322 Export-EntraUsers.ps1
-a--- 13.10.2026 13:44 8522 Export-EntraUsers.Tests.ps1
-a--- 14.10.2026 23:08 7241 ExportedUsersAdjustedDepartments.csv
-a--- 14.10.2026 23:04 26095 Update-EntraUsers.ps1
-a--- 14.10.2026 23:04 11592 Update-EntraUsers.Tests.ps1
PS C:\My Local Documents\GitHub\Customers\NIC> .\Update-EntraUsers.ps1 -InputPath .\ExportedUsersAdjustedDepartmer
-dryrun
Dryrun summary
Report also saved to file

PowerShell 7 (x64)

```
s '')
[DIFF] User 'BiancaP@M365x49603954.OnMicrosoft.com' (2f6c752e-2df2-4289-b93a-3e2e55eb8935): department -> 'Sales'
Sale')
[DIFF] User 'BrianJ@M365x49603954.OnMicrosoft.com' (ea4eb98c-447b-4335-bfde-f036996aefc2): department -> 'Unassign
as '')
[DIFF] User 'Crystal@M365x49603954.OnMicrosoft.com' (7f49cb5b-0b2f-4da6-a6c4-54e8892b4f0e): department -> 'Unassig
was '')
[DIFF] User 'GerhartM@M365x49603954.OnMicrosoft.com' (85eea74e-173f-4337-a48e-fd727d53e512): department -> 'Unassi
(was '')
[DIFF] User 'Hood@M365x49603954.OnMicrosoft.com' (7e9ee33f-9fa0-4646-ae5b-181f965b15e7): department -> 'Unassigned
 '')
[DIFF] User 'JohannaL@M365x49603954.OnMicrosoft.com' (bc2226b9-ff60-46d7-b41d-ff961a18ca49): department -> 'Engine
 (was 'Engineers')
[DIFF] User 'MalloryC@M365x49603954.OnMicrosoft.com' (0f3bcd4b-f2a3-4dd3-a56f-eb04a8c2a259): department -> 'Unassi
(was '')
[DIFF] User 'MeganB@M365x49603954.OnMicrosoft.com' (d2aa7374-29ee-4d93-83dd-5123622311ad): department -> 'Marketin
s 'Markets')
[DIFF] User 'Rainier@M365x49603954.OnMicrosoft.com' (e4b7337e-2fb2-4abb-91d1-1c1ab48a8235): department -> 'Unassig
was '')
[DIFF] User 'Stevens@M365x49603954.OnMicrosoft.com' (77c6e59f-aba1-4b8e-ad44-039788987761): department -> 'Unassig
was '')
------------------------------------------------
Summary Analysis:
  Total users examined: 34
  Users requiring updates: 14
  Users unchanged: 20
------------------------------------------------
Dry run completed successfully. No changes were applied.
Dry run diff audit log saved to: C:\My Local Documents\GitHub\Customers\NIC\update_audit_20261014_230858.csv
PS C:\My Local Documents\GitHub\Customers\NIC>
```

Summary of upcoming changes
Verify before change is performed

PowerShell 7 (x64)

```
[DIFF] User 'BrianJ@M365x49603954.OnMicrosoft.com' (ea4eb98c-447b-4335-bfde-f036996aefc2): department -> 'Unassigr
as '')
[DIFF] User 'Crystal@M365x49603954.OnMicrosoft.com' (7f49cb5b-0b2f-4da6-a
was '')
[DIFF] User 'GerhartM@M365x49603954.OnMicrosoft.com' (85eea74e-173f-4337-
(was '')
[DIFF] User 'Hood@M365x49603954.OnMicrosoft.com' (7e9ee33f-9fa0-4646-ae5b
 '')
[DIFF] User 'JohannaL@M365x49603954.OnMicrosoft.com' (bc2226b9-ff60-46d7-b41d-ff961a18ca49): department -> 'Engine
 (was 'Engineers')
[DIFF] User 'MalloryC@M365x49603954.OnMicrosoft.com' (0f3bcd4b-f2a3-4dd3-a56f-eb04a8c2a259): department -> 'Unassi
(was '')
[DIFF] User 'MeganB@M365x49603954.OnMicrosoft.com' (d2aa7374-29ee-4d93-83dd-5123622311ad): department -> 'Marketir
s 'Markets')
[DIFF] User 'Rainier@M365x49603954.OnMicrosoft.com' (e4b7337e-2fb2-4abb-91d1-1c1ab48a8235): department -> 'Unassig
was '')
[DIFF] User 'Stevens@M365x49603954.OnMicrosoft.com' (77c6e59f-aba1-4b8e-ad44-039788987761): department -> 'Unassig
was '')
------------------------------------------------
Summary Analysis:
  Total users examined: 34
  Users requiring updates: 14
  Users unchanged: 20
------------------------------------------------

Confirm
Are you sure you want to perform this action?
Performing the operation "Update 14 user(s) with attributes from CSV" on target "Entra ID Tenant
'81524fbc-5983-4b32-a3b8-e563b2e2ee98'".
[Y] Yes  [A] Yes to All  [N] No  [L] No to All  [S] Suspend  [?] Help (default is "Y"):
```

Pre-change CSV for rollback saved
info on updated users
Audit log csv of changes saved

File Edit Selection View Go Run Terminal Help

Customers

Update

Explorer

Cust...

NIC

entra_departm...

entra_users_202... U

Export-EntraUs... M

Export-EntraUsers.T...

ExportedUsersA... U

rollback_users_... U

update_audit_2... U

update_audit_2... U

Update-EntraU...

Update-EntraU...

Outline

entra_department_copilot_reporting_change_plan.md

Update-EntraUsers.Tests.ps1

rollback_users_20261014_231226.csv U

NIC > rollback_users_20261014_231226.csv > data

```
1  "id";"userPrincipalName";"department";"companyName";"jobTitle";"officeLocation";"usageLocation";"employeeId";"employeeType";"employ
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3  "a0186f0a-14c7-47f4-b16f-45fd86cb092a";"admin@m365x49603954.onmicrosoft.com";"";"";"";"";"US";"";"";""
4  "3e19959c-2aba-4a04-9d11-8238812f3f34";"AutomateB@M365x49603954.OnMicrosoft.com";"";"";"";"";"US";"";"";""
5  "f605f47a-8736-4985-ac9c-7c1c1086b635";"Baker@M365x49603954.OnMicrosoft.com";"";"";"";"";"";"";"";""
6  "2f6c752e-2df2-4289-b93a-3e2e55eb8935";"BiancaP@M365x49603954.OnMicrosoft.com";"Sale";"";"Seller";"20/2046";"US";"";"";""
7  "ea4eb98c-447b-4335-bfde-f036996aefc2";"BrianJ@M365x49603954.OnMicrosoft.com";"";"";"";"";"";"";"";""
8  "7f49cb5b-0b2f-4da6-a6c4-54e8892b4f0e";"Crystal@M365x49603954.OnMicrosoft.com";"";"";"";"";"";"";"";""
9  "85eea74e-173f-4337-a48e-fd727d53e512";"GerhartM@M365x49603954.OnMicrosoft.com";"";"";"Manager";"20/2008";"US";"";"";""
10 "7e9ee33f-9fa0-4646-ae5b-181f965b15e7";"Hood@M365x49603954.OnMicrosoft.com";"";"";"";"";"";"";"";""
11 "bc2226b9-ff60-46d7-b41d-ff961a18ca49";"JohannaL@M365x49603954.OnMicrosoft.com";"Engineers";"";"Senior Engineer";"23/2102";"US";"";
12 "0f3bcd4b-f2a3-4dd3-a56f-eb04a8c2a259";"MalloryC@M365x49603954.OnMicrosoft.com";"";"";"";"20/2046";"US";"";"";""
13 "d2aa7374-29ee-4d93-83dd-5123622311ad";"MeganB@M365x49603954.OnMicrosoft.com";"Markets";"";"Marketing Manager";"12/1110";"US";"";""
14 "e4b7337e-2fb2-4abb-91d1-1c1ab48a8235";"Rainier@M365x49603954.OnMicrosoft.com";"";"";"";"";"";"";"";""
15 "77c6e59f-aba1-4b8e-ad44-039788987761";"Stevens@M365x49603954.OnMicrosoft.com";"";"";"";"";"";"";"";""
16
```
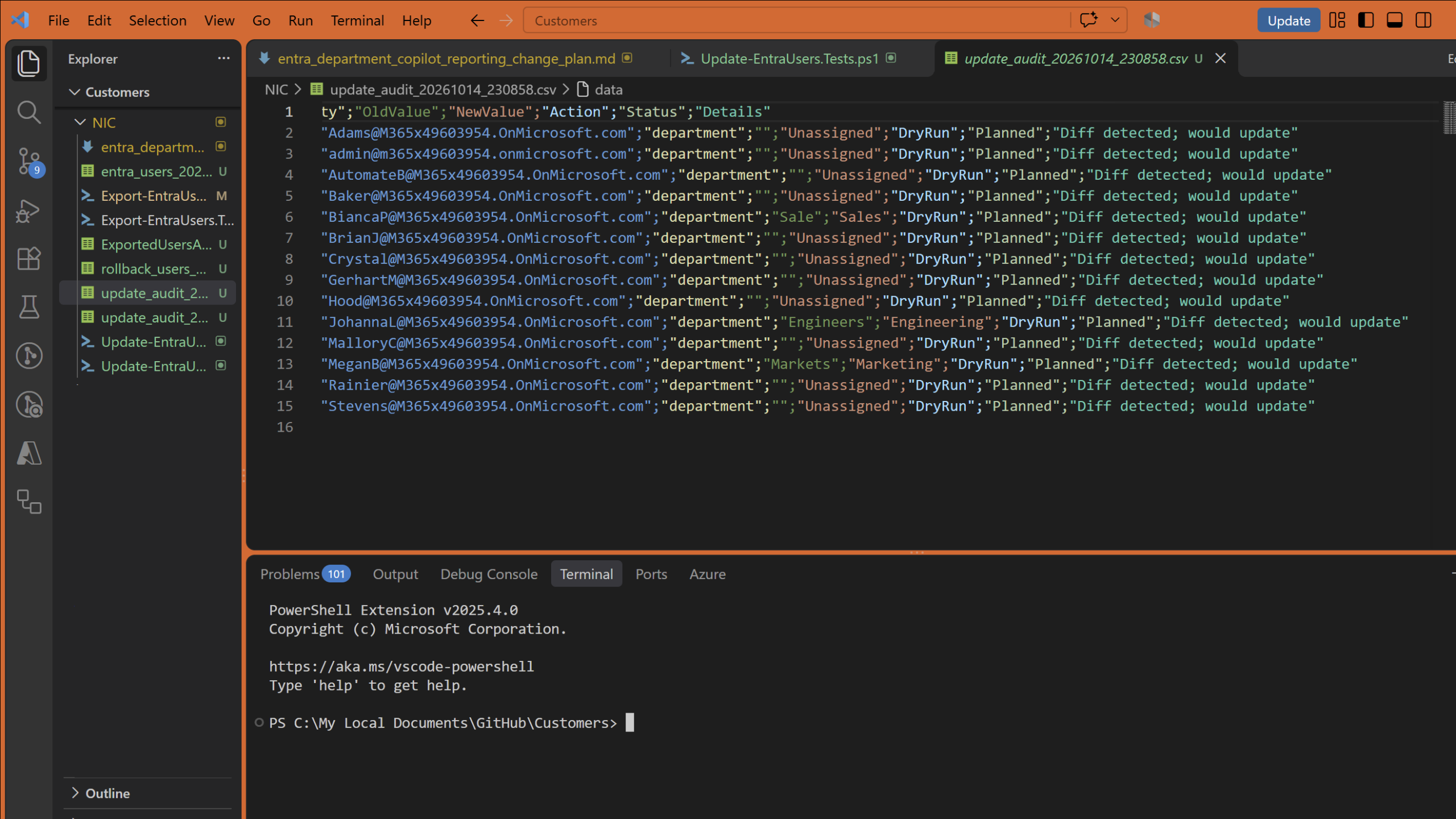

Problems 101 Output Debug Console Terminal Ports Azure

```
PowerShell Extension v2025.4.0
Copyright (c) Microsoft Corporation.

https://aka.ms/vscode-powershell
Type 'help' to get help.

PS C:\My Local Documents\GitHub\Customers>
```


File Edit Selection View Go Run Terminal Help
Customers
Update
Explorer
Customers
NIC
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entra_users_202... U
Export-EntraUs... M
Export-EntraUsers.T...
ExportedUsersA... U
rollback_users_... U
update_audit_2... U
update_audit_2... U
Update-EntraU...
Update-EntraU...
Outline
entra_department_copilot_reporting_change_plan.md
Update-EntraUsers.Tests.ps1
update_audit_20261014_230858.csv U
NIC > update_audit_20261014_230858.csv > data
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"Adams@M365x49603954.OnMicrosoft.com";"department";"";"Unassigned";"DryRun";"Planned";"Diff detected; would update"
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"Baker@M365x49603954.OnMicrosoft.com";"department";"";"Unassigned";"DryRun";"Planned";"Diff detected; would update"
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"BrianJ@M365x49603954.OnMicrosoft.com";"department";"";"Unassigned";"DryRun";"Planned";"Diff detected; would update"
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"Hood@M365x49603954.OnMicrosoft.com";"department";"";"Unassigned";"DryRun";"Planned";"Diff detected; would update"
"JohannaL@M365x49603954.OnMicrosoft.com";"department";"Engineers";"Engineering";"DryRun";"Planned";"Diff detected; would update"
"MalloryC@M365x49603954.OnMicrosoft.com";"department";"";"Unassigned";"DryRun";"Planned";"Diff detected; would update"
"MeganB@M365x49603954.OnMicrosoft.com";"department";"Markets";"Marketing";"DryRun";"Planned";"Diff detected; would update"
"Rainier@M365x49603954.OnMicrosoft.com";"department";"";"Unassigned";"DryRun";"Planned";"Diff detected; would update"
"Stevens@M365x49603954.OnMicrosoft.com";"department";"";"Unassigned";"DryRun";"Planned";"Diff detected; would update"
Problems 101
Output
Debug Console
Terminal
Ports
Azure
PowerShell Extension v2025.4.0
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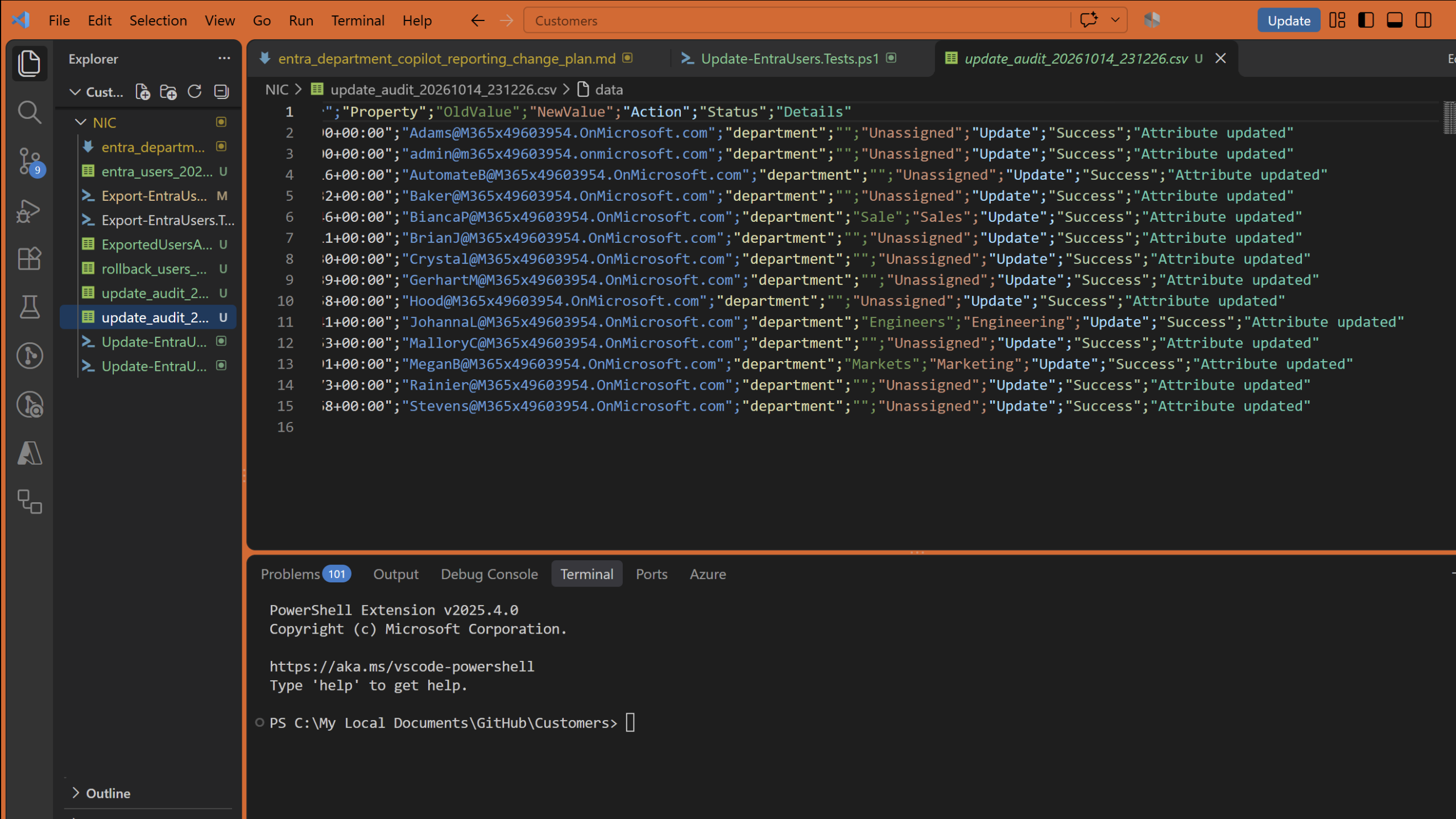
Type 'help' to get help.
PS C:\My Local Documents\GitHub\Customers>


File Edit Selection View Go Run Terminal Help
Customers
Update
Explorer
Cust...
NIC
entra_departm...
entra_users_202... U
Export-EntraUs... M
Export-EntraUsers.T...
ExportedUsersA... U
rollback_users_... U
update_audit_2... U
update_audit_2... U
Update-EntraU...
Update-EntraU...
Outline
entra_department_copilot_reporting_change_plan.md
Update-EntraUsers.Tests.ps1
update_audit_20261014_231226.csv U
NIC > update_audit_20261014_231226.csv > data
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2 0+00:00";"Adams@M365x49603954.OnMicrosoft.com";"department";"";"Unassigned";"Update";"Success";"Attribute updated"
3 0+00:00";"admin@m365x49603954.onmicrosoft.com";"department";"";"Unassigned";"Update";"Success";"Attribute updated"
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5 2+00:00";"Baker@M365x49603954.OnMicrosoft.com";"department";"";"Unassigned";"Update";"Success";"Attribute updated"
6 6+00:00";"BiancaP@M365x49603954.OnMicrosoft.com";"department";"Sale";"Sales";"Update";"Success";"Attribute updated"
7 .1+00:00";"BrianJ@M365x49603954.OnMicrosoft.com";"department";"";"Unassigned";"Update";"Success";"Attribute updated"
8 0+00:00";"Crystal@M365x49603954.OnMicrosoft.com";"department";"";"Unassigned";"Update";"Success";"Attribute updated"
9 9+00:00";"GerhartM@M365x49603954.OnMicrosoft.com";"department";"";"Unassigned";"Update";"Success";"Attribute updated"
10 8+00:00";"Hood@M365x49603954.OnMicrosoft.com";"department";"";"Unassigned";"Update";"Success";"Attribute updated"
11 1+00:00";"JohannaL@M365x49603954.OnMicrosoft.com";"department";"Engineers";"Engineering";"Update";"Success";"Attribute updated"
12 3+00:00";"MalloryC@M365x49603954.OnMicrosoft.com";"department";"";"Unassigned";"Update";"Success";"Attribute updated"
13 1+00:00";"MeganB@M365x49603954.OnMicrosoft.com";"department";"Markets";"Marketing";"Update";"Success";"Attribute updated"
14 3+00:00";"Rainier@M365x49603954.OnMicrosoft.com";"department";"";"Unassigned";"Update";"Success";"Attribute updated"
15 8+00:00";"Stevens@M365x49603954.OnMicrosoft.com";"department";"";"Unassigned";"Update";"Success";"Attribute updated"
16
Problems 101
Output
Debug Console
Terminal
Ports
Azure
PowerShell Extension v2025.4.0
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Type 'help' to get help.
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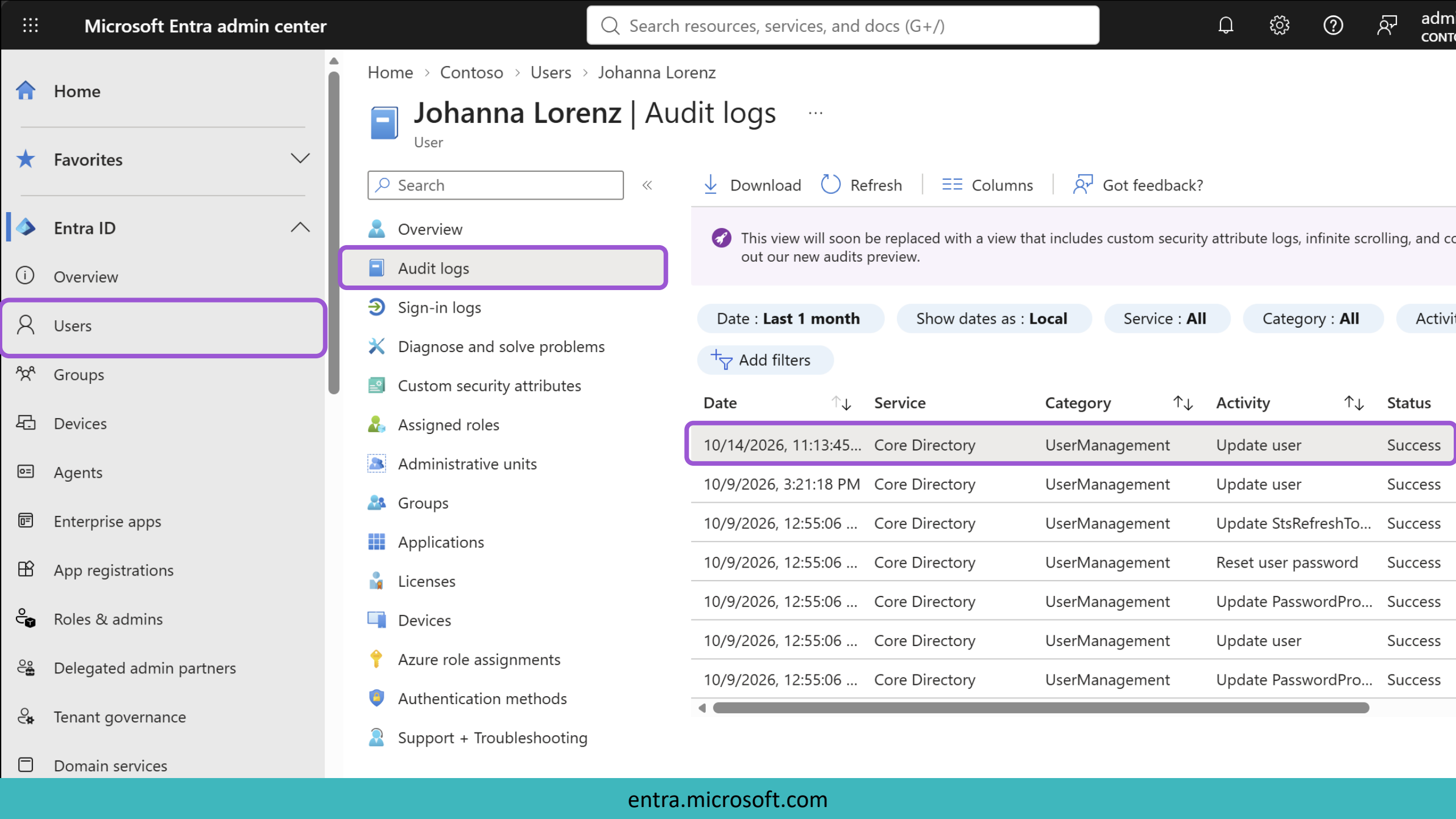

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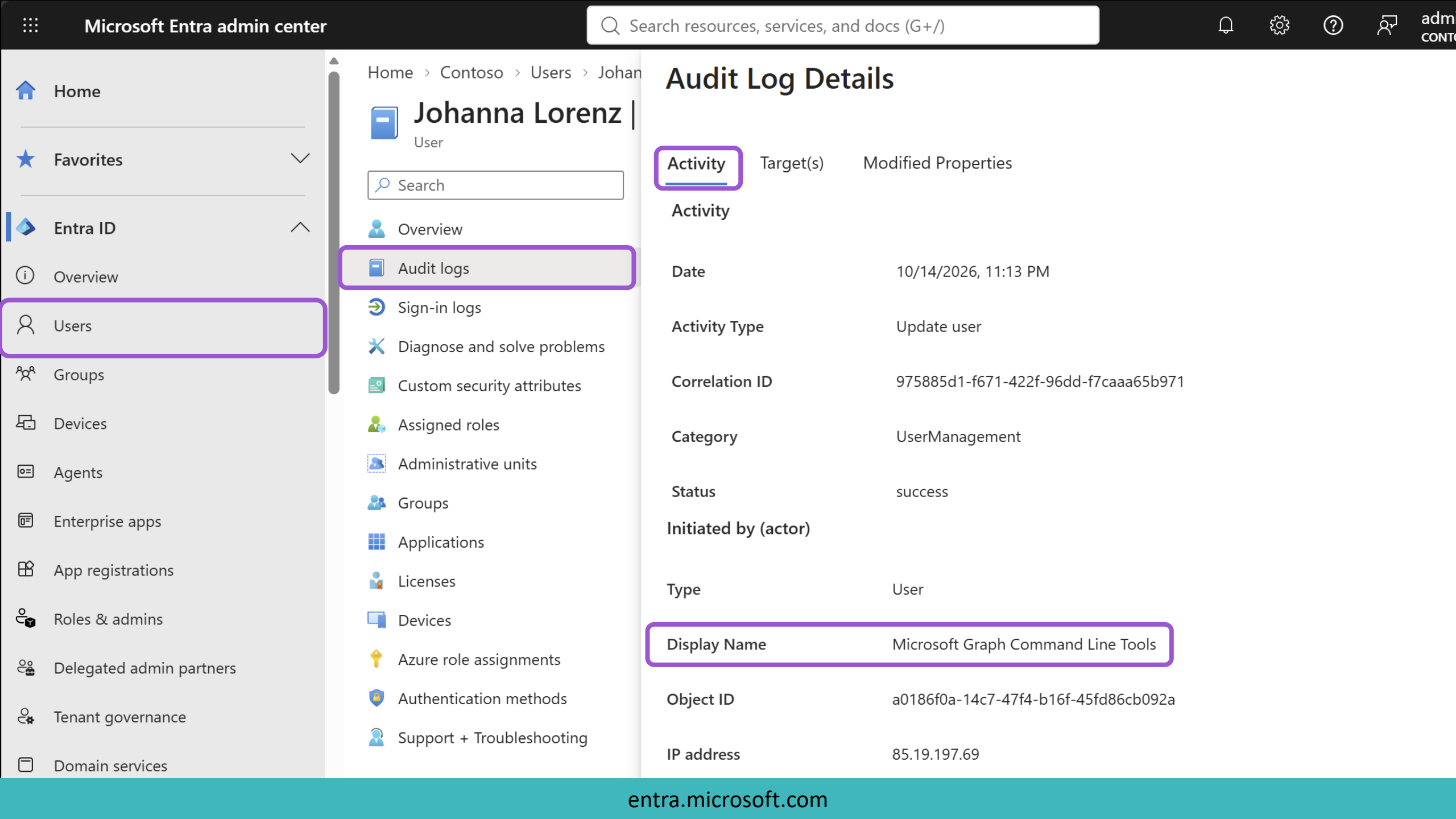

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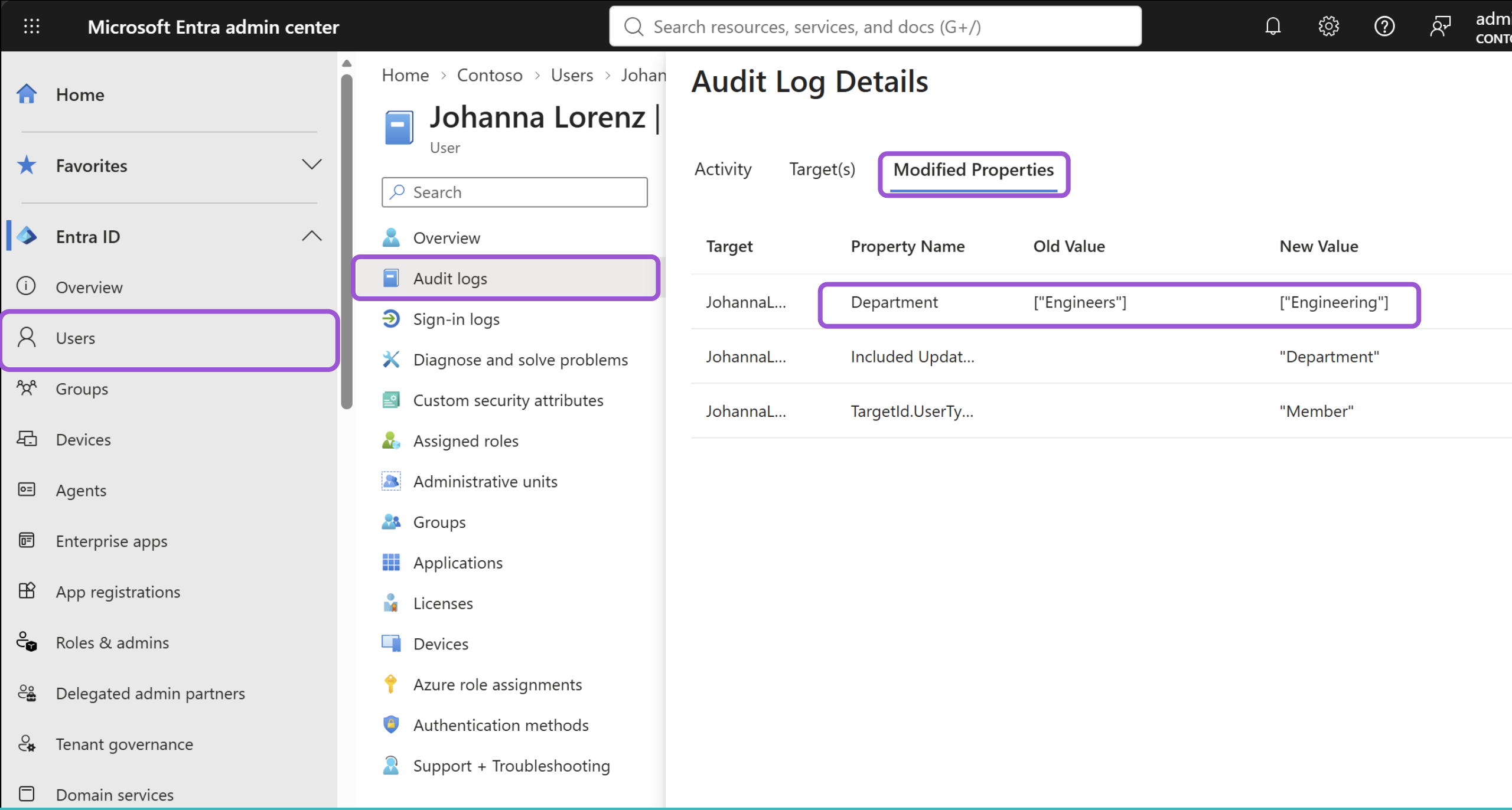

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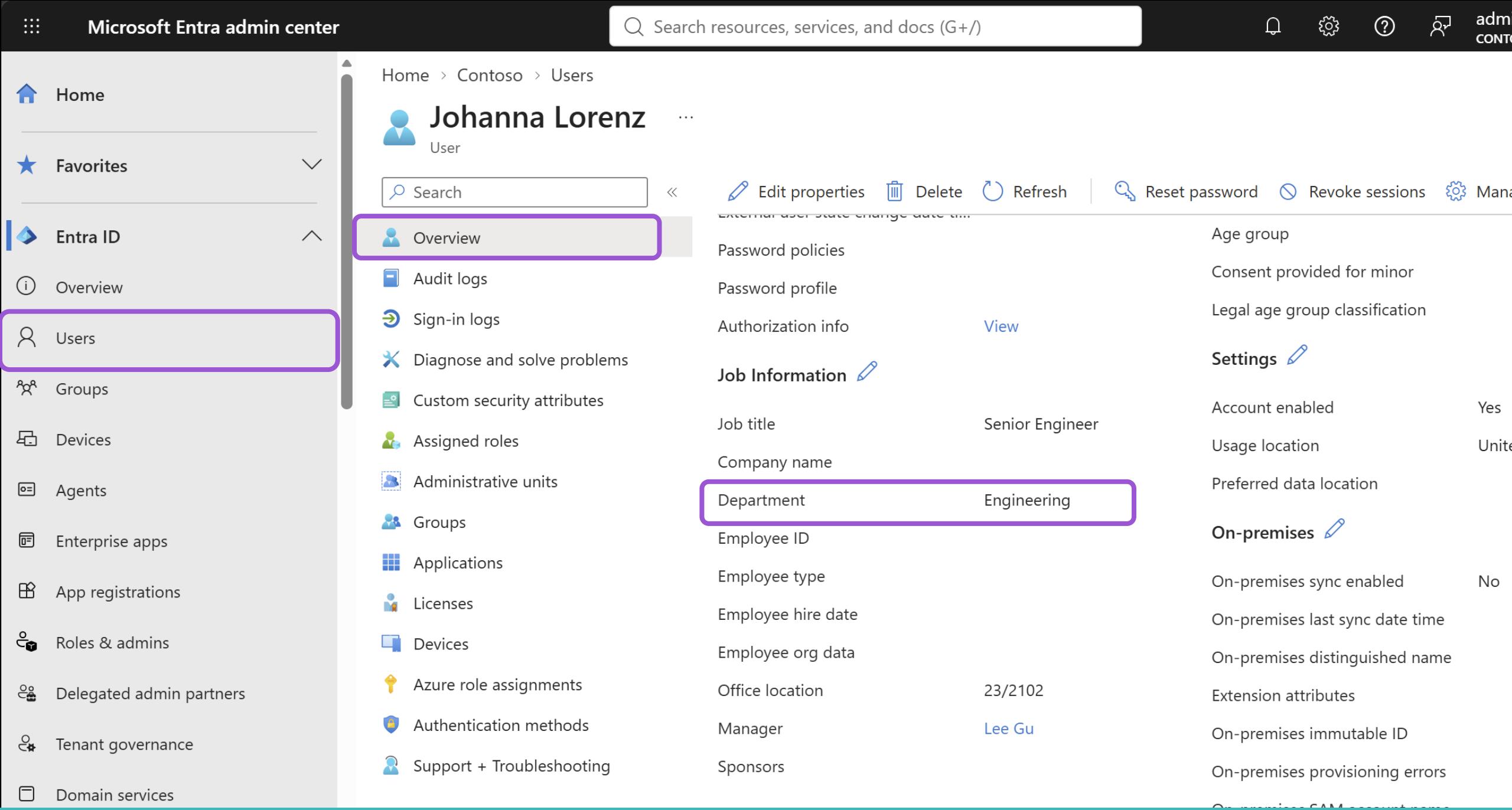
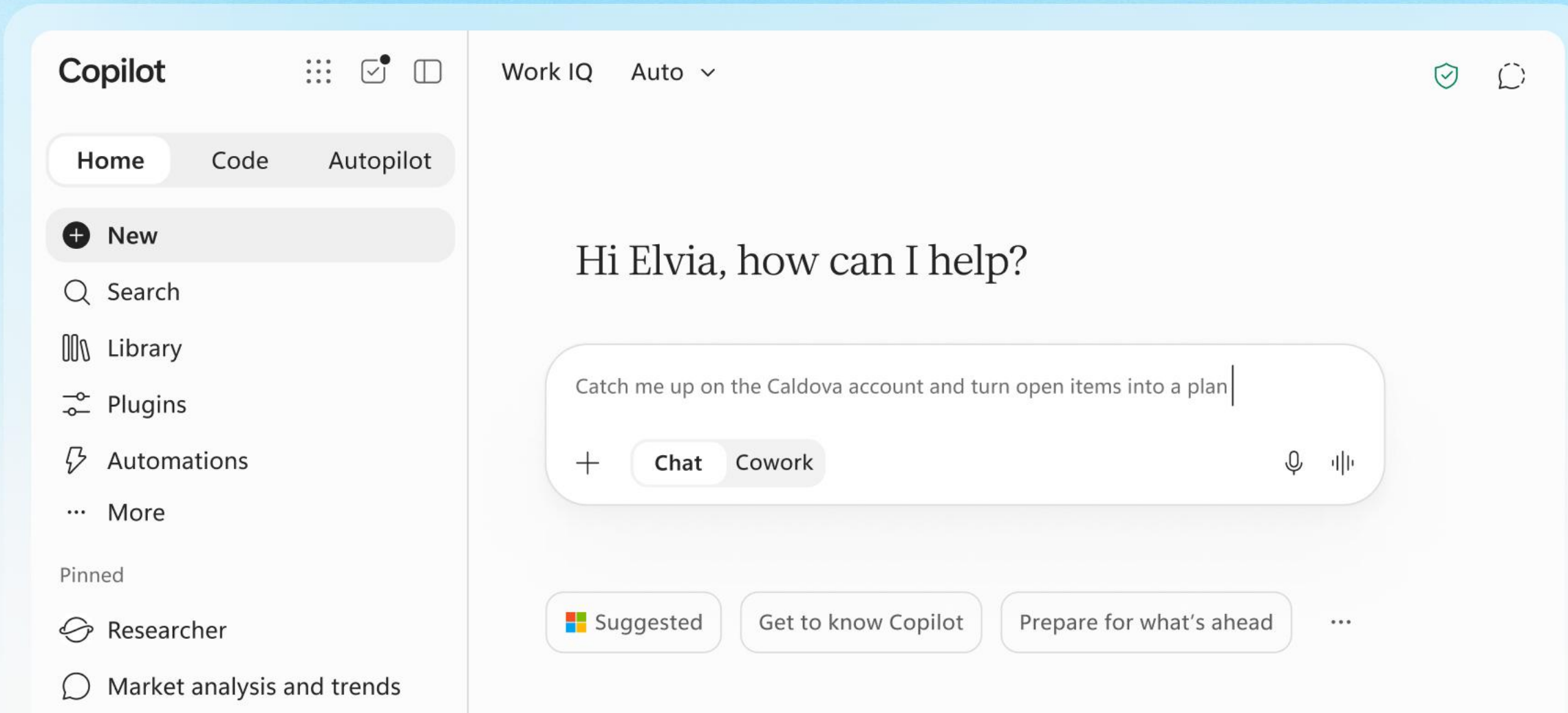
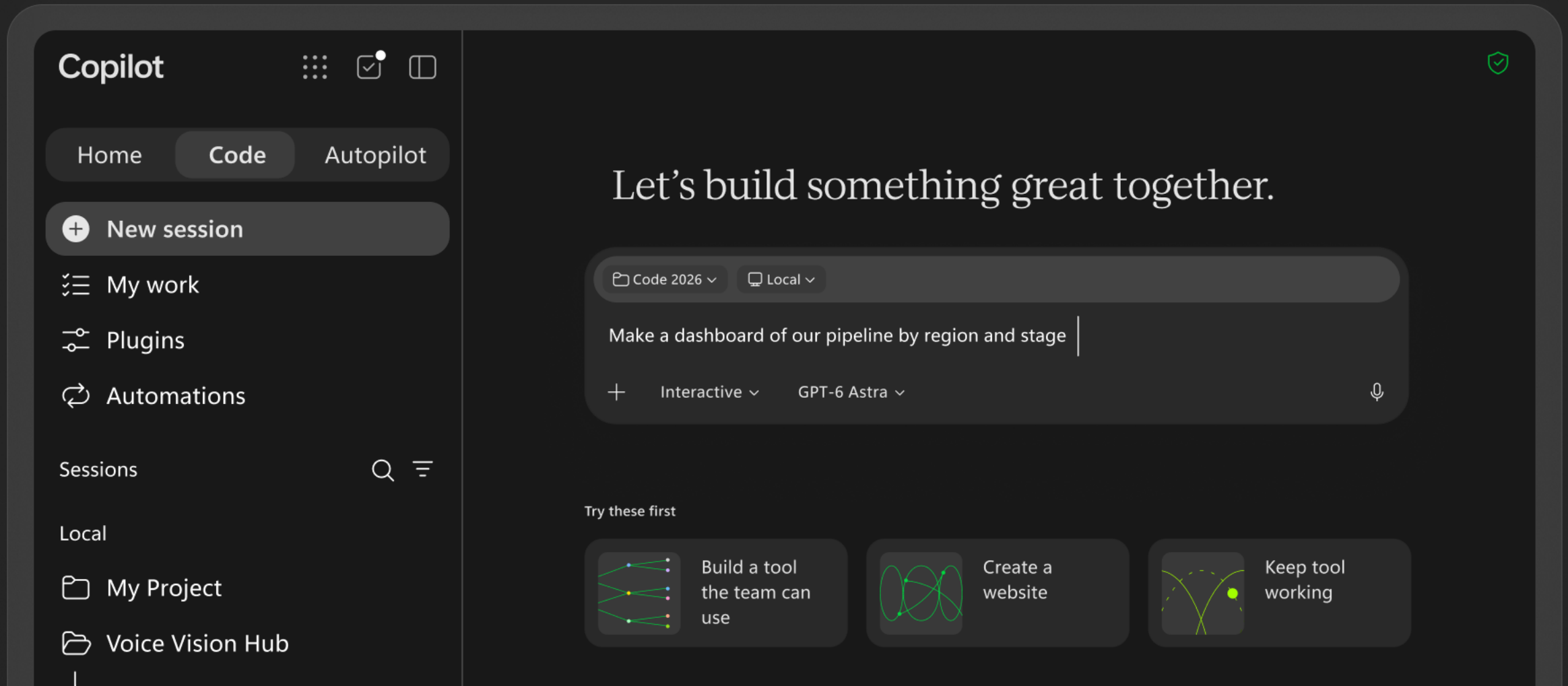

Microsoft Entra admin center
Search resources, services, and docs (G+/)
Home
Favorites
Entra ID
Overview
Users
Groups
Devices
Agents
Enterprise apps
App registrations
Roles & admins
Delegated admin partners
Tenant governance
Domain services
Home > Contoso > Users
Johanna Lorenz
User
Search
Overview
Audit logs
Sign-in logs
Diagnose and solve problems
Custom security attributes
Assigned roles
Administrative units
Groups
Applications
Licenses
Devices
Azure role assignments
Authentication methods
Support + Troubleshooting
Edit properties
Delete
Refresh
Reset password
Revoke sessions
Password policies
Password profile
Authorization info
View
Job Information
Job title
Senior Engineer
Company name
Department
Engineering
Employee ID
Employee type
Employee hire date
Employee org data
Office location
23/2102
Manager
Lee Gu
Sponsors
Age group
Consent provided for minor
Legal age group classification
Settings
Account enabled
Yes
Usage location
Preferred data location
On-premises
On-premises sync enabled
No
On-premises last sync date time
On-premises distinguished name
Extension attributes
On-premises immutable ID
On-premises provisioning errors

Change is rolling out now

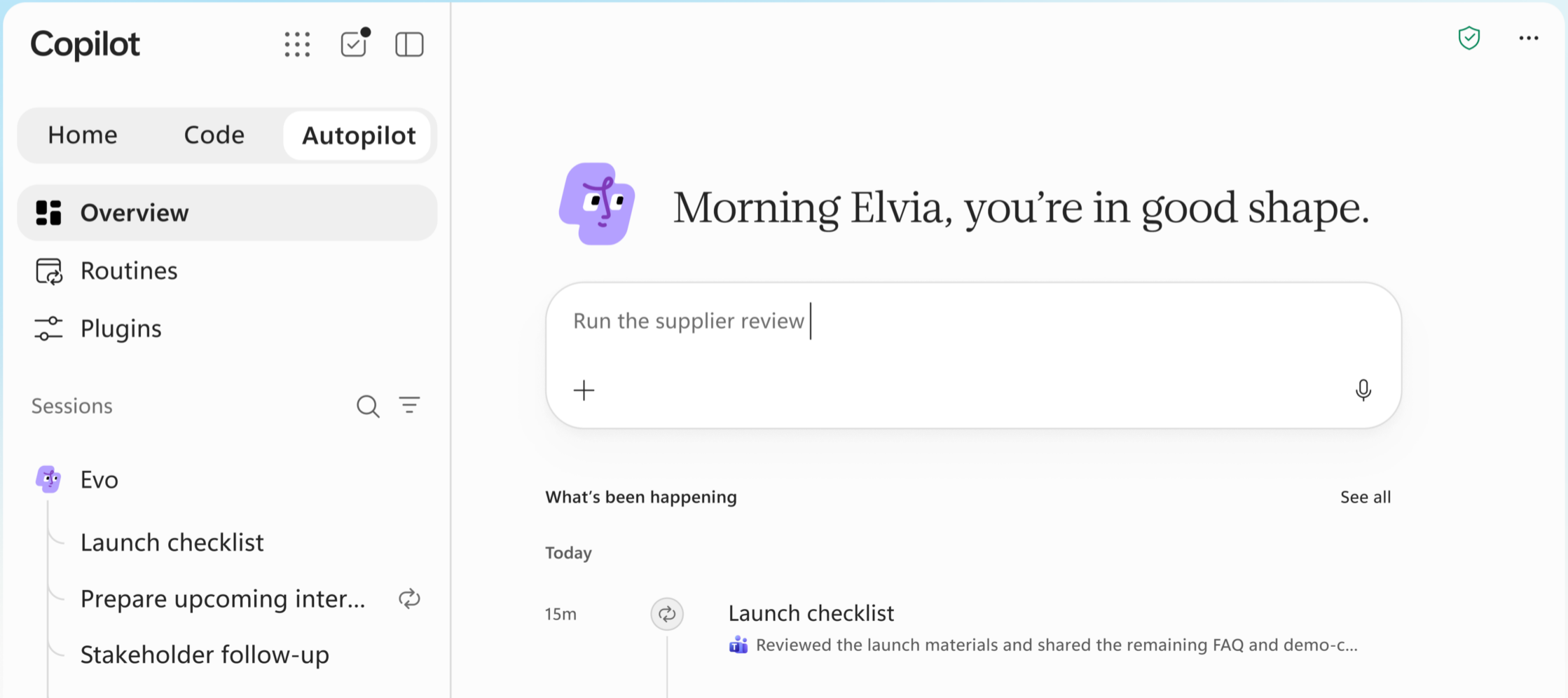
Home


Copilot
Work IQ
Auto
Home
Code
Autopilot
New
Search
Library
Plugins
Automations
More
Pinned
Researcher
Market analysis and trends
Hi Elvia, how can I help?
Catch me up on the Caldova account and turn open items into a plan
Chat
Cowork
Suggested
Get to know Copilot
Prepare for what's ahead

</> Code


Copilot
Home
Code
Autopilot
New session
My work
Plugins
Automations
Sessions
Local
My Project
Voice Vision Hub
Let’s build something great together.
Code 2026
Local
Make a dashboard of our pipeline by region and stage
Interactive
GPT-6 Astra
Try these first
Build a tool the team can use
Create a website
Keep tool working

Autopilot


Copilot
Home
Code
Autopilot
Overview
Routines
Plugins
Sessions
Evo
Launch checklist
Prepare upcoming inter...
Stakeholder follow-up
Morning Elvia, you're in good shape.
Run the supplier review
What's been happening
See all
Today
15m
Launch checklist
Reviewed the launch materials and shared the remaining FAQ and demo-c...

Food for thought

Ståle Hansen

CEO, Principal Cloud Architect
Copilot MVP & RD

Simon Skotheimsvik

Senior Cloud Architect
Intune MVP

CloudWay

NIC26

Thanks for your attention!

Ståle Hansen

CEO, Principal Cloud Architect
Copilot MVP & RD

Simon Skotheimsvik

Senior Cloud Architect
Intune MVP

CloudWay

NIC26

Work Ståle
M365 Copilot (Premium)

proving that Ombre was established in England before the arrival of Queen Catherine of Portugal. Furthermore, when discussing pack sizes, Chatto identifies Ombre as a **40-card game**, distinguishing it from games played with the full 52-card pack.

Modern Corrective

Modern scholarship largely confirms Chatto's linguistic argument. The World of Playing Cards traces the true historical breakthrough not to Ombre but to **fifteenth-century Italy**, where the game of **Trionfi** (later Tarot or Tarocchi) introduced a permanent series of trump cards into an otherwise ordinary playing-card pack. Historians now generally regard this as the earliest documented trump mechanism in Europe. Ombre therefore was **not one of the first games to use trumps**, nor was it originally a 52-card game. Instead, it represents a later and highly influential seventeenth-century development in which a selected trump suit played a central role. The historical sequence now appears to be: Italian **Trionfi** and tarot trumps in the fifteenth century, followed by games known as **Triomphe**, **Ruff**, **Trump**, and **Triumph**, with Ombre emerging later as one of the most successful trick-taking games built upon the established trump principle. wopc +1

I would actually consider this a particularly good "Chatto paragraph" because it shows both what he **did know** and what he **did not know**, which is very characteristic of his approach. He correctly identified the linguistic connection between Trump, Triomphe, and Triunfo, but lacked the later documentary evidence connecting the origin of trump cards to fifteenth-century Italian Trionfi and tarot.

Sources

Verify the dates and sources in this section

Message Copilot

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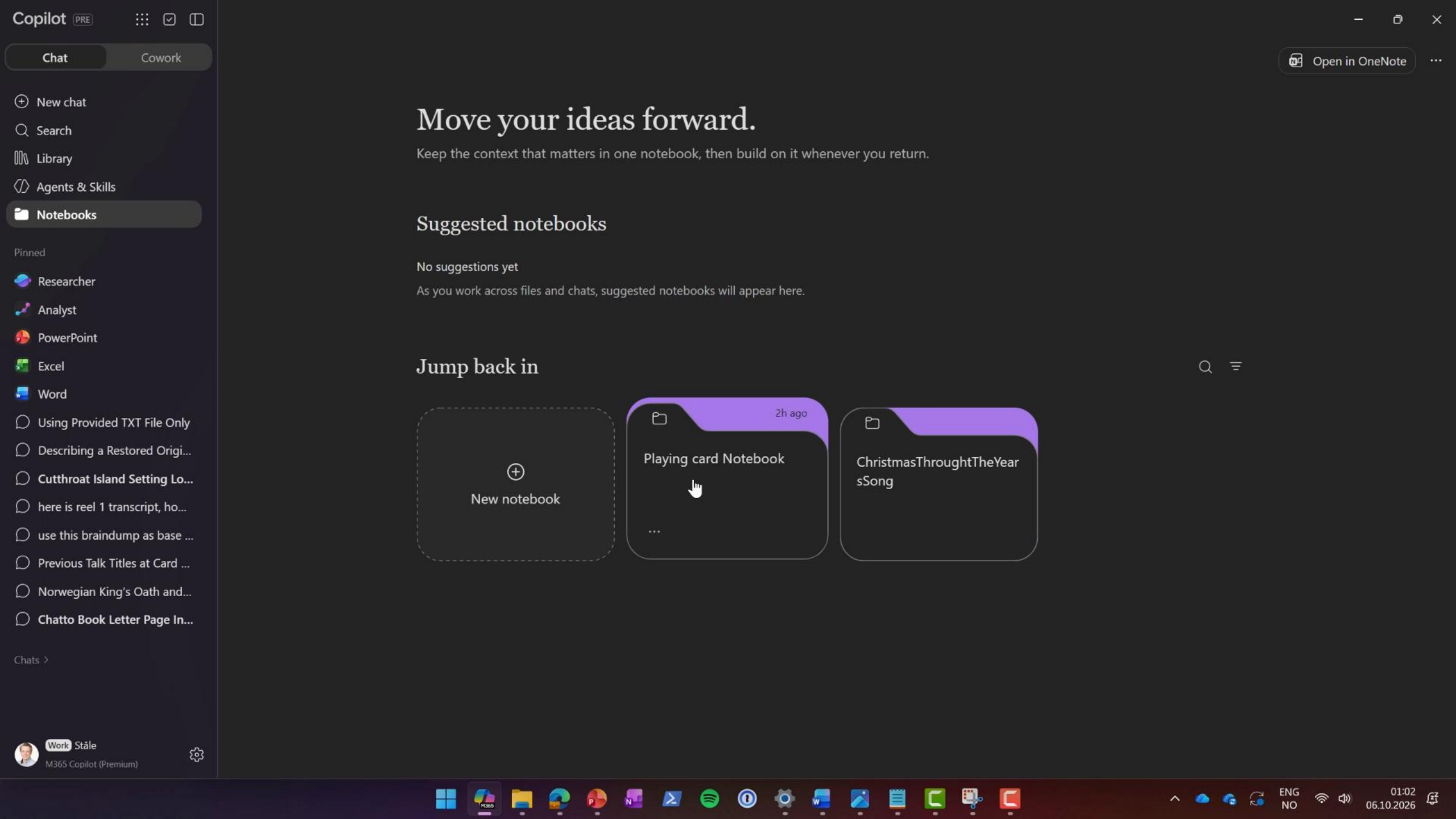
Sources

Citations

DiscussionOfTheBookHistoryOfPlayingCards1848
Ståle Hansen modified on 1/26/2026

Perspectives on the History of Tarot
wopc.co.uk

History of Tarot and Playing Cards: Timeline & Ori...
wopc.co.uk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Move your ideas forward.
Keep the context that matters in one notebook, then build on it whenever you return.
Suggested notebooks
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As you work across files and chats, suggested notebooks will appear here.
Jump back in
New notebook
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Playing card Notebook

Auto

1

Open in OneNote

Move chats

Send feedback

Let's refine *Playing card Notebook*

Message Copilot

Add references to Notebook

Surfaces references worth pulling into this notebook.

Create a progress tracker

Turns your open work into a tracker you can watch.

Generate a status deck

Shapes where things stand into a deck to share.

Chat history

Content

New blank

Report

Audio overview

Study guide

In Playing card Notebook

Notebook Insights

From Zero to IPCS - Playing Card Jour...

MalmoTalkResearch

Facts and Speculations

Chatto and how he did research

August 2026 Amerikaner History Discu...

August 2026 Learning Hombre from th...

Playing Cards

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Let's refine *Playing card Notebook*

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Surfaces references worth pulling into this notebook.

Create a progress tracker
Turns your open work into a tracker you can watch.

Generate a status deck
Shapes where things stand into a deck to share.

Chat history

Maker Space

New page
Presentation
Document
Workbook
Audio overview
Frontier
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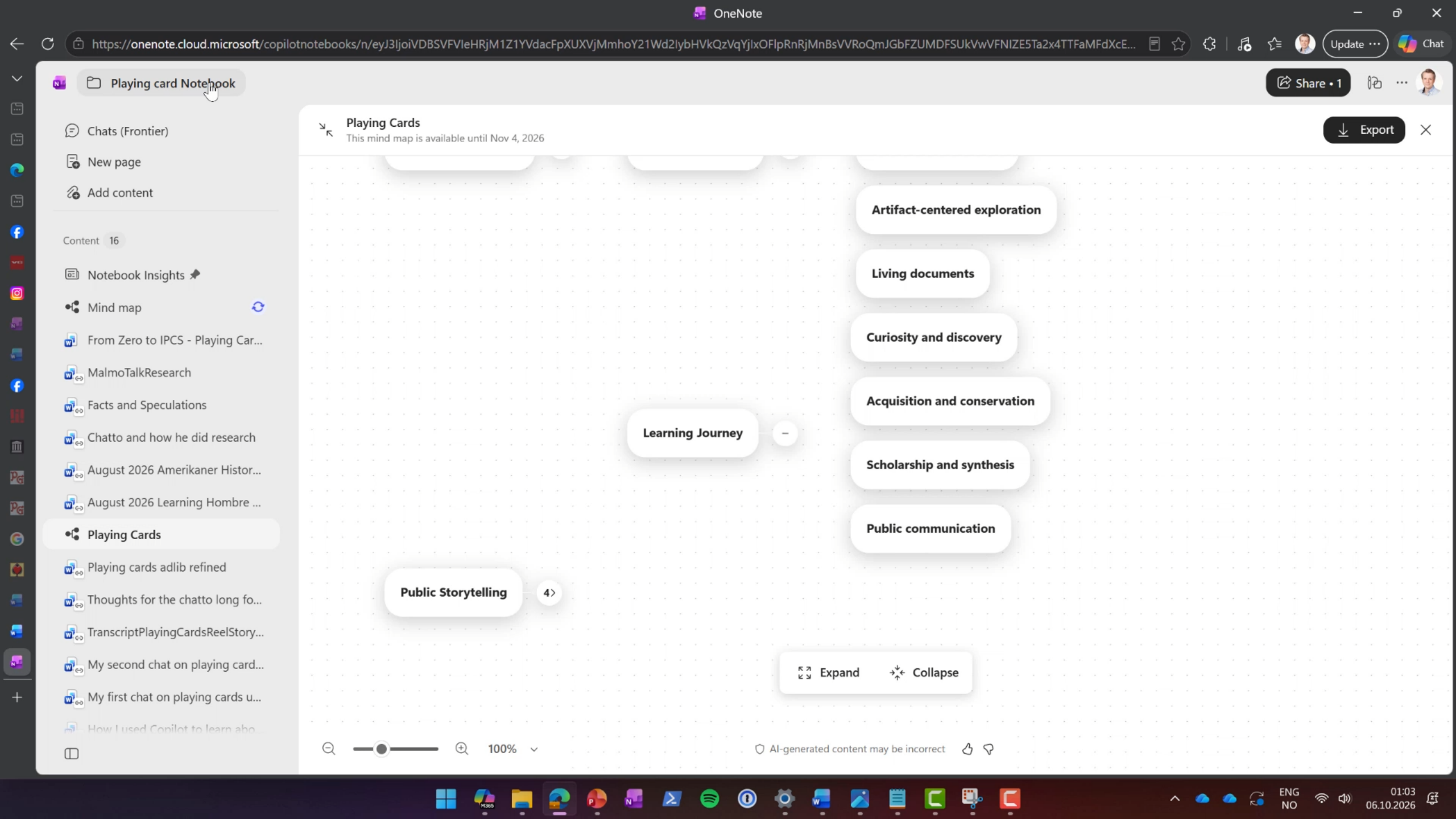
Explore your notebook as a mind map.
Study guide
Infographic

Suggested Spreadsheet
Playing-Card Evidence Matrix
Provide a traceable fact-checking system for the talk and reel series.
3 references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Living documents
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Acquisition and conservation
Learning Journey
Scholarship and synthesis
Public communication
Public Storytelling
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How I used Copilot to learn abo...

From Zero to IPCS - Playing Card Journey with Copilot

Created and added by **Ståle Hansen**

Auto

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From Zero to IPCS - Pl...

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systems, tarot, moral commentary and illustrations.

Evolution: One broad prompt had now produced a named author, a dated primary text and a digital book I could interrogate. Copilot had moved from recommendation engine to orientation guide.

January–spring 2026 — the questions became more specific

The early conversations show a sequence of small questions rather than one large research request. I asked about the English tax on the Ace of Spades, who Chatto was, the traditional identities of the kings, the French Queen, the name "Ace," the alleged hanging of a tax-stamp forger, and what happened to playing cards after 1848.

Examples of the prompt progression

- "Who is Chatto?"
- "Extract the relevant passage."
- "Is there a book that has a more complete history?"
- "Can you give me the titles in a bullet list?"
- "With publication year."
- "Is the 1848 book a good book? Is it renowned?"
- "What did he get wrong?"

These prompts show the first major evolution. I was no longer satisfied with being told that a book existed. I wanted a corpus, dates, comparative value, limitations and extracts. The questions moved from identification to evaluation.

The physical book changed the project

I found a damaged 1848 first edition, evaluated its completeness with Copilot, discussed restoration, and commissioned a custom leather binding. The cover itself became part of the

Auto

Let's refine Playing card Notebook

From Zero to IPCS - Pl...

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Notebook

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OneNote

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Notebook Insights

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Auto

Today

From Zero to IP...

give me a one line summary of this document

Work completed

The document traces how I evolved during 2026 from using Copilot to find a playing-card book into a source-critical researcher who challenged claims, learned through experience, created evidence-led stories, and developed an original interpretation for IPCS.

From Zero to IPC...

Notebook
Copilot will only use content from this notebook

Notebooks + Work data
Copilot will use this notebook and Work IQ

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Notebook

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Playing-card journey with Copilot Page 2

Auto

Today

evaluating Chatto's 1848 work, investigating specific cards and symbols, and demanding exact passages and evidence. From Zero to IPC...

- **From answers to source-controlled learning:** I learned to challenge plausible claims, define which primary sources Copilot could use, build and revise a living Word knowledge base, and test historical instructions by reconstructing and playing Hombre. From Zero to IPC...
- **From learning to contributing:** I turned the research into evidence-led reels, using public storytelling as a test of accuracy, and ultimately developed the "networked antiquarian" interpretation of Chatto's research for an IPCS audience. From Zero to IPC...

6 Oct, 1:05 AM

Sources

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Chats (Frontier)
New page
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Content 17
Notebook Insights
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My second chat on playing card...
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Summary of my jo...

- **From collecting to questioning:** I began by asking Copilot to find an old book about playing cards, then progressed to evaluating Chatto's 1848 work, investigating specific cards and symbols, and demanding exact passages and evidence. 【1-4812d059】
- **From answers to source-controlled learning:** I learned to challenge plausible claims, define which primary sources Copilot could use, build and revise a living Word knowledge base, and test historical instructions by reconstructing and playing Hombre. 【1-4812d059】
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Just start typing...

Auto

Today

evidence. From Zero to IPC...

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Sources

Summary of my journey

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Thanks for your attention!

Ståle Hansen

CEO, Principal Cloud Architect
Copilot MVP & RD

Simon Skotheimsvik

Senior Cloud Architect
Intune MVP

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